## Five must-read papers: semi-synthetic XAI benchmarks

Main text only. References, appendices, checklists and author biographies are cut. Pages are cropped to their text and scaled to fill A4. Each paper starts on a new sheet when printed double-sided. Printed 2026-10-24.

**1. XAI-BENCH** (pages 3-12 here; original pages 1-10)

Liu, Khandagale, White, Neiswanger. Synthetic Benchmarks for Scientific Research in Explainable Machine Learning. NeurIPS 2021 Datasets and Benchmarks.

*Look for:* The standard synthetic tabular benchmark. Section 5.3 imitates real data (Gaussian features with the real covariance, labels from a k-nearest-neighbour model); Table 1 shows how weakly proxy metrics track ground-truth Shapley values.

**2. OpenXAI** (pages 13-22 here; original pages 1-10)

Agarwal et al. OpenXAI: Towards a Transparent Evaluation of Model Explanations. NeurIPS 2022 Datasets and Benchmarks.

*Look for:* The standard toolkit and leaderboard: what a released benchmark is expected to contain. Look for how the synthetic generator keeps features independent so that data-level and model-level ground truth agree.

**3. ITErpretability** (pages 23-32 here; original pages 1-10)

Crabbe, Curth, Bica, van der Schaar. Benchmarking Heterogeneous Treatment Effect Models through the Lens of Interpretability. NeurIPS 2022 Datasets and Benchmarks.

*Look for:* Possibly the closest overlap with the core idea (real feature rows, synthetic outcomes, known important features). Check exactly what it keeps real, what it simulates, and what its ground truth is.

**4. Through the Looking Glass** (pages 33-52 here; original pages 1-20)

Velmurugan, Ouyang, Sindhgatta, Moreira. Through the looking glass: evaluating post hoc explanations using transparent models. International Journal of Data Science and Analytics, 2023.

*Look for:* Transparent models fitted to real datasets, used as the reference for post hoc explainers. Check how the reference explanation is defined (coefficients, rankings) and whether it is exact.

**5. Synthetic Ground-Truth Framework** (pages 53-68 here; original pages 1-13, 17-19)

Miro-Nicolau, Spinnato, Guidotti. A Synthetic Ground-Truth Framework for the Evaluation of Explainable AI Methods. arXiv 2609.30397, 2026.

*Look for:* The newest framework: ground truth defined by interventions on the trained model, across images, time series and tabular data. Its conclusion names hybrids with real data as future work. Figures sit after the references in this preprint and are kept (last three pages).

# Synthetic Benchmarks for Scientific Research in Explainable Machine Learning

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## Abstract

As machine learning models grow more complex and their applications become more high-stakes, tools for explaining model predictions have become increasingly important. This has spurred a flurry of research in model explainability and has given rise to feature attribution methods such as LIME and SHAP. Despite their widespread use, evaluating and comparing different feature attribution methods remains challenging: evaluations ideally require human studies, and empirical evaluation metrics are often data-intensive or computationally prohibitive on real-world datasets. In this work, we address this issue by releasing XAI-BENCH: a suite of synthetic datasets along with a library for benchmarking feature attribution algorithms. Unlike real-world datasets, synthetic datasets allow the efficient computation of conditional expected values that are needed to evaluate ground-truth Shapley values and other metrics. The synthetic datasets we release offer a wide variety of parameters that can be configured to simulate real-world data. We demonstrate the power of our library by benchmarking popular explainability techniques across several evaluation metrics and across a variety of settings. The versatility and efficiency of our library will help researchers bring their explainability methods from development to deployment. Our code is available at https://github.com/abacusai/xai-bench.

## 1 Introduction

The last decade has seen a rapid increase in applications of machine learning in a wide variety of high-stakes domains, such as credit scoring, fraud detection, criminal recidivism, and loan repayment [46, 11, 47, 9]. With the widespread deployment of machine learning models in applications that impact human lives, research on model explainability has become increasingly important. The applications of model explainability include debugging, legal obligations to give explanations, recognizing and mitigating bias, data labeling, and faster adoption of machine learning technologies [41, 69, 7, 21]. Many different methods for explainability are actively being explored, including logic rules [26, 63, 56], hidden semantics [68], feature attribution [51, 41, 50, 15, 61], and explanation by example [38, 13]. The most common type of explainers are post-hoc, local feature attribution methods [69, 41, 1, 51, 50, 15], which output a set of weights corresponding to the importance of each feature for a given datapoint and model prediction. Although various feature attribution methods are being deployed in different use cases today, currently there are no widely adopted methods to easily

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35th Conference on Neural Information Processing Systems (NeurIPS 2021) Track on Datasets and Benchmarks.

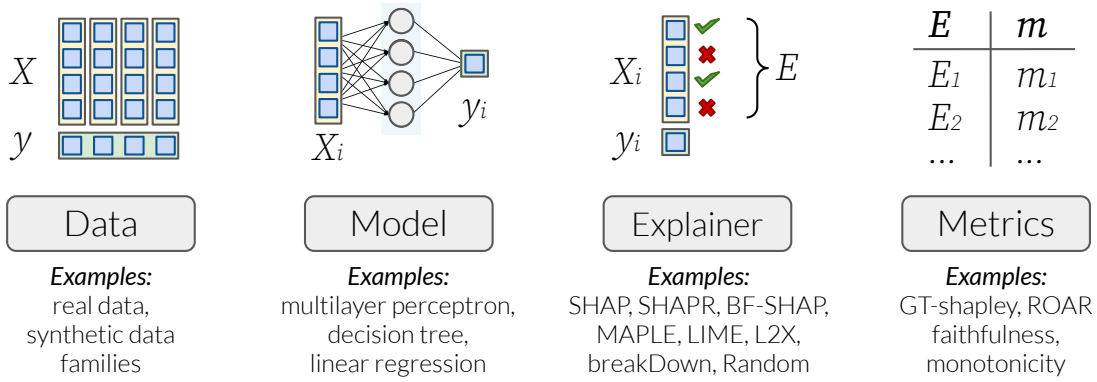



Figure 1: Overview of the main components in XAI-BENCH.

*evaluate and/or compare* different feature attribution algorithms. Indeed, evaluating the effectiveness of explanations is an intrinsically human-centric task that ideally requires human studies. However, it is often desirable to develop new explainability techniques using empirical evaluation metrics before the human trial stage. Although empirical evaluation metrics have been proposed, many of these metrics are either computationally prohibitive or require strong assumptions, to compute on real-world datasets. For example, a popular method for feature attribution is to approximate Shapley values [41, 19, 39, 61], but computing the distance to ground-truth Shapley values requires estimating exponentially many conditional feature distributions, which is not possible to compute unless the dataset contains sufficiently many datapoints across exponentially many combinations of features.

In this work, we overcome these challenges by releasing a suite of synthetic datasets, which make it possible to efficiently benchmark feature attribution methods. The use of synthetic datasets, for which the ground-truth distribution of data is known, makes it possible to exactly compute the conditional distribution over any set of features, thus enabling computations of many feature attribution evaluation metrics such as distance to ground-truth Shapley values [41], remove-and-retrain (ROAR) [31], faithfulness [4], monotonicity [43], and infidelity [67]. Our synthetic datasets offer a wide variety of parameters which can be configured to simulate real-world data and have the potential to identify subtle failures, such as the deterioration of performance on datasets with high feature correlation. We give examples of how real datasets can be converted to similar synthetic datasets, thereby allowing explainability methods to be benchmarked on realistic synthetic datasets.

We showcase the power of our library by benchmarking popular explainers such as SHAP [41], LIME [51], MAPLE [50], SHAPR [1], L2X[15], and breakDown [60], on a broad set of evaluation metrics, across a variety of axes of comparison, such as feature correlation, model type, and data distribution type. Our library is designed to substantially reduce the time required for researchers and practitioners to move their explainability algorithms from development to deployment. Our code, API docs, and raw experimental results are available at `https://github.com/abacusai/xai-bench`. We welcome contributions and hope to grow the repository to handle a wide variety of use-cases.

**Our contributions.** We summarize our main contributions below.

- We release a set of synthetic datasets with known ground-truth distributions, along with a library that makes it possible to efficiently evaluate feature attribution techniques with respect to popular evaluation metrics. Our synthetic datasets offer a number of parameters that can be configured to simulate real-world applications.
- We demonstrate the power of our library by benchmarking popular explainers such as SHAP [41], LIME [51], MAPLE [50], SHAPR [1], L2X[15], and breakDown [60].

## 2 Related Work

Model explainability in machine learning has seen a wide range of approaches, and multiple taxonomies have been proposed to classify the different types of approaches. Zhang et al. [69] describe three dimensions of explainability techniques: passive/active, type of explanation, and local/global explainations. The types of explanations they identified are logic rules [26, 63, 56], hidden semantics [68], feature attribution [51, 41, 50, 15, 61, 1], and explanation by example [38, 13]. Other surveys on explainable AI include Arrieta et al. [6], Adadi and Berrada [2], and Došilović et al. [24].

Techniques for feature attribution include approximating Shapley values [41, 19, 39, 61], approximating the model locally with a more explainable model [51], and approximating the mutual information of each feature with the label [15]. Other work has also identified failure modes for some explanation techniques. For example, recent work has shown that explanation techniques are susceptible to adversarial feature perturbations [23, 58, 30], high feature correlations [35], and small changes in hyperparameters [27, 8].

### 2.1 Benchmarking Explainability Techniques

One recent work [33] gave an experimental survey of explainability methods, testing SHAP [41], LIME [51], Anchors [52], Saliency Maps [57], Grad-CAM++ [12], and their proposed ExMatchina on image, text, audio, and sensory datasets. They use human labeling via Mechanical Turk as an evaluation metric. Another work [7] gave an experimental survey of several algorithms including local/global, white-box/black-box, and supervised/unsupervised techniques. The only feature attribution algorithms they tested were SHAP and LIME. Other recent work gives a benchmark on explainability for time-series classification [25], or for natural language processing (NLP) [21]. Finally, concurrent work [5] releases a library with several evaluation metrics for local linear explanation methods and uses the library to compare LIME and SHAP. To the best of our knowledge, no prior work has released a library with five different evaluation metrics or released a set of synthetic datasets for explainability with more than one tunable parameter.

### 2.2 Metrics

While the "correctness" of feature attribution methods may be subjective or application-specific [66], comparisons between methods are often based on human studies [34, 53, 55]. However, human studies are not always possible, and several empirical (non-human) evaluation metrics have been proposed. Faithfulness [4, 7, 3, 22, 36], infidelity [67, 10, 54], and monotonicity [43, 7, 18] are popular explainability metrics which measure whether each feature's susceptibility to change the model output is aligned with each feature's attribution weight. Another popular metric, remove-and-retrain (ROAR) [31, 28, 29, 44], measures these statistics by retraining the model each time relevant features are removed, in order to avoid inaccuracies due to distribution shift. In the next section, we give the formal definition and a discussion for each metric.

## 3 Evaluation Metrics

### 3.1 Preliminaries

We first give definitions and background information used throughout the next three sections. Given a distribution $\mathcal{D}$, each datapoint is of the form $(\boldsymbol{x}, y) \sim \mathcal{D}$, where $\boldsymbol{x}$ denotes the set of features, and $y$ denotes the label. We assume that $\boldsymbol{x} \in [0,1]^D$, yet all of the concepts we discuss can be generalized to arbitrary categorical and real-valued feature distributions. Assume we have a training set $\mathcal{D}_{\text{train}}$ and a test set $\mathcal{D}_{\text{test}}$, both drawn from $\mathcal{D}$. For the case of regression, we train a model $f : [0,1]^D \to [0,1]$ on the training set. We also implement classification using cross-entropy loss. Common choices for $f$ include a neural network or a decision tree.

A *feature attribution method* is a function $g$ which can be used to estimate the importance of each feature in making a prediction. That is, given a model $f$ and a datapoint $\boldsymbol{x}$, then $g(\boldsymbol{x}, f) = \boldsymbol{w} \in \mathcal{R}^D$, where each output weight $w_i$ corresponds to the relative importance of feature $i$ when making the prediction $f(\boldsymbol{x})$. Common choices for $g$ include SHAP [41] or LIME [51].

### 3.2 Metrics

In this section, we formally define popular evaluation metrics for explainability methods. Each evaluation metric has pros and cons and may be more or less appropriate depending on the application and problem instance. We provide a guide to choosing metrics in Section 3.3.

A *feature attribution evaluation metric* is a function that evaluates the weights of a feature attribution method on a datapoint $\boldsymbol{x}$. For example, given a datapoint $\boldsymbol{x}$ and a set of feature weights $\boldsymbol{w} = g(\boldsymbol{x}, f)$, then a value near or below zero indicates that $g$ did not provide an accurate feature attribution estimate for $\boldsymbol{x}$, while a value near one indicates that $g$ did provide an accurate feature attribution estimate.

Many evaluation metrics involve evaluating the change in performance of the model when a subset of features of a datapoint are removed. In order to measure the true marginal improvement for a set of

features $S$, one approach is to evaluate the model when replacing the features $S$ with their expected values conditioned on the remaining features [19, 62, 14, 32]. Formally, given a datapoint $\boldsymbol{x} \sim \mathcal{D}$ and a set of indices $S \subseteq \{1, \cdots, D\}$, we define $\mathcal{D}(\boldsymbol{x}_S)$ as the conditional probability distribution $\boldsymbol{x}' \sim \mathcal{D}$ such that $x'_i = x_i$ for all $i \in S$. In other words, given $\boldsymbol{x}$ and $S$, we have

$$p(\boldsymbol{x}' \sim \mathcal{D}(\boldsymbol{x}_S)) = p(\boldsymbol{x}' \sim \mathcal{D} \mid x'_i = x_i \text{ for all } i \in S). \tag{1}$$

By this definition, $\mathcal{D}(\boldsymbol{x}_\emptyset) = \mathcal{D}$, and if we define $F = \{1, \cdots, D\}$, then $\boldsymbol{x}' \sim \mathcal{D}(\boldsymbol{x}_F)$ is equal to $\boldsymbol{x}$ with probability 1. Later in this section, we discuss other popular choices such as interventional conditional distributions [42, 1]. Given a datapoint $\boldsymbol{x}$, a model $f$, and a weight vector $\boldsymbol{w}$, the first evaluation metric, **faithfulness** [4], is defined as follows:

$$\text{faithfulness} = \text{Pearson}\left(\left|\mathbb{E}_{\boldsymbol{x}' \sim \mathcal{D}(\boldsymbol{x}_{F \setminus i})}[f(\boldsymbol{x}')] - f(\boldsymbol{x})\right|_{1 \le i \le D}, [w_i]_{1 \le i \le D}\right). \tag{2}$$

Intuitively, faithfulness computes the Pearson correlation coefficient [65] between the weight vector $\boldsymbol{w}$ and the approximate marginal contribution $\left|\mathbb{E}_{\boldsymbol{x}' \sim \mathcal{D}(\boldsymbol{x}_{F \setminus i})}[f(\boldsymbol{x}')] - f(\boldsymbol{x})\right|$ for each feature $i$. Faithfulness is a lightweight metric that is especially useful for comparing which feature would have the most impact on the model output when individually changed.

The next metric computes the marginal improvement of each feature ordered by the weight vector $\boldsymbol{w}$ *without replacement*, and then computes the fraction of indices $i$ such that the marginal improvement for feature $i$ is greater than the marginal improvement for feature $i+1$. This makes it useful when comparing the effect of features as they are added sequentially. Formally, define $S^+(\boldsymbol{w}, i)$ as the set of $i$ most important weights, and let $S^+(\boldsymbol{w}, 0) = \emptyset$. Given a datapoint $\boldsymbol{x}$, a model $f$, and a weight vector $\boldsymbol{w}$, **monotonicity** [43] is defined as follows:

$$\text{monotonicity} = \frac{1}{D-1} \sum_{i=0}^{D-2} \mathbb{I}_{|\delta_i^+| \le |\delta_{i+1}^+|}, \tag{3}$$

$$\text{where } \delta_i^+ = \mathbb{E}_{\boldsymbol{x}' \sim \mathcal{D}(\boldsymbol{x}_{S^+(\boldsymbol{w}, i+1)})}[f(\boldsymbol{x}')] - \mathbb{E}_{\boldsymbol{x}' \sim \mathcal{D}(\boldsymbol{x}_{S^+(\boldsymbol{w}, i)})}[f(\boldsymbol{x}')]. \tag{4}$$

The types of metrics discussed so far all evaluate weight vectors by comparing an estimate of the marginal improvement of a set of features to their corresponding weights. Estimating the marginal improvement requires computing $f$ on different combinations of features, and it is possible that these combinations of features have very low density in $\mathcal{D}$, and are therefore unlikely to occur in $\mathcal{D}_{\text{train}}$. This is especially true for structured data or data where there are large low-density regions in $\mathcal{D}$ that may make the evaluations on $f$ unreliable. To help mitigate this issue, another paradigm of explainability evaluation metrics was proposed: remove-and-retrain (**ROAR**) [31]. In this paradigm, in order to evaluate the marginal improvement of sets of features, the model is retrained using a new dataset with the features removed. For example, rather than computing $|\mathbb{E}_{\boldsymbol{x}' \sim \mathcal{D}(\boldsymbol{x}_{F \setminus i})}[f(\boldsymbol{x}')] - f(\boldsymbol{x})|$, we would compute $|f^*(\mathbb{E}_{\boldsymbol{x}' \sim \mathcal{D}(\boldsymbol{x}_{F \setminus i})}[\boldsymbol{x}']) - f(\boldsymbol{x})|$, where $f^*$ denotes a model that has been trained on a modification of $\mathcal{D}_{\text{train}}$ where each datapoint has its $i$ features with highest weight removed. The original work plots the retrained model performance versus the number of features ablated [31], removing features in order of decreasing importance. Then feature attribution methods are compared by inspecting the steepness of these plots. Follow-up work has compressed the ROAR statistic into a scalar value by computing the area-under-the-curve (AUC) [28, 44]. We use this AUC version in Section 5, to be consistent with the other metrics that only output a single value. Note that to compute ROAR on all datapoints in the test set, the explainer must evaluate all datapoints in the training set to construct $D+1$ ablated datasets, and then the model must be retrained for each of these datasets. We give the formal definition in Appendix E.

A caveat for all of the aforementioned metrics is that they evaluate each feature weight by computing the effect of removing the feature from a single set of features $S$. While this evaluation is sufficient in many cases, it may lead to unreliable measurements for e.g. highly nonlinear models. Furthermore, the explicit goal of a popular line of explainability methods is to obtain fast and accurate approximations of *Shapley values* [41, 1, 40, 19, 39, 61]. To address this, we consider a metric based on Shapley values, **GT-Shapley**, which computes the Pearson correlation coefficient [65] of the feature weights to the ground-truth Shapley values. Shapley values take into account the marginal improvement of a feature $i$ across *all possible* exponentially many sets with and without $i$.

Next, we consider the **infidelity** metric [67]. This metric is computed by considering the effects of replacing each feature with a *noisy* baseline conditional expectation. Instead of computing the correlation between the feature importances and the change in function values (as in faithfulness and GT-Shapley), infidelity computes the difference between the change in function value and the dot product of the change in feature value with the feature importance vector, in expectation over the noise. Note that if we were to only add noise to one feature at a time, this would be similar in spirit to faithfulness (since the dot product would be equal to the weight of the feature which had noise added). Similar to prior work [67], we consider perturbations based on Gaussian noise. Therefore, infidelity can pick up nonlinear trends in feature importances better than faithfulness or monotonicity.

Finally, while Equation (1) defines "observational" conditional expectations [41, 1], we also implement "interventional" conditional expectations [19, 62], which are defined by assuming the features in $S$ are independent of the remaining features. This can be applied to all metrics defined in this section. The best choice of conditional expectations depends on the application [14], and we discuss the tradeoffs in the next section.

### 3.3 A guide to choosing metrics

All of the metrics listed above may be used for evaluating and comparing different feature attribution techniques. However, each metric has strengths and weaknesses, and choosing the most useful metric for a given situation depends on the use case, dataset, feature attribution technique, and computational constraints. We discuss strengths, weaknesses, and example use cases of each metric type.

For the ROAR paradigm, retraining the model with the most important features removed is especially important when the original model is not calibrated for out-of-distribution predictions [31], such as in high-dimensional applications like computer vision [44, 29, 59]. However, retraining might fail to give an accurate evaluation in the presence of high feature correlations [48]. Furthermore, retraining the model incurs a much larger computational cost.

For some feature attribution algorithms, the explicit goal is to efficiently approximate the Shapley values [41, 1, 40, 19, 39, 61], and the GT-Shapley metric is the best choice to determine which technique gives the best approximations to the true Shapley values. However, evaluating the ground-truth Shapley values has a computational cost that is exponential in the number of features. Therefore, the GT-Shapley metric is slow to evaluate on high-dimensional datasets.

Faithfulness, monotonicity, and infidelity are far less computationally intensive compared to ROAR and GT-Shapley. The main difference between faithfulness and monotonicity is that faithfulness considers subsets of features by iteratively removing the most important features *with replacement*, while monotonicity does this *without replacement*. Therefore, the former is better for applications where the main question is which features would individually change the output of the model on a given datapoint (and therefore may be better on datasets with less correlated features). The latter is better for applications where the main goal is to see the cumulative effect of adding features (and therefore performs comparatively better in the presence of correlated features).

The main difference between infidelity and faithfulness (as well as monotonicity) is that infidelity considers ablations of subsets of features, while faithfulness only considers ablating a single feature at a time. Therefore, infidelity may be more appropriate for models with highly nonlinear feature interactions, compared to faithfulness and monotonicity.

Finally, we discuss using interventional versus observational conditional expectations. As pointed out in prior work [14], interventional conditional expectations are better for applications that require being "true to the model", while observational conditional expectations are better for applications that require being "true to the data", because observational conditional expectations tend to spread out importance among correlated features (even features that are not used by the model). For example, interventional conditional expectations are more appropriate in explaining why a model caused a loan to be denied, while observational conditional expectations are more appropriate in explaining the causal features in the drug response to RNA sequences [14].

## 4 Synthetic Datasets

In this section, we describe the synthetic datasets used in our library. We start by discussing the benefits of synthetic datasets when evaluating feature attribution methods, and then describe the feature distributions implemented for these datasets.

### 4.1 The case for synthetic data

As shown in Section 3.2, for multiple metrics it is key to compute the conditional expectation $\mathbb{E}_{\boldsymbol{x}' \sim \mathcal{D}(\boldsymbol{x}_S)}[f(\boldsymbol{x}')]$ for a subset $S$, datapoint $\boldsymbol{x}$, and trained model $f$. On real-world datasets, the conditional distribution $\mathcal{D}(\boldsymbol{x}_S)$ can only be approximated, and the approximation may be very poor when the conditional distribution defines low-density regions of the feature space. Since all evaluation metrics require computing $\Theta(D)$ or $\Theta(2^D)$ expectations for each datapoint $\boldsymbol{x}$, is is likely that some evaluations will make use of a poor approximation. However, for the synthetic datasets that we define, the conditional distributions are known, allowing exact computation of the evaluation metrics.

Additionally, as we show in Section 5, synthetic datasets allow one to explicitly control all attributes of the dataset, which allows for targeted experiments, for example, investigating explainer performance as a function of feature correlation. For explainers such as SHAP [41] which assume feature independence, this type of experiment may be very beneficial. Finally, synthetic datasets can be used to simulate real datasets, which enables fair benchmarking of explainers with quantitative metrics.

### 4.2 Synthetic feature distributions

Now we describe the synthetic datasets in our library. In general, the datasets are expressed as $y = h(\boldsymbol{x})$, with y as label and $\boldsymbol{x}$ as feature vector. The generation is split into two parts, generating features $\boldsymbol{x}$, and defining a function to generate labels $y$ from $\boldsymbol{x}$. We implement multiple families of synthetic distributions in our library, including multivariate Gaussian, mixture of Gaussians, and multinomial feature distributions.

To give a concrete example, we describe here how to generate and use multivariate Gaussian synthetic features. The multivariate normal distribution of a $D$-dimensional random vector $\boldsymbol{X} = (X_1, ..., X_D)^T$ can be written as $\boldsymbol{X} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, where $\boldsymbol{\mu}$ is the $D$-dimensional mean vector, and $\boldsymbol{\Sigma}$ is the $D \times D$ covariance matrix. Without loss of generality, we can partition the $D$-dimensional vector $\boldsymbol{x}$ as $\boldsymbol{X} = (\boldsymbol{X}_1, \boldsymbol{X}_2)^T$. To compute the distribution of $\boldsymbol{X}_1$ conditional on $\boldsymbol{X_2} = \boldsymbol{x_2^*}$ where $\boldsymbol{x_2^*}$ is a $K$-dimensional vector with $0 < K < D$, we can then partition $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ accordingly:

$$\boldsymbol{\mu} = \begin{bmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{bmatrix}, \quad \boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix}.$$

Then the conditional distribution is a new multivariate normal $(\boldsymbol{X}_1 | \boldsymbol{X}_2 = \boldsymbol{x}_2^*) \sim \mathcal{N}(\boldsymbol{\mu}^*, \boldsymbol{\Sigma}^*)$ where

$$\boldsymbol{\mu}^* = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\boldsymbol{x}_2^* - \boldsymbol{\mu}_2), \quad \boldsymbol{\Sigma}^* = \boldsymbol{\Sigma}_{11} + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}. \tag{5}$$

For any $\boldsymbol{x}_2^* \in \mathbb{R}^K$, one can compute $\boldsymbol{\mu}^*$ and $\boldsymbol{\Sigma}^*$ and then generate samples from the conditional distribution. Parameter $\boldsymbol{\mu}$ can take any value, and $\boldsymbol{\Sigma}$ must be symmetric and positive definite. Similarly, we also give the derivation for additional distribution families in Appendix D, including mixtures of multivariate Gaussians, and multinomial features.

### 4.3 Labels

After defining a distribution of features via one of the above distribution families, we can then define a distribution over labels. The distributions we implement are `linear`, `piecewise constant`, `nonlinear additive`, and `piecewise linear`.

Data labels are computed in two steps: *(1)* raw labels are computed from features, i.e. $y_{\text{raw}} = \sum_{n=1}^{D} \Psi_n(x_n)$ where $\Psi_n$ is a function that operates on feature $n$, and *(2)* final labels are normalized to have zero mean and unit variance. The normalization ensures that a baseline ML model, which always predicts the mean of the dataset, has an MSE of 1. This allows results derived from different types of datasets to be comparable at scale.

For `linear` datasets, $\Psi_n(x_n)$ are scalar weights, and we can rewrite the raw labels as $y_{raw} = \boldsymbol{w}^T\boldsymbol{x}$. In our experiments in Section 5, we set $\boldsymbol{w} = [0, 1, \ldots, d-1]$. `piecewise linear` datasets are similar to `linear`, but a different weight vector is used in different parts of the feature space. In our experiments in Section 5, on the datasets with continuous features, we set $\boldsymbol{w} = [0, 1, \ldots, d-1]$ when the sum of the feature values is positive, and $\boldsymbol{w} = [d-1, d-2, \ldots, 0]$ otherwise. For `piecewise constant` datasets, $\Psi_n(x_n)$ are piecewise constant functions made up of different threshold values (similar to Aas et al. [1]). For `nonlinear additive` datasets, $\Psi_n(x_n)$ are nonlinear functions including *absolute*, *cosine*, and *exponent* function adapted from Chen et al. [15]. Detailed specifications can be found in Appendix F.

# 5 Experiments

We show experiments on several popular feature attribution methods across synthetic datasets.

## 5.1 Feature attribution methods

We compare eight different feature attribution methods: SHAP [41], SHAPR [1], brute-force Kernel SHAP (BF-SHAP) [41], LIME [51], MAPLE [50], L2X [15], breakDown [60], and the baseline RANDOM, which outputs random weights drawn from a standard normal distribution. We ran light hyperparameter tuning on all datasets. See Appendix E for details and descriptions for all methods. We report the mean and standard deviation from ten trials for all experiments.

## 5.2 Parameterized synthetic data experiments

We first show experiments using multivariate Gaussian datasets described in Section 4. Without loss of generality, we can assume that the feature set is normalized (in other words, $\boldsymbol{\mu}$ is set to 0, and the diagonal of $\boldsymbol{\Sigma}$ is set to 1). In all sections except Section 5.3, we set the non-diagonal terms of $\boldsymbol{\Sigma}$ to $\boldsymbol{\rho}$, which allows for the convenient parameterization of a global level of feature dependence [1].

We run experiments that compare eight feature attribution methods on the five evaluation metrics defined in Section 3.1 across several datasets and ML models. We conduct experiments by varying one or two of these dimensions at a time while holding the other dimensions fixed (for example, we compare different datasets while keeping the ML model fixed) and in Appendix G, we give the exhaustive set of experiments. Throughout this section, we will identify different types of failure modes, for example, failures for some explainability techniques over specific metrics (Table 1) or failures for some techniques on datasets with high levels of feature correlation (Figures 2 and 3).

**Performance across metrics** As shown in Table 1, the relative performance of explainers varies dramatically across metrics for a fixed multilayer perceptron trained on a `nonlinear additive` dataset with $\rho = 0.5$. Since $\rho = 0.5$ implies that the features are fairly correlated, we find that SHAPR outperforms SHAP on GT-Shapley, which is consistent with the fact that SHAPR was designed to outperform SHAP in the presence of dependent features [1]. SHAPR achieved the top performance for three metrics, but MAPLE had the most consistent performance across all five metrics. One possible explanation for this is that MAPLE draws on ideas from three different areas of explainability: example-based, local, and global explanations [50], which helps it achieve steady performance across many metrics. Finally, while breakDown achieves the worst score for GT-Shapley, it achieves the best score for monotonicity. Note that breakDown works by greedily choosing the features with the greatest effect on the model output, *with replacement*, making it particularly well-suited for the monotonicity metric, which checks whether replacing features sorted by importance with their background value with replacement monotonically decreases the change in model output.

Table 1: Explainer performance across metrics. All performance numbers are from explaining a multilayer perceptron trained on the Gaussian nonlinear additive dataset with $\rho = 0.5$.

| | RANDOM | SHAP | SHAPR | LIME | MAPLE | L2X | BREAKDOWN |
|---|---|---|---|---|---|---|---|
| faithfulness(↑) | $0.002_{\pm 0.034}$ | $0.651_{\pm 0.051}$ | $\mathbf{0.799}_{\pm 0.036}$ | $0.524_{\pm 0.06}$ | $0.478_{\pm 0.061}$ | $0.000_{\pm 0.075}$ | $0.110_{\pm 0.049}$ |
| monotonicity(↑) | $0.525_{\pm 0.017}$ | $0.537_{\pm 0.014}$ | $0.550_{\pm 0.025}$ | $0.517_{\pm 0.022}$ | $0.543_{\pm 0.026}$ | $0.535_{\pm 0.022}$ | $\mathbf{0.562}_{\pm 0.021}$ |
| ROAR(↑) | $0.380_{\pm 0.051}$ | $0.455_{\pm 0.054}$ | $\mathbf{0.465}_{\pm 0.054}$ | $0.432_{\pm 0.051}$ | $0.432_{\pm 0.059}$ | $0.365_{\pm 0.053}$ | $0.329_{\pm 0.057}$ |
| GT-Shapley(↑) | $0.004_{\pm 0.049}$ | $0.810_{\pm 0.023}$ | $\mathbf{0.930}_{\pm 0.012}$ | $0.711_{\pm 0.032}$ | $0.530_{\pm 0.128}$ | $-0.014_{\pm 0.068}$ | $-0.127_{\pm 0.066}$ |
| infidelity(↓) | $0.114_{\pm 0.058}$ | $0.050_{\pm 0.023}$ | $0.036_{\pm 0.013}$ | $0.053_{\pm 0.016}$ | $\mathbf{0.019}_{\pm 0.011}$ | $0.025_{\pm 0.010}$ | $0.126_{\pm 0.057}$ |

**Performance across dataset types and feature correlations** Next, we explore how the type of dataset and feature correlation affects performance of explainers on a multilayer perceptron with the faithfulness metric. As shown in Figure 2, a general trend is that explainers become less faithful as feature correlation increases. Explainers such as Kernel SHAP assume feature independence [1, 45] and tend to perform well when features are indeed independent ($\rho = 0$). This is especially apparent with the `linear` dataset, where the performance of most methods cluster above 0.9 at $\rho = 0$. However, LIME's performance drops as much as $\sim 90\%$ when features are almost perfectly correlated ($\rho = 0.99$). On the other hand, for both the `nonlinear additive` and `piecewise constant` datasets, MAPLE's performance stayed relative stable across values of $\rho$. For experiments on the `piecewise linear` dataset, see Appendix G.

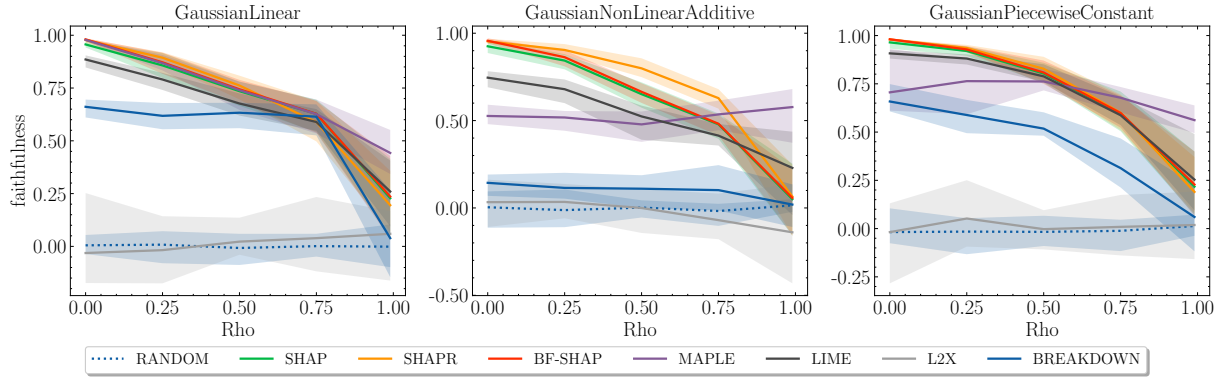



Figure 2: Results for faithfulness on a multilayer perceptron trained on three different datasets.

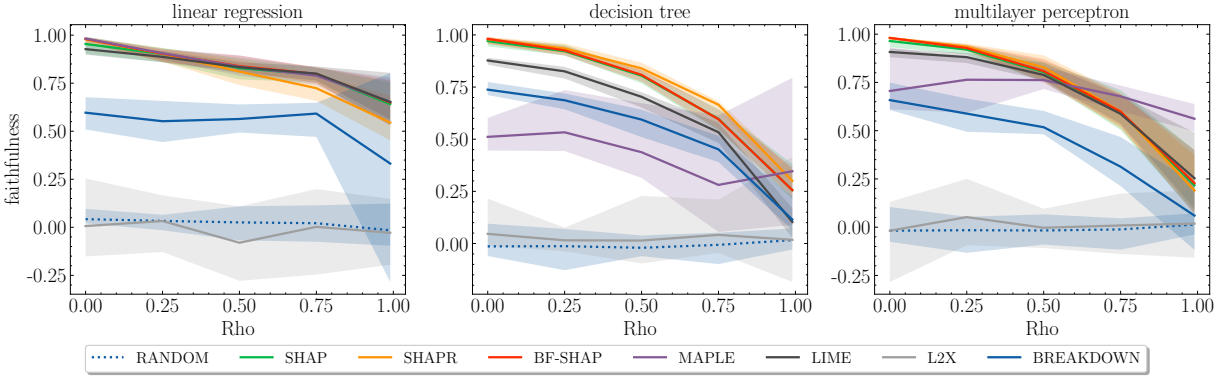



Figure 3: Results for faithfulness for three types of ML models—linear regression, decision tree, and multilayer perceptron—trained on a Gaussian piecewise constant dataset.

**Performance across ML models** Next, we train three ML models: linear regression, decision tree, and multilayer perceptron, with a `piecewise constant` dataset and compare faithfulness. Figure 3 shows that as in Figure 2, explainer performance drops as features become more correlated. Most explainers perform well for linear regression up to $\rho = 0.75$. The performance of SHAP, SHAPR, and LIME remain relatively consistent across ML models. In contrast, MAPLE performs significantly worse on the decision tree model.

### 5.3 Simulating real datasets

In this section, we demonstrate the power and flexibility of synthetic datasets by simulating two popular datasets: the wine quality dataset [16, 60] and the forest fire dataset [17] with synthetic features so that they can be used to efficiently benchmark feature attribution methods.

**Wine quality dataset** The wine dataset has 11 continuous features ($\boldsymbol{x}_{\text{real}}$) and one integer quality rating ($y_{\text{real}}$) between 0 and 10. In this section, it is formulated as a regression task, but it can also be formulated as a multi-class classification task. The features are first normalized to have zero mean and unit variance, then an empirical covariance matrix is computed (Appendix Figure 5), which is then used as the input covariance matrix to generate synthetic multivariate Gaussian features ($\boldsymbol{x}_{\text{sim}}$). Simulated wine quality ($y_{\text{sim}}$) is labeled by a $k$-nearest neighbor model based on real datapoints ($\boldsymbol{x}_{\text{real}}, y_{\text{real}}$).

We evaluate how close the simulated dataset is to the real one in two steps. First, we compute the Jensen-Shannon Divergence (JSD) [64] of the real and synthetic wine datasets. JSD measures the similarity between two distributions; it is bounded between 0 and 1, and lower JSD suggests higher similarity between two distributions. The JSD of marginal distributions between the real empirical features and the synthetic Gaussian features has a mean of 0.20, and the JSD of real and synthetic targets is 0.23, suggesting a good fit. Second, we train three types of ML models on both simulated and real wine datasets and compare the MSE of explanations on a common held-out real test set. As shown in Appendix Table 5, consistent low MSE across ML models and explainers suggest that the simulated dataset is a good proxy for the original wine dataset for evaluating explainers.

Next, we compute evaluation metrics for seven different explainers on the synthetic wine dataset. Note that computing these metrics accurately is not possible on the real wine dataset, as the conditional distribution is unknown. As shown in Table 2, SHAPR performs well on GT-Shapley, consistent with Table 1. SHAP and SHAPR both outperform LIME and MAPLE on faithfulness.

**Forest fire dataset** The forest fire dataset has 12 continuous features and one real-valued label indicating the area of burned forest. Again, we normalize the features to have zero mean and unit variance, and then we compute the covariance matrix, which is used to generate the synthetic dataset (the same way as the wine quality dataset above).

For the forest fire dataset, the JSD of marginal distributions between the real empirical features and the synthetic Gaussian features has a mean of $0.17$, and the JSD of real and synthetic targets is $0.15$, suggesting a good fit. We compute evaluation metrics for six different explainers on the synthetic forest fire dataset. See Table 3. SHAP achieved top performance on three of the five metrics.

Table 2: Explainer performance on the simulated wine dataset across metrics. All performance numbers are from explainers for a decision tree.

| | RANDOM | SHAP | SHAPR | LIME | MAPLE | L2X | BREAKDOWN |
|---|---|---|---|---|---|---|---|
| faithfulness (↑) | $-0.007_{\pm 0.005}$ | $\mathbf{0.534}_{\pm 0.045}$ | $0.528_{\pm 0.032}$ | $0.368_{\pm 0.031}$ | $0.034_{\pm 0.033}$ | $-0.030_{\pm 0.018}$ | $-0.042_{\pm 0.011}$ |
| monotonicity (↑) | $0.529_{\pm 0.008}$ | $0.549_{\pm 0.009}$ | $\mathbf{0.551}_{\pm 0.009}$ | $0.547_{\pm 0.007}$ | $0.520_{\pm 0.014}$ | $0.522_{\pm 0.005}$ | $0.493_{\pm 0.014}$ |
| ROAR (↑) | $0.698_{\pm 0.031}$ | $0.780_{\pm 0.016}$ | $0.549_{\pm 0.031}$ | $0.738_{\pm 0.026}$ | $\mathbf{0.818}_{\pm 0.022}$ | $0.664_{\pm 0.02}$ | $0.625_{\pm 0.002}$ |
| GT-Shapley (↑) | $0.004_{\pm 0.013}$ | $0.825_{\pm 0.006}$ | $\mathbf{0.945}_{\pm 0.002}$ | $0.745_{\pm 0.015}$ | $0.685_{\pm 0.008}$ | $-0.108_{\pm 0.029}$ | $-0.064_{\pm 0.02}$ |
| infidelity (↓) | $0.353_{\pm 0.174}$ | $0.234_{\pm 0.124}$ | $\mathbf{0.212}_{\pm 0.146}$ | $0.234_{\pm 0.126}$ | $0.234_{\pm 0.132}$ | $0.285_{\pm 0.115}$ | $0.365_{\pm 0.133}$ |

Table 3: Explainer performance on the simulated forest fires dataset across metrics. All performance numbers are from explainers for a decision tree.

| | RANDOM | SHAP | LIME | MAPLE | L2X | BREAKDOWN |
|---|---|---|---|---|---|---|
| faithfulness (↑) | $0.022_{\pm 0.034}$ | $\mathbf{0.571}_{\pm 0.023}$ | $0.449_{\pm 0.007}$ | $0.080_{\pm 0.056}$ | $0.001_{\pm 0.008}$ | $0.158_{\pm 0.032}$ |
| monotonicity (↑) | $0.537_{\pm 0.02}$ | $0.591_{\pm 0.007}$ | $\mathbf{0.598}_{\pm 0.002}$ | $0.561_{\pm 0.002}$ | $0.527_{\pm 0.01}$ | $0.575_{\pm 0.012}$ |
| ROAR (↑) | $0.575_{\pm 0.002}$ | $0.615_{\pm 0.011}$ | $0.616_{\pm 0.008}$ | $\mathbf{0.696}_{\pm 0.024}$ | $0.534_{\pm 0.018}$ | $0.604_{\pm 0.019}$ |
| GT-Shapley (↑) | $0.012_{\pm 0.06}$ | $\mathbf{0.870}_{\pm 0.005}$ | $0.779_{\pm 0.027}$ | $0.804_{\pm 0.011}$ | $0.031_{\pm 0.12}$ | $0.105_{\pm 0.013}$ |
| infidelity (↓) | $0.207_{\pm 0.125}$ | $\mathbf{0.075}_{\pm 0.074}$ | $0.077_{\pm 0.075}$ | $0.077_{\pm 0.079}$ | $0.091_{\pm 0.07}$ | $0.117_{\pm 0.076}$ |

### 5.4 Recommended usage

In Section 5, we gave a sample of the types of experiments that can be performed with our library (recall that comprehensive experiments are in Appendix G). For researchers looking to develop new explainability techniques, we recommend benchmarking new algorithms across all metrics using our synthetic datasets with different values of $\rho$. These datasets give a good initial picture of the efficacy of new techniques. For researchers with a dataset and application in mind, we recommend converting the dataset into a synthetic dataset using the technique described in Section 5.3. Note that converting to a synthetic dataset also gives the ability to evaluate explainability techniques on perturbations of the original covariance matrix, to simulate robustness to distribution shift. Finally, researchers can decide on the evaluation metric that is most suitable to the application at hand. See Section 3.3 for a guide to choosing the best metric based on the application.

## 6 Societal Impact

Machine learning models are more prevalent now than ever before. With the widespread deployment of models in applications that impact human lives, explainability is becoming increasingly important for the purposes of debugging, legal obligations, and mitigating bias [41, 69, 7, 21]. Given the importance of high-quality explanations, it is essential that explainability methods are reliable across all types of datasets. Our work seeks to speed up the development of explainability methods, with a focus on catching edge cases and failure modes, to ensure that new explainability methods are robust before they are used in the real world. Of particular importance are improving the reliability of explainability methods intended to recognize biased predictions, for example, ensuring that the features used to predict criminal recidivism are not based on race or gender [37]. Frameworks for evaluating and comparing explainability methods are an important part of creating inclusive and

unbiased technology. As pointed out in prior work [20], while methods for explainability or debiasing are important, they must be part of a larger, socially contextualized project to examine the ethical considerations of the machine learning application.

## 7 Conclusions and Limitations

In this work, we released a set of synthetic datasets along with a library for benchmarking feature attribution algorithms. The use of synthetic datasets with known ground-truth distributions makes it possible to exactly compute the conditional distribution over any set of features, enabling accurate computations of several explainability evaluation metrics, including ground-truth Shapley values, ROAR, faithfulness, and monotonicity. Our synthetic datasets offer a variety of parameters which can be configured to simulate real-world data and have the potential to identify failure modes of explainability techniques, for example, techniques whose performance is negatively correlated with dataset feature correlation. We showcase the power of our library by benchmarking several popular explainers with respect to five evaluation metrics across a variety of settings.

Despite the fact that the synthetic datasets aim to cover a broad range of feature distributions, correlations, scales, and target generation functions, there is almost certainly a gap between synthetic and real-world datasets. However, as discussed before, it is often the case that we do not know the ground truth generative model of real datasets, thus making it impossible to compute many objective metrics. Hence, there is a trade-off between data realism and ground truth availability.

Note that our library is **not** meant to be a replacement for human interpretability studies. Since the goals of explainability methods are inherently human-centric, the only foolproof method of evaluating explanation methods are to use human trials. Rather, our library is meant to substantially speed up the process of development, refinement, and identifying failures, before reaching human trials.

Overall, we recommend developing new explainability methods in this library, and then conducting human trials on real data. Our library is designed to substantially accelerate the process of moving new explainability algorithms from development to deployment. With the release of API documentation, walkthroughs, and a contribution guide, we hope that the scope of our library can increase over time.

## Acknowledgments and Disclosure of Funding

Work done while the first three authors were working at Abacus.AI. WN was supported by U.S. Department of Energy Office of Science under Contract No. DE-AC02-76SF00515.

# OpenXAI: Towards a Transparent Evaluation of Post hoc Model Explanations

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## Abstract

While several types of post hoc explanation methods have been proposed in recent literature, there is very little work on systematically benchmarking these methods. Here, we introduce OpenXAI, a comprehensive and extensible open-source framework for evaluating and benchmarking post hoc explanation methods. OpenXAI comprises of the following key components: (i) a flexible synthetic data generator and a collection of diverse real-world datasets, pre-trained models, and state-of-the-art feature attribution methods, (ii) open-source implementations of twenty-two quantitative metrics for evaluating faithfulness, stability (robustness), and fairness of explanation methods, and (iii) the first ever public XAI leaderboards to readily compare several explanation methods across a wide variety of metrics, models, and datasets. OpenXAI is easily extensible, as users can readily evaluate custom explanation methods and incorporate them into our leaderboards.

Overall, OpenXAI provides an automated end-to-end pipeline that not only simplifies and standardizes the evaluation of post hoc explanation methods, but also promotes transparency and reproducibility in benchmarking these methods. While the first release of OpenXAI supports only tabular datasets, the explanation methods and metrics that we consider are general enough to be applicable to other data modalities. OpenXAI datasets and data loaders, implementations of state-of-the-art explanation methods and evaluation metrics, as well as leaderboards are publicly available at `https://open-xai.github.io/`. OpenXAI will be regularly updated to incorporate text and image datasets, other new metrics and explanation methods, and welcomes inputs from the community.

## 1 Introduction

As predictive models are increasingly deployed in critical domains (e.g., healthcare, law, and finance), there has been a growing emphasis on explaining the predictions of these models to decision makers (e.g. doctors, judges) so that they can understand the rationale behind model predictions, and determine if and when to rely on these predictions. To this end, various techniques have been proposed in recent literature to generate post hoc explanations of individual predictions made by complex ML models. Several of such *local explanation methods* output the influence of each of the features on the model's prediction, and are therefore referred to as *local feature attribution methods*. Due to their generality, feature attribution methods are increasingly being utilized to explain complex models in medicine, finance, law, and science [23, 34, 78]. Thus, it is critical to ensure that the explanations generated by these methods are *reliable* so that relevant stakeholders and decision makers are provided with credible information about the underlying models [6].

36th Conference on Neural Information Processing Systems (NeurIPS 2022) Track on Datasets and Benchmarks.

Prior works have studied several notions of explanation reliability such as *faithfulness* (or fidelity) [81, 51, 33], *stability* (or robustness) [7, 4], and *fairness* [18, 9], and proposed metrics for quantifying these notions. Many of these works also demonstrated through small-scale experiments or qualitative analysis that certain explanation methods are not effective w.r.t. specific notions of reliability. For instance, Alvarez-Melis and Jaakkola [7] visualized the explanations generated by some of the popular gradient based explanation methods [67, 66, 70, 72] for MNIST images, and showed that they are not robust to small input perturbations. However, it is unclear if such findings generalize beyond the settings studied. More broadly, one of the biggest open questions which has far-reaching implications for the progress of explainable AI (XAI) research is: *which explanation methods* are effective w.r.t. *what notions of reliability* and *under what conditions*? [43]. A first step towards answering this question involves systematically benchmarking explanation methods in a reproducible and transparent manner. However, the increasing diversity of explanation methods, and the plethora of evaluation settings and metrics outlined in existing research without standardized open-source implementations make it rather challenging to carry out such benchmarking efforts.

In this work, we address the aforementioned challenges by introducing OpenXAI, a comprehensive and extensible open-source framework for systematically and efficiently benchmarking explanation methods in a transparent and reproducible fashion. More specifically, our work makes the following key contributions:

1. We introduce the OpenXAI framework, an *open-source ecosystem* designed to support systematic, reproducible, and efficient evaluations of post hoc explanation methods. OpenXAI unifies the existing scattered repositories of datasets, models, and evaluation metrics, and provides a *simple and easy-to-use API* that enables researchers and practitioners to benchmark explanation methods using just a few lines of code (Section 2).
2. Our OpenXAI framework currently provides open-source implementations and ready-to-use API interfaces for *seven state-of-the-art feature attribution methods* (LIME, SHAP, Vanilla Gradients, Gradient x Input, SmoothGrad, and Integrated Gradients), and *twenty-two quantitative metrics* to evaluate the faithfulness, stability, and fairness of feature attribution methods. In addition, it includes a comprehensive collection of *seven real-world datasets* spanning diverse real-world domains, and *sixteen different pre-trained models*. OpenXAI also introduces *a novel and flexible synthetic data generator* to synthesize datasets of varying sizes, complexity, and dimensionality which facilitate the construction of reliable ground truth explanations (Section 2).
3. As part of our OpenXAI framework, we also develop the first-ever *public XAI leaderboards* (shown in Figure 1) to promote transparency, and to allow users to easily compare the performance of multiple explanation methods across a wide variety of synthetic and real-world datasets, evaluation metrics, and predictive models.
4. OpenXAI framework is *easily extensible* i.e., researchers and practitioners can readily incorporate custom explanation methods, datasets, predictive models, and evaluation metrics into our framework and leaderboards (Section 2).
5. Lastly, using our proposed OpenXAI framework, we *perform rigorous empirical benchmarking* of the aforementioned state-of-the-art feature attribution methods to determine which methods are effective w.r.t. what notions of reliability across a wide variety of datasets and predictive models (Section 3).

Overall, our OpenXAI framework provides an end-to-end pipeline that unifies, simplifies, and standardizes several existing workflows to evaluate explanation methods. By enabling systematic and efficient evaluation and benchmarking of existing and new explanation methods, our OpenXAI framework can inform and accelerate new research in the emerging field of XAI. OpenXAI will be regularly updated and welcomes input from the community.

**Related Work.** Our work builds on the vast literature in explainable AI. Here, we discuss closely related works and their connections to our benchmark. A more detailed discussion of the related work is included in the Appendix.

*Evaluation Metrics for Post hoc Explanations*: Prior research has studied several notions of explanation reliability, namely, *faithfulness* (or fidelity), *stability* (or robustness), and *fairness* [51, 81, 7, 18]. While the faithfulness notion captures how faithfully a given explanation captures the true behavior of the underlying model [81, 51, 33], stability ensures that explanations do not change

drastically with small perturbations to the input [29, 7]. The fairness notion, on the other hand, ensures that there are no group-based disparities in the faithfulness or stability of explanations [18]. To this end, prior works [51, 69, 81, 7, 18, 33] proposed various evaluation metrics to quantify the aforementioned notions. For instance, Petsiuk et al. [59] measured the change in the probability of the predicted class when important features (as identified by an explanation) are deleted from or introduced into the data instance. A sharp change in the probability implies a high degree of explanation faithfulness. Alvarez-Melis and Jaakkola [7] loosely quantified stability as the maximum change in the resulting explanations when small perturbations are made to a given instance. Dai et al. [18] quantified unfairness of explanations as the difference between the faithfulness (or stability) metric values averaged over instances in the majority and the minority subgroups.

*XAI Libraries and Benchmarks:* Prior works have introduced a few XAI libraries and benchmarks, the most popular among them being Captum [42], Quantus [32], XAI-Bench [51], and SHAP Benchmark. Below, we provide a brief description of each of these, and detail how our work differs from them.

While ***Captum library*** [42] is an open-source library which provides implementations and APIs for various state-of-the-art explanation methods, its focus is not on evaluating and/or benchmarking these methods which is the main goal of our work. ***Quantus library*** [32], on the other hand, provides implementations of certain evaluation metrics to measure the faithfulness and stability/robustness of explanation methods. However, it does not focus on benchmarking explanation methods or providing public dashboards to compare the performance of these methods. Furthermore, the stability/robustness measures [7] supported by Quantus are somewhat outdated and have been superseded by recently proposed metrics [4]. In addition, Quantus does not support any fairness metrics to evaluate disparities in the quality of explanations which is very important in real-world settings such as healthcare, criminal justice, and policy. In contrast, OpenXAI not only subsumes popular faithfulness and stability/robustness metrics supported by Quantus but also supports 19 new metrics to measure the faithfulness, stability/robustness, as well as the fairness of explanation methods [4, 18, 43]. In addition, OpenXAI focuses on systematically benchmarking state-of-the-art explanation methods and providing public dashboards to readily compare these methods.

***SHAP benchmark*** [2] only focuses on evaluating and comparing different variants of SHAP [54] via certain faithfulness metrics which are similar to the Prediction Gap on Important (PGI) and Unimportant (PGU) feature perturbation metrics outlined in our work. Note that the SHAP benchmark does not include any stability/robustness or fairness metrics. In contrast, OpenXAI not only includes 20 new metrics to evaluate the stability/robustness and fairness of explanation methods but also benchmarks various other methods (e.g., LIME, Gradient-based methods).

***XAI-Bench*** [51] constructed synthetic datasets with ground truth explanations to evaluate the faithfulness of a few explanation methods (e.g., LIME, SHAP, MAPLE). However, recent research argued that their evaluation is unreliable, and predictive models learned using their synthetic datasets may not adhere to the ground truth explanations [24]. In addition, the aforementioned evaluation is rather limited in scope as synthetic datasets may not even be representative of real-world data [24]. In contrast, our work not only proposes a novel synthetic data generator that addresses the shortcomings of the synthetic datasets constructed in XAI-Bench but also facilitates the evaluation and benchmarking of the faithfulness, stability, as well as the fairness of 7 state-of-the-art explanation methods on 7 real-world datasets with no ground truth explanations.

In summary, our work is significantly different from existing libraries and benchmarks, and makes the following key contributions:

- We provide implementations and easy-to-use API interfaces for 22 metrics to evaluate the faithfulness, stability, and fairness of explanation methods. 18 out of the 22 state-of-the-art metrics included in OpenXAI have not been implemented in any prior libraries or benchmarks – e.g., faithfulness metrics such as Feature Agreement (FA), Rank Agreement (RA), Sign Agreement (SA), Signed Rank Agreement (SRA), Pairwise Rank Agreement (PRA), stability metrics such as Relative Representation Stability (RRS), Relative Output Stability (ROS), and all fairness metrics.
- We also introduce a novel and flexible synthetic data generator to synthesize datasets of varying sizes, complexity, and dimensionality to facilitate the construction of reliable ground truth explanations in order to evaluate state-of-the-art explanation methods. Our synthetic data generator addresses the shortcomings of the prior synthetic benchmark (XAI-

Bench) by generating synthetic datasets which encapsulate certain key properties, namely, unambiguously defined local neighborhoods, a clear description of feature importances in each local neighborhood, and feature independence. These properties, in turn, allow us to theoretically guarantee that any accurate model trained on our synthetic datasets will adhere to the ground truth explanations of the underlying data.

- We perform rigorous empirical benchmarking of 7 state-of-the-art feature attribution methods using our OpenXAI framework to determine which methods are effective w.r.t. each of the 22 evaluation metrics across 8 real-world and synthetic datasets, and 16 different predictive models. Note that none of the previously proposed libraries or benchmarks carry out such exhaustive benchmarking efforts across such a wide variety of metrics, models, and datasets. We also introduce the first ever public XAI leaderboards with such a wide variety of explanation methods, metrics, models, and datasets, to promote transparency and showcase the results of our benchmarking efforts.

## 2 Overview of OpenXAI Framework

OpenXAI provides a comprehensive programmatic environment with synthetic and real-world datasets, data processing functions, explainers, and evaluation metrics to rigorously and efficiently benchmark explanation methods. Below, we discuss each of these components in detail.

**1) Datasets and Predictive Models.** The current release of our OpenXAI framework includes a collection of eight different synthetic and real-world datasets. While synthetic datasets allow us to construct ground truth explanations which can then be used to evaluate explanations output by state-of-the-art methods, real-world datasets (where it is typically hard to construct ground truth explanations) help us benchmark these methods in a more realistic manner suitable for practical applications [51]. We would like to note that OpenXAI includes datasets that are widely employed in XAI research to evaluate the efficacy of newly proposed methods and study the behavior of existing methods [9, 18–20, 38, 69, 73].

*Synthetic Datasets*: While prior research [51, 41] proposed methods to generate synthetic datasets and corresponding ground truth explanations, they all suffer from a significant drawback as demonstrated by Faber et al. [24] – there is no guarantee that the models trained on these datasets will adhere to the ground truth explanations of the underlying data. This, in turn, implies that evaluating post hoc explanations using the above ground truth explanations would be incorrect since post hoc explanations are supposed to reliably explain the behavior of the underlying model, and not that of the underlying data. To illustrate, let us consider the case where we use aforementioned methods to construct a synthetic dataset with features $A, B$, $C$, and $D$ such that the ground truth labels only depend on features $A$ and $B$ i.e., the ground truth explanation of the underlying data indicates that features $A$ and $B$ are most important. If we train a model on this data and if features $A$ and $B$ are correlated with $C$ and $D$ respectively, then the resulting model may base its predictions on $C$ and $D$ (and not $A$ and $B$) and still be very accurate. If a post hoc explanation of this model then (correctly) indicates that the most important features of the model are $C$ and $D$, this explanation may be deemed incorrect if we compare it against the ground truth explanation of the underlying data. This problem further exacerbates as we increase the complexity of the ground truth labeling function [24].

To address the aforementioned challenges, we develop a novel synthetic data generation mechanism, *SynthGauss*, which encapsulates three key properties, namely, feature independence, unambiguously-defined local neighborhoods, and a clear description of feature influence in each local neighborhood. Intuitively, this approach generates $K$ well-separated clusters where points in each cluster $k \in \{1, 2, \cdots, K\}$ are sampled from a Gaussian distribution $\mathcal{N}(\mu_k, \Sigma_k)$ where $u_k \in R^d$ is the mean and $\Sigma_k \in R^{d \times d}$ is the covariance matrix. While this parameterization is general enough to support the construction of synthetic datasets of $K$ clusters with varying means and covariances, we set the means of all the clusters such that the intracluster distances are significantly smaller than the intercluster distances, and we set the covariance matrices of all the clusters to identity. This ensures that all the features are independent, and local neighborhoods (clusters) are unambiguously defined.

We then generate ground truth labels for instances by first randomly sampling feature mask vectors $m_k \in \{0, 1\}^d$ (vectors comprising of 0s and 1s) for each cluster $k$. The vector $m_k$ determines which features influence the ground truth labeling process for instances in cluster $k$ (a value of 1 indicates that the corresponding feature is influential). We then randomly sample feature weight

vectors $w_k \in R^d$ which capture the relative importance of each of the features in the labeling process of instances in each cluster $k$. The ground truth labels of instances in each cluster $k$ are then computed as a function (e.g., sigmoid) of the feature values of individual instances, and the dot product of the corresponding cluster's feature mask vector and weight vector i.e., $m_k \odot w_k$. Complete pseudocode and other details of this generation process are included in the Appendix. Note that $m_k$ corresponds to the ground truth explanation for all instances in cluster $k$. Since our generation process is designed to encapsulate feature independence, unambiguous definitions of local neighborhoods, and clear descriptions of feature influences, any accurate model trained on the resulting dataset will adhere to the ground truth explanations of the underlying data (See Theorem 1 in Appendix).

*Real-world Datasets*: In the current release of OpenXAI, we include seven real-world datasets that are highly diverse in terms of several key properties. They comprise of data spanning multiple real-world domains (e.g., finance, lending, healthcare, and criminal justice), varying dataset sizes (e.g., small vs. large-scale), dimensionalities (e.g., low vs. high dimensional), class imbalance ratios, and feature types (e.g., continuous vs. discrete). We focus on tabular data in this release as such data is commonly encountered in real-world applications where explainability is critical [75], and has also been widely studied in XAI literature [51]. Table 1 provides a summary of the real-world datasets currently included in OpenXAI. See Section E.1 in the Appendix for detailed descriptions of individual datasets. While these real-world datasets are primarily drawn from prior research and existing repositories, OpenXAI provides comprehensive data loading and pre-processing capabilities to make these datasets *XAI-ready* (more details below). We also plan to expand our collection of real-world datasets in the next iteration. Adding a new dataset into our collection is as simple as uploading a .csv file or a .zip folder. Users can also submit requests to incorporate new datasets into the OpenXAI framework by filling a simple form and providing links to the datasets (See Appendix).

**Table 1: Summary of currently available datasets in OpenXAI.** Here, "feature types" denotes whether features in the dataset are discrete or continuous, "feature information" describes what kind of information is captured in the dataset, and "balanced" denotes whether the dataset is balanced w.r.t. the predictive label.

| Dataset | Size | # features | Feature types | Feature information | Balanced |
|---|---|---|---|---|---|
| Synthetic Data | 5,000 | 20 | continuous | synthetic | ✓ |
| German Credit [22] | 1,000 | 20 | discrete, continuous | demographic, personal, financial | ✗ |
| HELOC [25] | 9,871 | 23 | continuous | demographic, financial | ✓ |
| COMPAS [36] | 18,876 | 7 | discrete, continuous | demographic, personal, criminal | ✗ |
| Adult Income [79] | 48,842 | 13 | discrete, continuous | demographic, personal, education/employment, financial | ✗ |
| Give Me Some Credit [27] | 102,209 | 10 | discrete, continuous | demographic, personal, financial | ✗ |
| Pima-Indians Diabetes [71] | 768 | 9 | discrete, continuous | demographic, healthcare | ✗ |
| Framingham heart study [1] | 4,240 | 16 | continuous | demographic, healthcare | ✗ |

*Data loaders and pre-trained models*: OpenXAI provides a `Dataloader` class that can be used to load the aforementioned collection of synthetic and real-world datasets as well as any other custom datasets, and ensures that they are *XAI-ready*. More specifically, this class takes as input the name of an existing OpenXAI dataset or a new dataset (name of the .csv file), and outputs a train set which can then be used to train a predictive model, a test set which can be used to generate local explanations of the trained model, as well as any ground-truth explanations (if and when available). If the dataset already comes with pre-determined train and test splits, this class loads train and test sets from those pre-determined splits. Otherwise, it divides the entire dataset randomly into train (70%) and test (30%) sets. Users can also customize the percentages of train-test splits. The code snippet below shows how to import the `Dataloader` class and load an existing OpenXAI dataset.

```
from OpenXAI import Dataloader
loader_train, loader_test = Dataloader.return_loaders(data_name='german',
download=True)
inputs, labels = iter(loader_test).next()
```

We also pre-trained two classes of predictive models (e.g., deep neural networks of varying degrees of complexity, logistic regression models etc.) and incorporated them into the OpenXAI framework so that they can be readily used for benchmarking explanation methods. The code snippet below shows how to load OpenXAI's pre-trained models using our `LoadModel` class.

```
from OpenXAI import LoadModel
model = LoadModel(data_name='german', ml_model='ann')
```

Adding additional pre-trained models into the OpenXAI framework is as simple as uploading a file with details about model architecture and parameters in a specific template. Users can also submit requests to incorporate custom pre-trained models into the OpenXAI framework by filling a simple form and providing details about model architecture and parameters (See Appendix).

**2) Explainers.** OpenXAI provides ready-to-use implementations of six state-of-the-art feature attribution methods, namely, LIME, SHAP, Vanilla Gradients, Gradient x Input, SmoothGrad, and Integrated Gradients. An implementation of a random baseline which randomly assigns importance values to each of the features, and returns these random assignments as explanations is also included. Our implementations of these methods build on other open-source libraries (e.g., Captum [42]) as well as their original implementations. While methods such as LIME and SHAP leverage perturbations of data instances and their corresponding model predictions to *learn* a local explanation model, they do not require access to the internals of the models or their gradients. On the other hand, Vanilla Gradients, Gradient x Input, SmoothGrad, and Integrated Gradients require access to the gradients of the underlying models but do not need to repeatedly query the models for their predictions (see Table 6 in Appendix for a brief summary of these methods). These differences influence the efficiency with which explanations can be generated by these methods. OpenXAI provides an abstract `Explainer` class which enables us to load existing explanation methods as well as integrate new explanation methods.

```
from OpenXAI import Explainer
exp_method = Explainer(method='LIME')
explanations = exp_method.get_explanations(model, X=inputs, y=labels)
```

All the explanation methods included in OpenXAI are readily accessible through the `Explainer` class, and users just have to specify the method name in order to invoke the appropriate method and generate explanations as shown in the above code snippet. Users can easily incorporate their own custom explanation methods into the OpenXAI framework by extending the `Explainer` class and including the code for their methods in the `get_explanations` function (see template below) of this class. They can then submit a request to incorporate their custom methods into OpenXAI library by filling a form and providing the GitHub link to their code as well as a summary of their explanation method (See Appendix).

**3) Evaluation Metrics.** OpenXAI provides implementations and ready-to-use APIs for a set of twenty-two quantitative metrics proposed by prior research to evaluate the faithfulness, stability, and fairness of explanation methods. OpenXAI is the first XAI benchmark to consider all the three aforementioned aspects of explanation reliability. More specifically, we include eight different metrics to measure explanation faithfulness (both with and without ground truth explanations) [43, 59], three different metrics to measure stability [4], and eleven different metrics to measure group-based disparities (unfairness) [18] in the values of the aforementioned faithfulness and stability metrics. The metrics that we choose are drawn from the latest works in explainable AI literature. Below, we briefly describe these metrics. Detailed descriptions of all the metrics along with notation and equations are included in the Appendix.

a) *Ground-truth Faithfulness*: Krishna et al. [43] recently proposed six evaluation metrics to capture the similarity between the top-K or a select set of features of any two feature attribution-based explanations. We leverage these metrics to capture the similarity between the explanations output by state-of-the-art methods and the ground-truth explanations constructed using our synthetic data generation process. These metrics and their definitions are given as follows: i) Feature Agreement (FA) which computes the fraction of top-K features that are common between a given post hoc explanation and the corresponding ground truth explanation, ii) Rank Agreement (RA) metric which measures the fraction of top-K features that are not only common between a given post hoc explanation and the corresponding ground truth explanation, but also have the same position in the respective rank orders, iii) Sign Agreement (SA) metric which computes the fraction of top-K

features that are not only common between a given post hoc explanation and the corresponding ground truth explanation, but also share the same sign (direction of contribution) in both the explanations, iv) Signed Rank Agreement (SRA) metric which computes the fraction of top-K features that are not only common between a given post hoc explanation and the corresponding ground truth explanation, but also share the same feature attribution sign (direction of contribution) and position (rank) in both the explanations, v) Rank Correlation (RC) metric which computes Spearman's rank correlation coefficient to measure the agreement between feature rankings provided by a given post hoc explanation and the corresponding ground truth explanation, and vi) Pairwise Rank Agreement (PRA) metric which captures if the relative ordering of every pair of features is the same for a given post hoc explanation as well as the corresponding ground-truth explanation.

b) *Predictive Faithfulness*: We leverage the metrics outlined by [59, 18] to measure the faithfulness of an explanation when no ground truth is available. This metric, referred to as Prediction Gap on Important feature perturbation (PGI), computes the difference in prediction probability that results from perturbing the features deemed as influential by a given post hoc explanation. Higher values on this metric imply greater explanation faithfulness. We also consider the converse of this metric, Prediction Gap on Unimportant feature perturbation (PGU), which perturbs the unimportant features and measures the change in prediction probability.

c) *Stability*: We consider the metrics introduced by Alvarez-Melis and Jaakkola [7], Agarwal et al. [4] to measure how robust a given explanation is to small input perturbations. More specifically, we leverage the metrics Relative Input Stability (RIS), Relative Representation Stability (RRS), and Relative Output Stability (ROS) which measure the maximum change in explanation relative to changes in the inputs, model parameters, and output prediction probabilities respectively.

d) *Fairness*: Following the work by Dai et al. [18], we measure the fairness of post hoc explanations by averaging all the aforementioned metric values across instances in the majority and minority subgroups, and comparing the two estimates. If there is a huge difference in the two estimates, then we consider this to be evidence for unfairness.

Invoking the aforementioned metrics to benchmark an explanation methods is quite simple and the code snippet below describes how to invoke the RIS metric. Users can easily incorporate their own custom evaluation metrics into OpenXAI by filling a form and providing the GitHub link to their code as well as a summary of their metric (See Appendix).

```
from OpenXAI import Evaluator
metric_evaluator = Evaluator(inputs, labels, model, explanations)
score = metric_evaluator.eval(metric='RIS')
```

***Benchmarking***: As can be seen from the code snippets in this section, OpenXAI allows end users to easily benchmark explanation methods using just a few lines of code. To summarize the benchmarking process, let us consider a scenario where we would like to benchmark a new explanation method using OpenXAI's pre-trained neural network model and the German Credit dataset. First, we use OpenXAI's `Dataloader` class to load the German Credit dataset. Second, we load the neural network model ('ann') using our `LoadModel` class. Third, we extend the `Explainer` class and incorporate the code for the new explanation method in the `get_explanation` function of this class. Finally, we evaluate the new explanation method using various metrics from the `Evaluator` class.

**4) Leaderboards.** OpenXAI introduces the first ever public XAI leaderboards to promote transparency, and enable users to easily compare the performance of multiple explanation methods across a variety of evaluation metrics, predictive models, and datasets. In the current release, we have six different leaderboards each corresponding to a particular dataset. A snapshot of one of our leaderboard pages is shown in Figure 1. Users can submit requests for their custom explanation methods to be featured on one of our leaderboards. To this end, they first need to following the aforementioned *benchmarking process* to develop and evaluate their explanation method.

## 3 Benchmarking Analysis

Next, we describe how we benchmark state-of-the-art explanation methods using our OpenXAI framework, and also discuss key findings of this benchmarking analysis. Code to reproduce all the results is available at `https://github.com/AI4LIFE-GROUP/OpenXAI`.

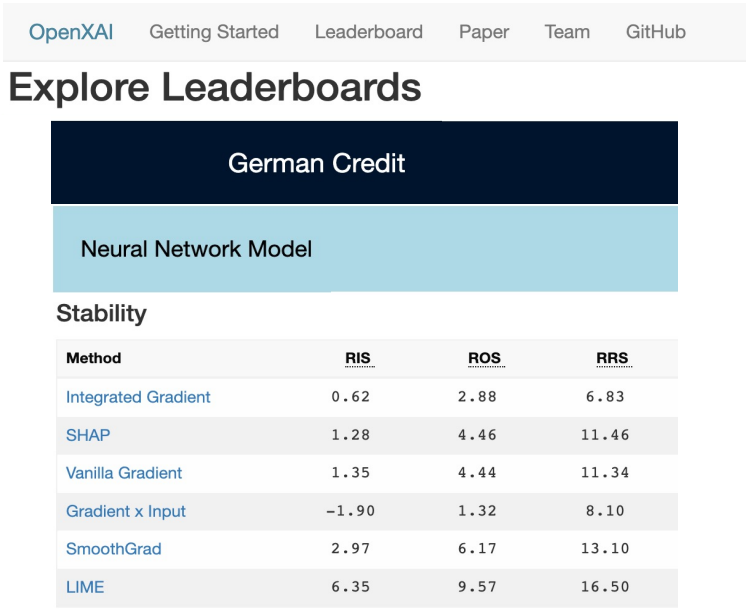



**Figure 1: A snapshot of the leaderboard page from OpenXAI public website.** We also provide interactive ranking functionality (arrow mark in the figure) which allows users to rank explanation methods based on metrics of their choice. Please visit the website to see leaderboards for other datasets.

**Experimental Setup.** We benchmark all the six state-of-the-art feature attribution methods currently available in our OpenXAI framework along with the random baseline, using the `openxai.Evaluator` module (See Section 2). We use default hyperparameter settings for all these methods following the guidelines outlined in the original implementations. Details about the hyperparameters used in our experiments are discussed in Section E.3 in the Appendix. Our OpenXAI framework currently has two pre-trained models, a logistic regression model and a deep neural network model, for each dataset. The neural network models have two fully connected hidden layers with 100 nodes in each layer, and they use ReLU activation functions and an output softmax layer. See Appendix E.4 for more details on model architectures, model training, and model performance.

**Table 2: Ground-truth and predicted faithfulness results on the Heloc dataset for all explanation methods with LR model.** Shown are average and standard error metric values computed across all instances in the test set. ↑ indicates that higher values are better, and ↓ indicates that lower values are better. Values corresponding to best performance are bolded.

| Method | PRA (↑) | RC (↑) | FA (↑) | RA (↑) | SA (↑) | SRA (↑) | PGU (↓) | PGI (↑) |
|---|---|---|---|---|---|---|---|---|
| Random | 0.500±0.00 | 0.005±0.01 | 0.498±0.00 | 0.043±0.00 | 0.251±0.00 | 0.022±0.00 | 0.033±0.00 | 0.035±0.00 |
| VanillaGrad | **1.0**±0.00 | **1.0**±0.00 | **0.957**±0.00 | **0.957**±0.00 | 0.469±0.01 | 0.469±0.01 | 0.034±0.00 | 0.036±0.00 |
| IntegratedGrad | 1.0±0.00 | 1.0±0.00 | 0.957±0.00 | 0.957±0.00 | 0.469±0.01 | 0.469±0.01 | 0.034±0.00 | 0.036±0.00 |
| Gradient x Input | 0.641±0.00 | 0.390±0.01 | 0.582±0.00 | 0.049±0.00 | 0.255±0.01 | 0.022±0.00 | 0.033±0.00 | 0.036±0.00 |
| SmoothGrad | 1.0±0.00 | 1.0±0.00 | 0.957±0.00 | 0.957±0.00 | 0.274±0.00 | 0.274±0.00 | **0.015**±0.00 | **0.047**±0.00 |
| SHAP | 0.645±0.00 | 0.384±0.01 | 0.586±0.00 | 0.054±0.00 | 0.269±0.01 | 0.024±0.00 | 0.033±0.00 | 0.036±0.00 |
| LIME | 0.982±0.00 | 0.994±0.00 | 0.932±0.00 | 0.671±0.00 | **0.929**±0.00 | **0.670**±0.00 | 0.042±0.00 | 0.029±0.001 |

**Faithfulness.** We evaluate the ground-truth and predictive faithfulness of explanations generated by state-of-the-art methods using both synthetic and real-world datasets.

*Ground-truth faithfulness*: We evaluate ground-truth faithfulness by calculating the similarity between the generated explanations and the ground-truth explanations using the metrics discussed in Section 2. Results for various ground-truth faithfulness metrics are shown in Tables 2, 3, 16, 17. Vanilla Gradients, SmoothGrad, and Integrated Gradients produce explanations that achieve perfect scores on four ground-truth faithfulness metrics, viz. pairwise rank agreement (PRA), feature agreement (FA), rank agreement (RA), and rank correlation (RC) metrics, for all datasets. However, on average, across all datasets, LIME outperforms other methods on the signed agreement (SA) [+61.6%] and signed-rank agreement (SRA) [+65.3%] metrics, whereas gradient-based explainers achieve relatively lower values. While illustrative in nature, these findings show how OpenXAI can help identify the limitations of existing explanation methods, which in turn can inform the design of new methods.

**Table 3: Ground-truth and predicted faithfulness results on the Adult Income dataset for all explanation methods with LR model.** Shown are average and standard error metric values computed across all instances in the test set. ↑ indicates that higher values are better, and ↓ indicates that lower values are better. Values corresponding to best performance are bolded.

| Method | PRA (↑) | RC (↑) | FA (↑) | RA (↑) | SA (↑) | SRA (↑) | PGU (↓) | PGI (↑) |
|---|---|---|---|---|---|---|---|---|
| Random | 0.499±0.00 | 0.0±0.00 | 0.496±0.00 | 0.068±0.00 | 0.250±0.00 | 0.037±0.00 | 0.053±0.00 | 0.06±0.00 |
| VanillaGrad | 1.±0.00 | 1.±0.00 | 0.923±0.00 | 0.921±0.00 | 0.138±0.00 | 0.136±0.00 | 0.07±0.001 | 0.039±0.001 |
| IntegratedGrad | 1.±0.00 | 1.±0.00 | **0.923**±0.00 | 0.923±0.00 | 0.138±0.00 | 0.138±0.00 | 0.07±0.001 | 0.039±0.001 |
| Gradient x Input | 0.580±0.00 | 0.281±0.00 | 0.567±0.00 | 0.075±0.00 | 0.070±0.00 | 0.003±0.00 | 0.043±0.00 | 0.073±0.00 |
| SmoothGrad | **1.**±0.00 | **1.**±0.00 | 0.923±0.00 | **0.923**±0.00 | 0.741±0.00 | **0.741**±0.00 | **0.008**±0.00 | **0.099**±0.001 |
| SHAP | 0.655±0.00 | 0.379±0.00 | 0.601±0.00 | 0.105±0.00 | 0.133±0.00 | 0.009±0.00 | 0.047±0.00 | 0.068±0.00 |
| LIME | 0.913±0.00 | 0.921±0.00 | 0.869±0.00 | 0.697±0.00 | **0.858**±0.00 | 0.689±0.00 | 0.014±0.00 | 0.094±0.001 |

*Predictive faithfulness*: Tables 2, 3, 16, 17 show results for the PGI and PGU metrics implemented in OpenXAI (see Section 2 and Appendix A). Overall, we find that SmoothGrad explanations are most faithful to the underlying model and, on average, across multiple datasets outperform other feature-attribution methods on PGU metric (+43.03%). However, results from the German credit dataset for the ANN model show that Gradient x Input produces considerably more faithful (+6.74%) explanations than other methods. Finally, this analysis confirms the finding by Krishna et al. [43] that explanations output by state-of-the-art methods do not necessarily align with each other. This finding further highlights the need for rigorous empirical and theoretical benchmarking of explanation methods.

**Stability.** Next, we examine the stability of explanation methods when the underlying models are LR models in Tables 4 and 5, and neural network models in Tables 19 and 20 in the Appendix. Due to space constraints, we focus on RIS and RRS metrics in the main paper and leave the other results to the Appendix. Overall, the relative stability varies considerably across different datasets, implying that no single explanation method is consistently the most stable. First, for the synthetic dataset in Table 4, we find that Gradient x Input, on average, outperforms feature-attribution methods in relative input stability (+93.5%, RIS) and relative representation stability (+59.2%, RRS). However, stability of Gradient x Input significantly degrades on real-world datasets (Table 5, 19, 20). Second, as shown in Tables 5, 19, and 20, there is no single explanation method that has the highest input and representation stability across all the real-world datasets. On average, across all real-world datasets, SmoothGrad achieves 63.2% higher RRS values compared to other methods, whereas no method performs consistently well when it comes to the RIS metric.

**Table 4: Stability of explanation methods on the Synthetic dataset with LR model.** Shown are average and standard error values across all test set instances. Values closer to zero are desirable, and the best performance is bolded.

| Method | RIS | RRS |
|---|---|---|
| Random | 6.868±0.013 | 6.687±0.015 |
| Vanilla Gradients | 6.133±0.011 | 6.144±0.006 |
| Integrated Gradients | 5.957±0.013 | 9.022±0.043 |
| Gradient x Input | **0.405**±0.015 | **3.422**±0.037 |
| SmoothGrad | 5.249±0.008 | 9.419±0.037 |
| SHAP | 5.673±0.012 | 8.751±0.035 |
| LIME | 9.355±0.008 | 13.564±0.036 |

**Table 5: Stability of explanation methods on the German Credit dataset with LR model.** Shown are average and standard error values across all test set instances. Values closer to zero are desirable, and the best performance is bolded.

| Method | RIS | RRS |
|---|---|---|
| Random | 6.274±0.104 | 16.448±0.124 |
| Vanilla Gradients | -1.384±0.112 | 5.241±0.024 |
| Integrated Gradients | -2.004±0.119 | **4.560**±0.029 |
| Gradient x Input | -0.906±0.104 | 9.437±0.124 |
| SmoothGrad | -4.780±0.117 | 4.931±0.122 |
| SHAP | **-0.230**±0.109 | 10.056±0.115 |
| LIME | -0.698±0.109 | 9.397±0.119 |

**Fairness.** To measure fairness of explanation methods, we compute the average metric values (for each of the aforementioned faithfulness and stability metrics) for different subgroups (e.g., male and female) in the dataset and compare them. Larger gaps between the metric values for different subgroups indicates higher disparities (unfairness). Without loss of generality, we present results using the PGU (see Section 2 and Appendix A) metric. Results for LR models in Figures 2 and 3 provide two key findings. First, the fairness analysis in Figures 2 and 3 shows that there are disparities in the faithfulness of explanations (see Section 2) output by several methods (Vanilla Gradients, Integrated Gradients, and SmoothGrad). Second, Gradient x Input results in the least amount of disparity across both the datasets. These results also suggest a trade-off between various evaluation metrics. For instance, Gradient x Input method underperforms on faithfulness and stability

metrics, but outperforms other methods (+8.9%) when it comes to fairness metrics. Given such trade-offs, practitioners can leverage the OpenXAI leaderboards (Figure 1) to select an explanation method that best meets application-specific needs. Results with NN models and other fairness metrics are included in the Appendix E.7.

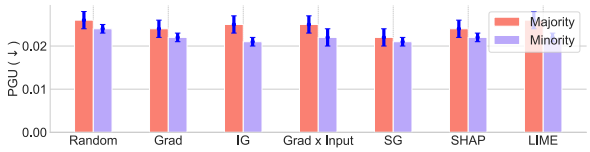



**Figure 2: Fairness analysis of PGU metric on the German Credit dataset with LR model.** Shown are average and standard error values for majority (male) and minority (female) subgroups. Larger gaps between the values of majority and minority subgroups (i.e, red and blue bars respectively) indicate higher disparities which are undesirable.

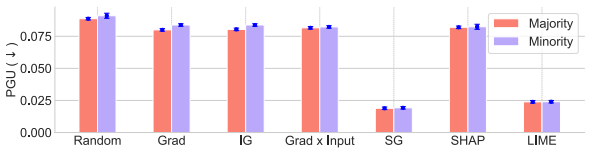



**Figure 3: Fairness analysis of PGU metric on the Adult Income dataset with LR model.** Shown are average and standard error values for majority (male) and minority (female) subgroups. Larger gaps between the values of majority and minority subgroups (i.e, red and blue bars respectively) indicate higher disparities which are undesirable.

## 4 Conclusions

As post hoc explanations are increasingly being employed to aid decision makers and relevant stakeholders in various high-stakes applications, it becomes important to ensure that these explanations are reliable. To this end, we introduce OpenXAI, an open-source ecosystem comprising of XAI-ready datasets, implementations of state-of-the-art explanation methods, evaluation metrics, leaderboards and documentation to promote transparency and collaboration around evaluations of post hoc explanations. OpenXAI can readily be used to benchmark new explanation methods as well as incorporate them into our framework and leaderboards. By enabling systematic and efficient evaluation and benchmarking of existing and new explanation methods, OpenXAI can inform and accelerate new research in the emerging field of XAI. OpenXAI will be regularly updated with new datasets, explanation methods, and evaluation metrics, and welcomes input from the community.

## Acknowledgments and Disclosure of Funding

The authors would like to thank the anonymous reviewers for their helpful feedback and all the funding agencies listed below for supporting this work. This work is supported in part by the NSF awards #IIS-2008461 and #IIS-2040989, and research awards from Google, JP Morgan, Amazon, Harvard Data Science Initiative, and D$^3$ Institute at Harvard. HL would like to thank Sujatha and Mohan Lakkaraju for their continued support and encouragement. The views expressed here are those of the authors and do not reflect the official policy or position of the funding agencies.

## References

[1] Framingham heart study dataset | kaggle. https://www.kaggle.com/datasets/aasheesh200/framingham-heart-study-dataset. (Accessed on 08/15/2022).

[2] Shap benchmark. URL https://shap.readthedocs.io/en/latest/index.html.

[3] Chirag Agarwal and Anh Nguyen. Explaining image classifiers by removing input features using generative models. In *ACCV*, 2020.

[4] Chirag Agarwal, Nari Johnson, Martin Pawelczyk, Satyapriya Krishna, Eshika Saxena, Marinka Zitnik, and Himabindu Lakkaraju. Rethinking stability for attribution-based explanations. In *ICLR 2022 Workshop on PAIR$^2$Struct*, 2022.

[5] Sushant Agarwal, Shahin Jabbari, Chirag Agarwal, Sohini Upadhyay, Steven Wu, and Himabindu Lakkaraju. Towards the unification and robustness of perturbation and gradient based explanations. In *ICML*, 2021.

[6] Ulrich Aivodji, Hiromi Arai, Olivier Fortineau, Sébastien Gambs, Satoshi Hara, and Alain Tapp. Fairwashing: the risk of rationalization. In *ICML*, 2019.

# Benchmarking Heterogeneous Treatment Effect Models through the Lens of Interpretability

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## Abstract

Estimating personalized effects of treatments is a complex, yet pervasive problem. To tackle it, recent developments in the machine learning (ML) literature on heterogeneous treatment effect estimation gave rise to many sophisticated, but opaque, tools: due to their flexibility, modularity and ability to learn constrained representations, neural networks in particular have become central to this literature. Unfortunately, the assets of such black boxes come at a cost: models typically involve countless nontrivial operations, making it difficult to *understand* what they have learned. Yet, understanding these models can be crucial – in a medical context, for example, discovered knowledge on treatment effect heterogeneity could inform treatment prescription in clinical practice. In this work, we therefore use post-hoc *feature importance* methods to identify features that influence the model's predictions. This allows us to evaluate treatment effect estimators along a new and important dimension that has been overlooked in previous work: We construct a benchmarking environment to empirically investigate the ability of personalized treatment effect models to identify *predictive covariates* – covariates that determine differential responses to treatment. Our benchmarking environment then enables us to provide new insight into the strengths and weaknesses of different types of treatment effects models as we modulate different challenges specific to treatment effect estimation – e.g. the ratio of prognostic to predictive information, the possible nonlinearity of potential outcomes and the presence and type of confounding.

## 1 Introduction

The need to estimate the effects of actions – such as treatments, policies and other interventions – is ubiquitous in many domains, ranging from economics to medicine. Many applications where treatment effects are of interest additionally operate under *high stakes*, for example treatment decisions in a hospital setting – making it particularly important that estimates leading to individual decisions are reliable. As interest in designing *personalized* treatments is growing across fields, a substantial literature on learning treatment effect heterogeneity has emerged in machine learning (ML) in recent years. In this context, a plethora of sophisticated methods for estimating *conditional average treatment*

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36th Conference on Neural Information Processing Systems (NeurIPS 2022) Track on Datasets and Benchmarks.

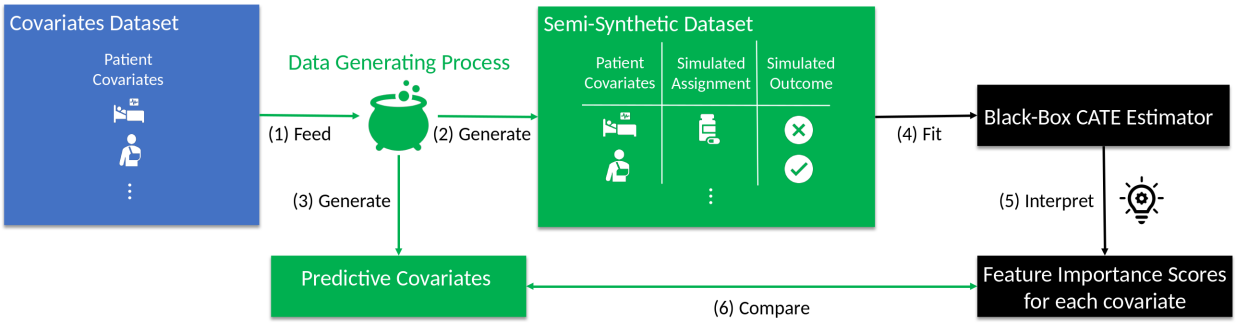



Figure 1: Illustration of the ITErpretability benchmarking environment. (1) Covariates are extracted from any dataset. (2) These covariates are labeled with a transparent data generating process for the assignments and the outcomes. This results in a semi-synthetic dataset $\mathcal{D}$. (3) Since the data generating process is transparent, we know the indices $\mathcal{I}_{\text{pred}}$ of predictive covariates. (4) We fit a black-box CATE estimator $\hat{\tau}$ on $\mathcal{D}$. (5) We then use a feature importance method to assign feature importance scores $a_i(\hat{\tau}, x)$ to each covariate $x_i$. (6) We evaluate each CATE estimator based on a new metric measuring the accordance between the most important features and the predictive covariates.

*effects* (CATE) and/or an individual's *potential outcomes* (POs) under different treatments have been proposed in the recent ML literature (see e.g. [1]), all aiming to improve the *precision* in estimating effects. In particular, recent work has produced both *model-agnostic* estimation strategies, which can incorporate *any* ML method into the estimation of effects [2–4], *and* model-specific strategies, which *adapt specific* ML methods to the treatment effect estimation problem – for example random forests [5, 6], gaussian processes [7, 8] and, most predominantly, neural networks [4, 9–13].

Despite the growing interest in such methods, [14] noted that their evaluation has been quite one-dimensional: Most, if not all, of this work has focussed on assessing performance of proposed algorithms in terms of the *Precision of Estimating Heterogeneous Effects* (PEHE) criterion of [15], which measures the root mean squared error (RMSE) of the estimated CATE function on a test-set. However, in many applications, black-box predictions of expected treatment effects do not suffice. In the context of drug development, for example, it is at least equally important to assess whether an algorithm *discovers the correct drivers of the underlying effect heterogeneity* or leads to the right *interpretation* thereof [16]: discovering effect-modifiers can be important both in exploratory clinical trial phases, e.g. to identify potential biomarkers which may help to explain an enhanced treatment effect for future drug development, and in confirmatory trial phases, e.g. to rule out that a drug is ineffective for some biomarker signatures or to otherwise adapt prescription criteria [16, 17]. Such interpretation could also be important in clinical practice e.g. when *explaining* treatment recommendations derived from an estimated CATE function. In this paper, we therefore leverage recent advances in *explainable artificial intelligence* [18–20] to interpret the discoveries of ML-based CATE estimators, and propose a benchmarking environment which we use to provide insight into the performance of different learning strategies in discovering drivers of heterogeneity.

**Related work.** How to interpret CATE estimators has received little attention in the ML literature thus far. Some work *implicitly* enables interpretation by relying on methods that are inherently more interpretable, e.g. linear regressions [21, 22] or tree-based models [23, 24]. Another recent stream of work *explicitly* focuses on interpretability, similarly proposing the use of methods that are inherently interpretable: [25] rely on decision lists, [26] construct interpretable hyper-boxes for matching, [27] use causal rule ensembles, [28] use mixture models with sparse components, [29] use the fused lasso for estimation of subgroup piece-wise constant treatment effects and [30] rely on an additive neural network architecture and [31] propose using transformer backbones that can be interpreted using their attention weights. With a goal similar to our work, [16] investigate how well different *forest-based* CATE estimation strategies discover effect modifiers, but they rely on variable importance scores inherent to random forests to do so. To the best of our knowledge, the only work that considers *post-hoc* interpretability of already fitted, arbitrary black-box CATE models uses *model distillation* approaches to do so: [32] use the fitted black-box CATE estimator as a "teacher" and a multi-task decision tree as an interpretable "student" and similarly, [33] propose to create interpretable CATE estimators by fitting an arbitrary interpretable model on top of the potential outcome predictions of a NN-based first stage estimator. Such approaches do not provide an interpretation of the black-box output directly and therefore need to rely on the interpretable student model being good enough to cover the complexities of the original estimator. In this paper, we take a different approach and consider the use of *post-hoc feature importance methods* to interpret black-box CATE estimators *directly*.

**Contributions & Outlook.** Our contributions are threefold: (i) we study how to *interpret* black-box CATE estimators and use this to *evaluate* them along a *new and important dimension* that has been overlooked in previous work – namely, their ability to correctly discover drivers of effect heterogeneity –, (ii) we propose a *benchmark environment* to do so, and (iii) we provide *new insights* into the performance of existing methods on this new task. We proceed as follows: We begin by recalling fundamentals of the CATE estimation setting and discuss its unique characteristics in Section 2. In Section 3, we review feature importance methods and discuss how they can be applied to interpret what CATE estimators have learned about treatment effect heterogeneity, with the ultimate goal to *compare different CATE estimators* on their ability to identify predictive covariates/features (i.e. covariates that determine differential responses to treatment and, hence, are the ones that truly matter when estimating treatment effects). In Section 4, we then introduce a new benchmark environment for evaluating black-box CATE estimators, supplemented with feature importance methods to interpret their output, on precisely this task. This *ITErpretability* benchmark, as illustrated in Figure 1, can be used with any covariates dataset and is semi-synthetic: it relies on real covariates with synthetic treatment assignment and outcomes, supplying us with otherwise unavailable *ground-truth access* to predictive covariates. Finally, in Section 5, we use this benchmark to investigate the performance of existing, neural network-based, CATE estimation strategies on this new task, and provide interesting insights into the impact of typical difficulties in CATE problems: the strength of predictive relative to prognostic information, the possible nonlinearity of potential outcomes and the presence of confounding.

## 2 Setting: The CATE Estimation Problem

We consider a standard treatment effect estimation setting as formalized within the Neyman-Rubin potential outcomes framework [34]. We assume access to an observational i.i.d. dataset $\mathcal{D} = \{(Y^n, X^n, W^n)\}_{n=1}^N$ Here, $Y^n \in \mathcal{Y} \subset \mathbb{R}$ is a binary or continuous outcome of interest, $X^n \in \mathcal{X} \subset \mathbb{R}^d$ is a vector of pre-treatment covariates, and $W^n \in \{0, 1\}$ is a binary treatment assigned according to a (usually unknown) propensity score $\pi(x) = P(W = 1 \mid X = x)$. Each individual has multiple potential outcomes (POs) $Y^n(w)$ associated with different treatment values, yet only the outcome associated with the received treatment is observed, i.e. $Y^n = Y^n(W^n)$. To investigate treatment effect heterogeneity, we focus on the conditional average treatment effect (CATE), computed as the expected difference between the two POs for an individual with covariates $X = x$:

$$\tau(x) = \mathbb{E}_P[Y(1) - Y(0) \mid X = x] = \mu_1(x) - \mu_0(x) \tag{1}$$

with $\mu_w(x) = \mathbb{E}_P[Y(w) \mid X = x]$ the expected potential outcome.

**What Makes CATE Estimation Special Relative to Standard Supervised Learning?** As we discuss in more detail in Appendix A, three characteristics are generally considered central to the CATE estimation problem (see e.g. [13, 14]): ① The need to rely on strong untestable assumptions to ensure *identifiability* of treatment effects from observational data (here: ignorability assumptions [35], as discussed further in Appendix A). ② The presence of *covariate shift* due to confounding (correlation between covariates and treatment assignment). ③ The *absence of the target label* of interest ($Y(1) - Y(0)$) as only one of the POs can be observed in practice, or conversely, that CATE can arise either as a single regression or as a *difference* between two functions

In our context, ① & ② are mainly important for the *construction of benchmarks*: as discussed in Section 4, characteristic ① (and to some extent ③) leads to the necessity to rely on (semi-)*synthetic* data as ground truth knowledge of the data-generating process is generally unavailable from real data (see also the discussion in [14]). Characteristic ② leads to a natural *experimental knob* to include in a benchmark to be modulated when evaluating models. As we discuss in detail in Section 3, it is ③ that leads to most interesting considerations when studying *how to interpret* CATE estimators. It also lead to the emergence of different CATE estimation strategies: generally, as discussed in [4], $\tau(x)$ can either be estimated *indirectly* using the difference between the PO regression estimates (i.e. as $\hat{\mu}_1 - \hat{\mu}_0$) as in most methods proposed in the recent ML literature [7, 8, 10, 12, 13] or *directly* by fitting a regression model using pseudo-outcome $Y_{\hat{\eta}}$ as a surrogate for the unobserved PO difference (relying on initial estimates of some of the nuisance functions $\eta = (\mu_0, \mu_1, \pi)$) as in e.g. [2–4]. We discuss different instantiations of these strategies considered in our experiments in Section 5.

## 3 Feature Importance in the Context of Treatment Effect Estimation

In this section, we discuss how to use feature importance to investigate what a CATE estimator (or: model) has learned about treatment effect heterogeneity. We start by describing a general feature importance formalism: We assume that we have access to a *black-box* model $\hat{\tau} : \mathbb{R}^d \to \mathbb{R}$ to estimate the CATE $\tau$. Feature importance methods permit to understand the prediction of this model by highlighting features (covariates) the model is sensitive to. Concretely, this is done by assigning an importance score $a_i(\hat{\tau}, x) \in \mathbb{R}$ to each feature $x_i$ contained in the vector $x$ with [2] $i \in [d]$. This score reflects the importance of $x_i$ to predict the CATE $\hat{\tau}(x)$. The score is such that the importance of a feature $x_i$ increases with $|a_i(\hat{\tau}, x)|$. When it comes to the sign, features with $a_i(\hat{\tau}, x) > 0$ tend to increase the CATE $\hat{\tau}(x)$ and features with $a_i(\hat{\tau}, x) < 0$ tend to decrease it.

There exist many methods to assign importance scores $a_i(\hat{\tau}, x)$ to the features. Different feature importance methods tend to attribute different relative importance between features [36, 37]. This is because they measure different characteristics of the black-box model: *gradient-based* methods compute scores based on the model's gradient with respect to the features [38–41]; *perturbation-based* methods compute scores based on the model's sensitivity to features perturbations [42–44] and some other methods rely on the neuron's activation to compute the importance scores [45, 46]. In Section 5, we use Integrated Gradients [39] as our main feature importance method. This is because it offers the best performances empirically and is typically more computationally efficient than the previous methods. A comparison between different feature importance methods is provided in Appendix B. We will now discuss the specificity of CATE models in an interpretability perspective.

**What Makes CATE Interpretability Special?** Above, we fixed the black-box to interpret to be the CATE estimate $\hat{\tau}$. However, CATE can of course also be written (and, as in most methods proposed in the recent ML literature, be estimated) in terms of the expected POs, i.e. $\tau = \mu_1 - \mu_0$, which have their own feature importance $a_i(\mu_0, x)$ and $a_i(\mu_1, x)$. In general, feature importance for CATE will differ from those of the POs, leading to different insights. This multiplicity of interpretation certainly distinguishes CATE models from interpreting standard supervised learners and deserves a discussion.

In order to attach meaning to those possible interpretations, it is useful to consider an important distinction between so-called *predictive* and *prognostic* covariates made in the medical literature [47, 48]. Prognostic covariates determine outcome regardless of treatment assignment – common risk factors such as age or gender may fall in this category. Such variables are taken into account by the two potential outcomes $\mu_w$ in a similar way. In this way, prognostic covariates correspond to features $x_i$ that are important for both POs $a_i(\mu_0, x) \neq 0$ and $a_i(\mu_1, x) \neq 0$. Predictive covariates, on the other end, are predictive of effect heterogeneity, i.e they determine differential responses to treatment – hormone receptor status in cancer patients is one such example [49]. In this way, predictive covariates correspond to features $x_i$ that are important for the CATE $a_i(\tau, x) \neq 0$. If we assume that the potential outcomes $\mu_0, \mu_1$ are differentiable with respect to the covariates, one can extend the previous definitions globally. In this case, a feature $x_i$ is *prognostic* if both potential outcomes depend on that feature: $\partial\mu_0/\partial x_i \neq 0$ and $\partial\mu_1/\partial x_i \neq 0$. Similarly, a feature $x_i$ is *predictive* if the CATE depends on that feature: $\partial\tau/\partial x_i \neq 0$. In the medical context, any measured patient covariate could be prognostic, predictive or both (or, of course, irrelevant).

Whereas [33] used a model-distillation approach to interpret only the PO models, we argue that it is most interesting in the CATE estimation context to interpret the learned CATE models *directly* by focusing on the discovery of predictive covariates. Identifying such predictive covariates can, for example, provide precious information to support exploratory analyses in clinical trials, which in turn allow pharmaceutical companies to refine the target population for a treatment, improving the likelihood of a successful later stage trial [16]. From that perspective, CATE models $\hat{\tau}$ that are better at identifying predictive covariates are clearly preferable. Due to the absence of treatment effect labels (Section 2), it is far from obvious whether all estimation strategies result in successful identification of predictive covariates. Furthermore, since indirect learners target the CATE indirectly through the POs, there is no guarantee that the resulting models can distinguish between prognostic and predictive covariates. With these simple observations, it is obvious that CATE interpretability comes with a unique set of challenges. We will now introduce a benchmark to study CATE models through this angle.

[2] In the following, $[n]$ denotes the set of natural numbers between 1 and $n \in \mathbb{N}^*$.

## 4 The ITErpretability Benchmark

Next, we describe our proposed *ITErpretability* benchmark that uses ideas from *interpretability* to measure the ability of *treatment effect* estimators to identify predictive covariates. We propose a framework that relies on a *semi-synthetic* data generating process (DGP), which is standard in the CATE estimation literature [14]: because identifiability assumptions are generally untestable, *simulating* outcomes and treatment assignments ensures that they hold; additionally it ensures that the underlying CATE function is *known*. In our context, we cannot rely on existing and established semi-synthetic benchmarks such as IHDP [15, 10] or ACIC2016 [50], most importantly because they did not *record* which covariates are predictive or prognostic. Further, the *experimental knobs* considered therein are not of primary interest in our setting[3]; instead we thus design our own DGPs that allow to us to obtain interesting new insights in our experiments. Below, we discuss our DGP and proposed metrics.

**DGP.** We would like to rely on a DGP that covers a range of realistic scenarios *and* for which we can clearly identify prognostic and predictive covariates. Since this last information is generally not available in *real* observational data, we use a semi-synthetic approach in which we reuse covariates $X^n$ from a real dataset and synthetically generate the treatment assignments $W^n$ and outcomes $Y^n$ for all $n \in [N]$. In this way, the resulting semi-synthetic dataset has realistic covariates *and* we have a full knowledge on how outcomes are generated. In particular, we can restrict to DGPs for which prognostic and predictive covariates are clearly distinct and identifiable. We implement this by selecting *non-overlapping* subsets $\mathcal{I}_{\text{prog}} \subset [d]$ of prognostic and two subsets $\mathcal{I}_0, \mathcal{I}_1 \subset [d]$ of predictive covariates. The prognostic covariates similarly contribute to both POs $Y(0), Y(1)$ through a function $x \mapsto \mu_{\text{prog}}(x_{\mathcal{I}_{\text{prog}}})$, where we let $x_{\mathcal{I}}$ denote the vector $(x_i)_{i \in \mathcal{I}}$. for a set $\mathcal{I} \subset [d]$. The predictive covariates, on the other hand, contribute to either only $Y(0)$ through a function $x \mapsto \mu_{\text{pred0}}(x_{\mathcal{I}_0})$ or to $Y(1)$ through a function $x \mapsto \mu_{\text{pred1}}(x_{\mathcal{I}_1})$.[4] We include a predictive scale $\omega_{\text{pred}} \in \mathbb{R}^+$ that permits to tune the relative strength between the prognostic and predictive contributions to the POs. This full process is detailed in Algorithm 1. All the experiments from Section 5 are produced by varying the inputs of this algorithm; in particular, we consider different types of outcome functions and propensity scores.

**Algorithm 1:** Semi-Synthetic Data Generating Process

**Input:** Covariates dataset $\{X^n \in \mathbb{R}^d\}_{n=1}^N$, Prognostic function $\mu_{\text{prog}} : \mathbb{R} \to \mathbb{R}$, Predictive functions $\mu_{\text{pred0}}, \mu_{\text{pred1}} : \mathbb{R} \to \mathbb{R}$, Propensity score $\pi : \mathcal{X} \to \mathbb{R}$, Feature sets size $n_{\mathcal{I}} \in \mathbb{N}^*$, Predictive scale $\omega_{\text{pred}} \in \mathbb{R}^+$, Noise level $\sigma \in \mathbb{R}^+$

**Output:** Semi-synthetic observational dataset $\mathcal{D} = \{(Y^n, X^n, W^n)\}_{n=1}^N$, Prognostic features $\mathcal{I}_{\text{prog}} \subset [d]$, Predictive features $\mathcal{I}_{\text{pred}} \subset [d]$

**Ensure:** $d > 3 \cdot n_{\mathcal{I}}$ /* Avoid overlap between $\mathcal{I}_{\text{prog}}$, $\mathcal{I}_0$ and $\mathcal{I}_1$ */

$\mathcal{I} \leftarrow$ Sample $3 \cdot n_{\mathcal{I}}$ elements from $[d]$ without replacement /* Get relevant features */

$\mathcal{I}_{\text{prog}}, \mathcal{I}_0, \mathcal{I}_1 \leftarrow$ Split $\mathcal{I}$ into 3 sets of size $n_{\mathcal{I}}$ /* Get prog. and pred. features */

$\mathcal{D} \leftarrow \emptyset$ /* Initialize dataset */

**for** $n \in [N]$ **do**

- $Y(0) \leftarrow \mu_{\text{prog}}(X^n_{\mathcal{I}_{\text{prog}}}) + \omega_{\text{pred}} \cdot \mu_{\text{pred0}}(X^n_{\mathcal{I}_0})$ /* Get untreated outcome */
- $Y(1) \leftarrow \mu_{\text{prog}}(X^n_{\mathcal{I}_{\text{prog}}}) + \omega_{\text{pred}} \cdot \mu_{\text{pred1}}(X^n_{\mathcal{I}_1})$ /* Get treated outcome */
- $W^n \sim \text{Bernoulli}[\pi(X^n)]$ /* Sample treatment assignment */
- $\epsilon \sim \mathcal{N}(0, \sigma)$ /* Sample noise */
- $Y^n = W^n \cdot Y(1) + (1 - W^n) \cdot Y(0) + \epsilon$ /* Get observed outcome */
- $\mathcal{D} \leftarrow \mathcal{D} \cup \{(Y^n, X^n, W^n)\}$ /* Append dataset */

**end**

**return** $\mathcal{D}$, $\mathcal{I}_{\text{prog}}$, $\mathcal{I}_{\text{pred}} = \mathcal{I}_0 \sqcup \mathcal{I}_1$

**Metrics.** After using Algorithm 1, we split the generated dataset into a training set $\mathcal{D}_{\text{train}}$ and a testing set $\mathcal{D}_{\text{test}}$ ($80\% - 20\%$ split). We fit a model $\hat{\tau}$ to estimate the CATE on the training set $\mathcal{D}_{\text{train}}$ and evaluate the model on the held-out test set $\mathcal{D}_{\text{test}}$. As aforementioned, our purpose is to assess if this

[3] In fact, due to the DGP used in IHDP *all* important variables are both predictive and prognostic, allowing for no interesting distinctions between discoveries.

[4] Note that the expected difference between the POs can be written as $\mathbb{E}[Y(1) - Y(0) \mid X = x] = \omega_{\text{pred}} \cdot [\mu_{\text{pred1}}(x_{\mathcal{I}_1}) - \mu_{\text{pred0}}(x_{\mathcal{I}_0})]$. In this way, the treatment effect depends only on the covariates indexed by $\mathcal{I}_{\text{pred}} = \mathcal{I}_0 \sqcup \mathcal{I}_1$. This indeed corresponds to the definition of predictive covariates given in Section 3.

model has correctly identified the predictive covariates. With our choice of DGP, we know that those predictive covariates correspond to the indices in $\mathcal{I}_{\text{pred}} = \mathcal{I}_0 \sqcup \mathcal{I}_1$. By recalling that the importance attributed to covariate $x_i$ for a prediction $\hat{\tau}(x)$ increases with the absolute value $|a_i(\hat{\tau}, x)|$, we can compute the average proportion of the attribution *correctly* allocated to the predictive covariates:

$$\text{Attr}_{\text{pred}} = \frac{1}{|\mathcal{D}_{\text{test}}|} \sum_{X \in \mathcal{D}_{\text{test}}} \frac{\sum_{i \in \mathcal{I}_{\text{pred}}} |a_i(\hat{\tau}, X)|}{\sum_{i=1}^{d} |a_i(\hat{\tau}, X)|}. \tag{2}$$

Note that $\text{Attr}_{\text{pred}} \in [0, 1]$, where $\text{Attr}_{\text{pred}} = 0$ corresponds to a model that does not identify any predictive covariate and $\text{Attr}_{\text{pred}} = 1$ corresponds to a model that only identifies predictive covariates. Ideally, predictive covariates should be the most important for a model that estimates the CATE; thus we expect good models to score high with respect to this metric. A similar metric $\text{Attr}_{\text{prog}}$ can be defined analogously to measure the fraction of the feature attribution *incorrectly* allocated to the prognostic covariates by replacing $\mathcal{I}_{\text{pred}} \mapsto \mathcal{I}_{\text{prog}}$ in (2). Ideally, $\text{Attr}_{\text{prog}}$ should be zero if all importance is correctly allocated to predictive variables. Finally, we will sometimes also report the standard PEHE metric, i.e. the RMSE of estimating CATE: $\text{PEHE} = \sqrt{n_{\text{test}}^{-1} \sum_{X \in \mathcal{D}_{\text{test}}} [\hat{\tau}(X) - \tau(X)]^2}$.

## 5 Experiments

In this section, we benchmark different types of CATE estimators on their ability to identify predictive covariates through feature importance scores. We study 3 different characteristics of the data generating process which we expect to impact this ability: (i) the relative strength between the prognostic contribution $\mu_{\text{prog}}$ and the predictive contributions $\mu_{\text{pred0}}, \mu_{\text{pred1}}$ to the POs $Y(0), Y(1)$ (Sec. 5.1); (ii) The presence of nonlinearities in the prognostic and predictive functions $\mu_{\text{prog}}, \mu_{\text{pred0}}, \mu_{\text{pred1}}$ (Sec. 5.2), and (iii) the fact that the treatment assignment might be biased according to a nontrivial propensity score $\pi$ (Sec. 5.3). All the experiments are done by varying the inputs of Algorithm 1. The code to reproduce the experiments is available at https://github.com/JonathanCrabbe/ITErpretability and https://github.com/vanderschaarlab/ITErpretability.

**Datasets.** We extract covariates to use in our our benchmarking environment from the following four datasets: TCGA [51], Twins [52], News [53] and ACIC2016 [50], which were selected due to their diverse characteristics in terms of the number of features, mixture of categorical/continuous features and population size. The number of covariates in these datasets range from $d = 39$ to $d = 100$ and we set $n_{\mathcal{I}} = \lfloor 0.2 \cdot d \rfloor$. Refer to Appendix C for details of the datasets.

**Learners.** We consider a number of CATE estimators based on neural networks throughout our experiments. As *direct* estimators, we use [3]'s DR-learner, which relies on a doubly robust pseudo-outcome, and [2]'s X-learner, which uses a weighted average of two direct (singly-robust) treatment effect estimates from both treatment groups. As *indirect* estimators, we use [2]'s T-learner, which fits *t*wo regression models $\hat{\mu}_w(x)$ (one for each treatment group) and sets $\hat{\tau}(x) = \hat{\mu}_1(x) - \hat{\mu}_0(x)$, and S-learner, which includes the treatment assignment variable $W$ as a standard feature in a *s*ingle regression $\hat{\mu}(x, w)$ and sets $\hat{\tau}(x) = \hat{\mu}(x, 1) - \hat{\mu}(x, 0)$. Finally, we consider [10]'s TARNet which can be seen as a *hybrid* between S- and T-learner [4]: it learns a representation $\Phi(x)$ *shared* between treatment groups, which is used by *treatment-specific* outcome heads $h_w(\Phi(x))$ so that $\hat{\tau}(x) = h_1(\Phi(x)) - h_0(\Phi(x))$. In the experiments with confounding, we also use [10]'s CFRNet, which differs from TARNet only in a regularization term that encourages the representation to be *balanced* (follow a similar distribution) across treatment groups. We discuss all models in more detail in Appendix C, where we also detail how we fix hyperparameters across all models to ensure similar capacity.

### 5.1 Experiment 1: Altering the Strength of Predictive Effects

**Setup.** We begin by investigating how the strength of predictive effects *relative* to prognostic effects influences the ability of different learners to discover predictive covariates. This is an interesting question, as in practice predictive signals are often assumed to be much *weaker* than prognostic ones [2, 13, 16]. Thus, ideally estimators should be able to correctly identify predictive covariates even when effects are weak; yet some of [16]'s empirical results comparing different random forest-based learners across DGPs with different predictive effects, show that this is not always the case. For our experiments, we thus continuously vary predictive effect size in a DGP with a linear parametrization for the prognostic and predictive functions:

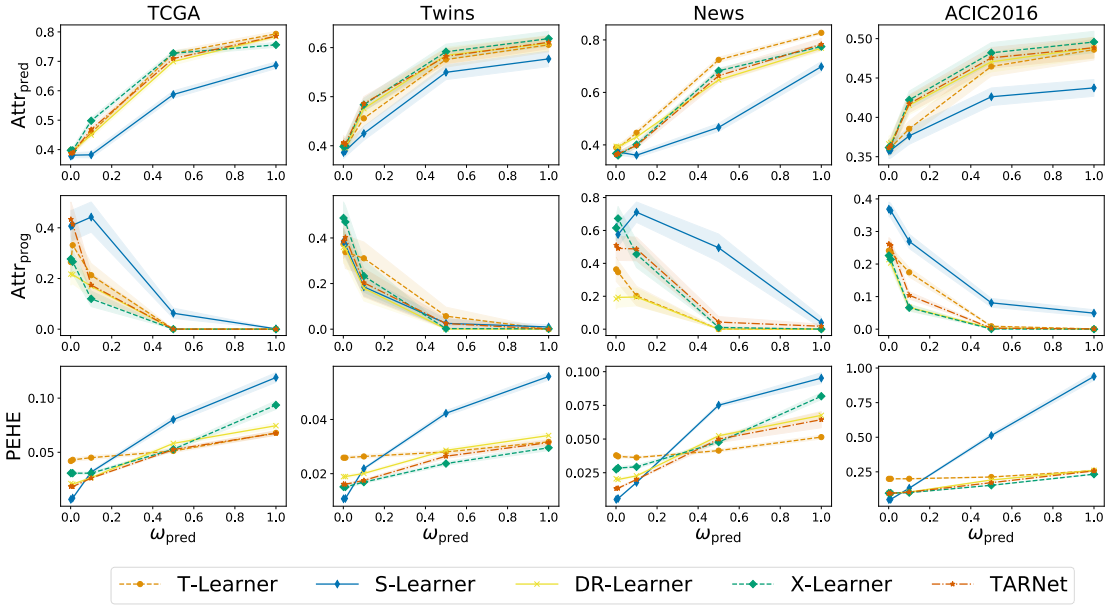



Figure 2: Performance comparison in terms of $\text{Attr}_{\text{pred}}$ (top, higher is better), $\text{Attr}_{\text{prog}}$ (middle, lower is better) and PEHE (bottom, lower is better) when varying the predictive scale, using four feature datasets (TGCA, Twins, News, ACIC). Averaged across multiple runs, shaded areas indicates one standard error.

$\mu_{\text{prog}}(x_{\mathcal{I}_{\text{prog}}}) = \alpha_{\text{prog}}^{\intercal}\, x_{\mathcal{I}_{\text{prog}}}$, $\mu_{\text{pred0}}(x_{\mathcal{I}_0}) = \alpha_{\text{pred0}}^{\intercal}\, x_{\mathcal{I}_0}$ and $\mu_{\text{pred1}}(x_{\mathcal{I}_1}) = \alpha_{\text{pred1}}^{\intercal}\, x_{\mathcal{I}_1}$ with weights sampled randomly $\alpha_{\text{prog}},\ \alpha_{\text{pred0}},\ \alpha_{\text{pred1}} \sim U([-1,1]^{n_{\mathcal{I}}})$, where $U$ denotes the uniform distribution. To vary the relative strength between the prognostic and the predictive contributions to the POs, we change the predictive scale $\omega_{\text{pred}} \in \{10^{-3}, 10^{-2}, 10^{-1}, 0.5, 1\}$. Here, treatments are assigned completely at random, i.e. $\pi(x) = 0.5$.

**Results.** In Fig. 2 we present results on correct attributions $\text{Attr}_{\text{pred}}$ and misattributions $\text{Attr}_{\text{prog}}$, as well as the standard PEHE metric for comparison, for all datasets. We make a number of interesting observations: ① *Attribution trends indeed vary with* $\omega_{\text{pred}}$. Correct predictive attributions $\text{Attr}_{\text{pred}}$ substantially increase as $\omega_{\text{pred}}$ increases for all learners and across all datasets considered – this confirms the intuition that the stronger the predictive effects are, the easier it is to identify their origin. Conversely, prognostic misattributions $\text{Attr}_{\text{prog}}$ decrease as $\omega_{\text{pred}}$ increases, indicating that one reason for the low $\text{Attr}_{\text{prog}}$ at low $\omega_{\text{pred}}$ is that learners *confuse* prognostic effects for predictive ones. ② *Comparing learners, S-Learners appear to struggle most to make correct attributions*. The most salient observation across all datasets is that the S-Learner does does substantially worse at $\text{Attr}_{\text{pred}}$. With the exception of the Twins dataset, this usually also translates into higher $\text{Attr}_{\text{prog}}$ than all other learners. We believe that this is because the S-Learner uniquely neither has a treatment-group specific component (like T-learner and TARNet do) nor models CATE directly (like DR- and X-learner do); learned treatment effect heterogeneity thus has to arise through learned interactions with the treatment indicator – which appears to lead to less reliable predictive covariate discoveries. All other methods perform very similar to each other. ③ *Using attribution metrics indeed leads to interesting new insights relative to considering only PEHE.* We observe that the S-learner does *best* in terms of PEHE when predictive strength is low, while the T-learner does worst (as discussed below, this is in line with expectations and empirical observations previously made in e.g. [2, 13]) – yet, as we saw above, this does not translate into better discoveries. Similarly, the better performance in terms of PEHE of some other strategies relative to T-learner when $\omega_{\text{pred}}$ is small also does not lead to better discoveries. We attribute this to the fact that when $\omega_{\text{pred}}$ is very small, PEHE will favour any method that outputs near-zero treatment effects; indeed all considered methods except the T-learner incorporate an implicit inductive bias that shrinks effects [13] – which appears to help only in terms of PEHE. Note also that PEHE is not directly comparable across different values of $\omega_{\text{pred}}$ as it naturally increases as the scale of CATE changes. Finally, we investigate performance on this task using a dataset with higher dimensional input in Appendix C.4.

### 5.2 Experiment 2: Incorporating Nonlinearities

**Description.** In practice, there is no particular reason to expect that POs are linear functions of the covariates. Next we therefore investigate how nonlinearities in the POs influence the ability of CATE

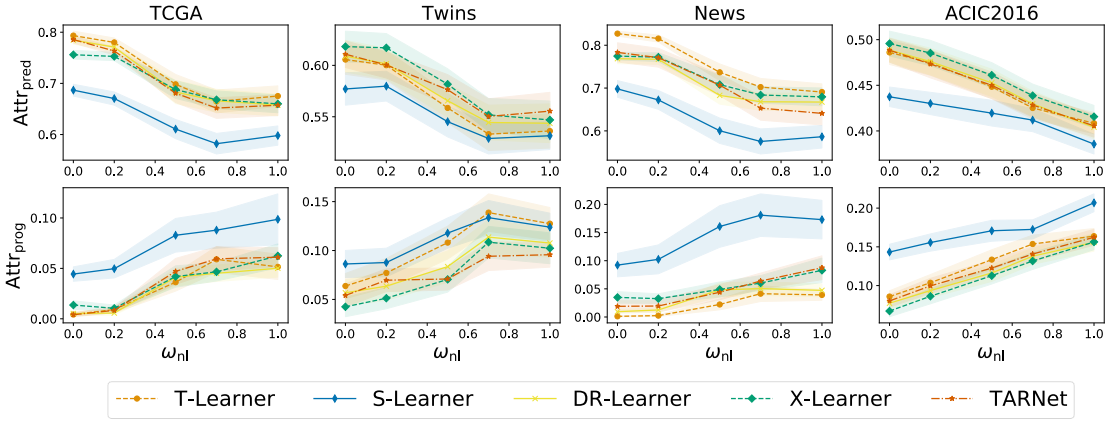



Figure 3: Performance comparison in terms of $\text{Attr}_{\text{pred}}$ (top, higher is better) and $\text{Attr}_{\text{prog}}$ (bottom, lower is better) when varying the nonlinearity scale, using four feature datasets (TGCA, Twins, News, ACIC). Averaged across multiple runs, shaded areas indicates one standard error.

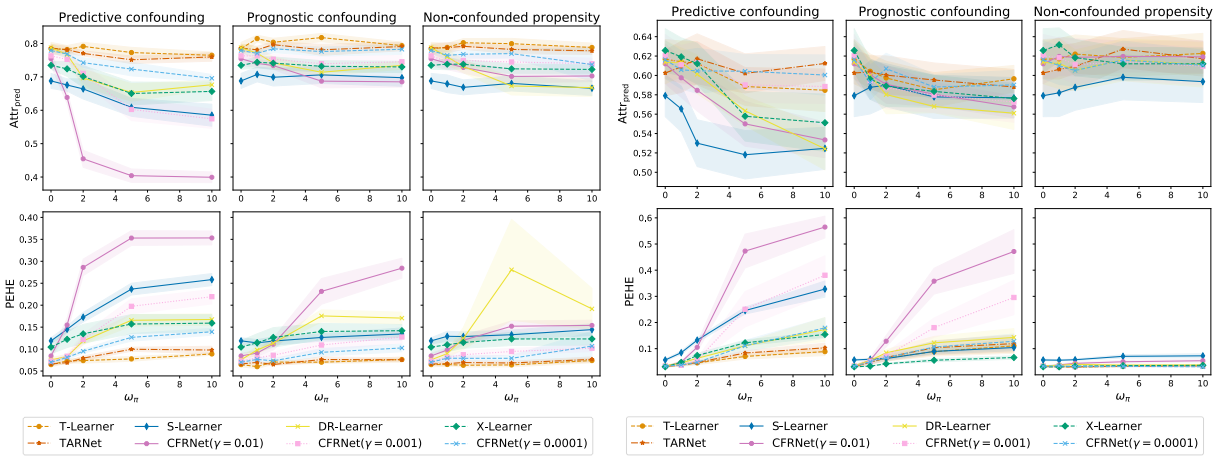
estimators to identify predictive covariates. To do so, we use a parametrization for the prognostic and predictive functions that allows us to control the strength of the nonlinearities through parameter $\omega_{\text{nl}}$: $\mu_{\text{prog}}(x_{\mathcal{I}_{\text{prog}}}) = (1-\omega_{\text{nl}}) \cdot \alpha_{\text{prog}}^{\intercal}\, x_{\mathcal{I}_{\text{prog}}} + \omega_{\text{nl}} \cdot \chi(\alpha_{\text{prog}}^{\intercal}\, x_{\mathcal{I}_{\text{prog}}})$, $\mu_{\text{pred0}}(x_{\mathcal{I}_0}) = (1-\omega_{\text{nl}}) \cdot \alpha_{\text{pred0}}^{\intercal}\, x_{\mathcal{I}_0} + \omega_{\text{nl}} \cdot \chi(\alpha_{\text{pred0}}^{\intercal}\, x_{\mathcal{I}_0})$ and $\mu_{\text{pred1}}(x_{\mathcal{I}_1}) = (1-\omega_{\text{nl}}) \cdot \alpha_{\text{pred1}}^{\intercal}\, x_{\mathcal{I}_1} + \omega_{\text{nl}} \cdot \chi(\alpha_{\text{pred1}}^{\intercal}\, x_{\mathcal{I}_1})$ with weights and nonlinearity sampled randomly $\alpha_{\text{prog}},\ \alpha_{\text{pred0}},\ \alpha_{\text{pred1}} \sim U([-1,1]^{n_{\mathcal{I}}}), \quad \chi \sim U(\mathcal{F})$, where $\mathcal{F}$ is a set of 10 nonlinear functions $\mathbb{R} \to \mathbb{R}$ (specified in Appendix C.5). To vary the strength of the nonlinearities, we let the nonlinearity scale vary $\omega_{\text{nl}} \in [0,1]$. We note that $\omega_{\text{nl}} = 0$ corresponds to linear function as in the previous experiment. On the other hand, $\omega_{\text{nl}} = 1$ corresponds to purely nonlinear functions. Here, we set $\omega_{\text{pred}} = 1$ – a setting for which all learners performed well in the previous experiment and $\pi(x) = 0.5$.

**Results.** We present attribution score results across all datasets in Fig. 3. We find that ① *Attribution trends vary with* $\omega_{\text{nl}}$. As $\omega_{\text{nl}}$ increases and the underlying DGP is dominated by the nonlinearity, hence becoming more difficult to learn, we observe that correct attribution ($\text{Attr}_{\text{pred}}$) decreases for all methods. Also here, we observe that this is mirrored by an *increase* in confusion of prognostic effects for predictive ones (as seen in the increasing $\text{Attr}_{\text{prog}}$). Further, we note that ② *Relative ordering of methods does not change.* The S-Learner continues to underperform compared to all other learners and the performance gap does not substantially change across values of $\omega_{\text{nl}}$.

### 5.3 Experiment 3: The Effect of Confounding

**Description.** Finally, we turn to examine the effect of *confounding*, i.e. covariate shift between treatment groups resulting from treatment assignment being based on observables – a problem that much of the ML literature proposing CATE estimators has focussed on (e.g. [9, 10, 12]). A popular solution to deal with said covariate shift has been to rely on *balancing* regularization that penalizes distributional distance (here: $\text{MMD}^2$) between treatment groups in representation space; here we therefore also consider [10]'s CFRNet which is identical to TARNet but includes a discrepancy-based regularization term controlled by hyperparameter $\gamma$. We note that if the covariates that determine treatment assignment are either predictive or prognostic – i.e. they are true *confounders* – it is generally not possible to remove *all* covariate shift without removing predictive/prognostic information. We therefore consider a final experiment where we structurally vary not only the degree of assignment bias (through propensity scale $\omega_\pi$) but also what *type of information* assignment is based on. We achieve this by modifying the propensity score: $\pi(x) = \texttt{Sigmoid}(\omega_\pi \cdot Z_{\text{score}}[\psi(x)])$, where $Z_{\text{score}}[\cdot]$ indicates normalization across the generated training dataset (this ensures well-behaved propensity scores centered at 0.5) and $\psi : \mathcal{X} \to \mathbb{R}$ controls the type of confounding.

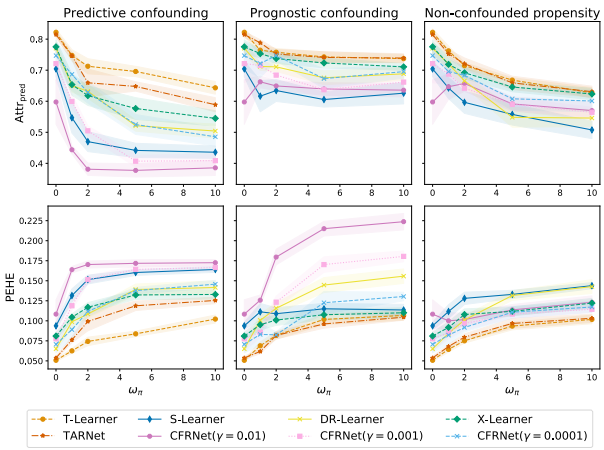
We consider 3 types of confounding, each corresponding to a different choice for $\psi$. *Predictive confounding* corresponds to setting $\psi = \mu_{\text{pred1}} - \mu_{\text{pred0}}$. It mimics a scenario where treatment assignment is biased towards those with characteristics making them most likely to respond well to it, e.g. a doctor assigning treatment with knowledge of CATE. *Prognostic confounding* corresponds to setting $\psi = \mu_{\text{prog}}$. It mimics the most classical confounding setting where treatment assignment is biased towards those with characteristics making them more likely to have a good outcome regardless of treatment, e.g. self-selection into a treatment program. Finally, we consider a *non-confounded*



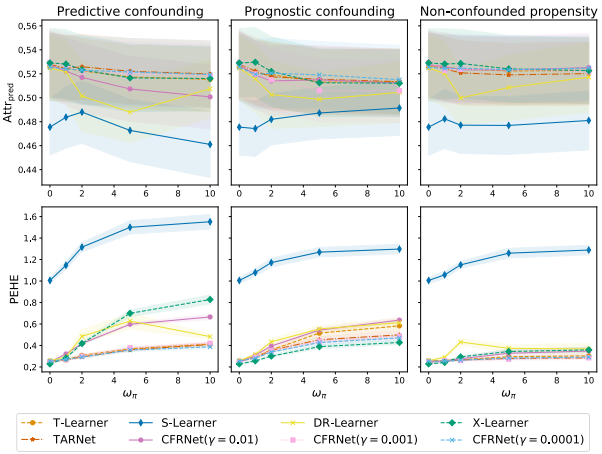

(a) TCGA




(b) Twins




(c) News




(d) ACIC2016

Figure 4: Performance comparison when increasing the propensity scale. Averaged across multiple runs, shaded areas indicates one standard error.

*propensity*: $\psi(x) = x_i$ for some irrelevant covariate $i \notin \mathcal{I}_{\text{prog}} \sqcup \mathcal{I}_{\text{pred}}$. In this case, the treatment selection is not based on a covariate that affects outcome; note that the distribution of covariates in $\mathcal{I}_{\text{pred}} \sqcup \mathcal{I}_{\text{prog}}$ might still differ across treatment groups if they are correlated with the chosen 'irrelevant'. Note that for $\omega_\pi = 0$, all settings are identical and reduce to the previous $\pi(x) = 0.5$. For the potential outcomes, we consider the previous setting with $\omega_{\text{pred}} = 1$ and $\omega_{\text{nl}} = 0$. We believe that such an explicit distinction between different confounding types has not yet been investigated in related work.

**Results.** We present results comparing $\text{Attr}_{\text{pred}}$ and PEHE across the three settings for each dataset in Fig. 4. We make numerous interesting observations: ① *Attribution trends indeed vary with* $\omega_\pi$. Also in this experiment attribution scores vary systematically as we change $\omega_\pi$; across the different settings the quality of attributions worsens the more biased treatment assignment is, albeit only marginally for some settings. ② *There are systematic differences across the three settings.* We observe that performance deteriorates the most in the setting with predictive confounding. We believe that this is a result of less observed variation across predictive covariates *within* a group due to the assignment bias, making it harder for any model to learn that treatment effect varies systematically across these covariates. Overall, we observe the least deterioration in attribution performance in the prognostic confounding setting. That is because, while the prognostic confounding and non-confounded setting are almost identical in terms of $\text{Attr}_{\text{pred}}$ for TCGA, Twins and ACIC, we observe an interesting difference in News: Perhaps surprisingly, we find that performance as measured by $\text{Attr}_{\text{pred}}$ in News deteriorates *less* in the prognostic confounding setting than in the non-confounded setting. We attribute this to two *competing* forces being at play here: First, having any sort of covariate imbalance, as in the non-confounded setting, appears to have the potential to lead to a decrease in $\text{Attr}_{\text{pred}}$. Second, in the prognostic confounding setting, there is a similar effect as in the predictive setting that can offset some of the initial performance decrease: as prognostic covariates now have less variation within a group, models are less likely to *misattribute* a predictive effect to them. ③ *CFRNet's balancing regularization has different effects across settings.* We find that

the addition of the balancing regularization term can lead to a very large drop in $\mathrm{Attr}_{\mathrm{pred}}$ in the predictive confounding setting – this is expected as overly aligning distributions should lead to a loss in explanatory power, which is also reflected in the PEHE. In the other two settings, no such effect is visible when considering $\mathrm{Attr}_{\mathrm{pred}}$; yet, we observe that PEHE does worsen considerably in the prognostic confounding setting. We believe that this is an effect of the prognostic component of the POs being estimated less accurately due to the balancing regularization, making the CATE estimate, the difference between the two estimated POs, less accurate overall.

# 6 Discussion

In this paper, we have introduced the ITErpretability benchmark, a new environment to benchmark CATE models with the help of feature importance methods. We empirically demonstrated on various datasets that this benchmark provides insights that are *not* accessible with the metrics and benchmarks considered standard in the CATE literature. We believe that this work opens up many interesting avenues for future research: First, our environment could be used to extend insights to many more of the CATE estimators proposed in recent literature. One could, for example, replicate our propensity experiments with learners tackling confounding with methods other than balancing, e.g. importance weighting as in [12], or compare performance across different classes of underlying ML methods – i.e. compare how discoveries differ across implementations relying on neural networks with e.g. random forests as in [16] or the gaussian processes of [7, 8]. Second, it may be worthwhile to further study possible failure modes of the considered learning strategies within and beyond our benchmarking environment. This could be achieved both by considering an even wider range of DGPs to investigate how our results generalize empirically, and by complementing our empirical findings with theoretical ones studying under which conditions what types of learners are *guaranteed* to discover the correct predictive covariates. Third, we believe that it would be interesting to consider how to *improve existing* or *develop new* CATE estimation strategies with the help of interpretability techniques or insights derived from experiments such as the ones from Sec. 5. Finally, note that we have exclusively focused on *feature importance* methods here. Another possible extension of this work would then be to perform a similar study with other type of explanation methods, such as example-based explanation methods like Influence Functions [55] and hybrid methods like SimplEx [56].


# Acknowledgments and Disclosure of Funding

The authors are grateful to Javier Abad Martinez and Bogdan Cebere for implementing and running preliminary, yet different, experiments in the early and explorative stages of the project. Jonathan Crabbé is funded by Aviva, Alicia Curth is funded by AstraZeneca, Ioana Bica is funded by the Alan Turing Institute (under the EPSRC grant EP/N510129/1) and Mihaela van der Schaar is funded by the Office of Naval Research (ONR under the grant NSF 1722516). The authors have no competing interests to disclose.

International Journal of Data Science and Analytics (2025) 20:615–635
https://doi.org/10.1007/s41060-023-00445-1

REGULAR PAPER

# Through the looking glass: evaluating post hoc explanations using transparent models

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Received: 13 April 2023 / Accepted: 9 August 2023 / Published online: 12 September 2023
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**Abstract**
Modern machine learning methods allow for complex and in-depth analytics, but the predictive models generated by these methods are often highly complex and lack transparency. Explainable Artificial Intelligence (XAI) methods are used to improve the interpretability of these complex "black box" models, thereby increasing transparency and enabling informed decision-making. However, the inherent fitness of these explainable methods, particularly the faithfulness of explanations to the decision-making processes of the model, can be hard to evaluate. In this work, we examine and evaluate the explanations provided by four XAI methods, using fully transparent "glass box" models trained on tabular data. Our results suggest that the fidelity of explanations is determined by the types of variables used, as well as the linearity of the relationship between variables and model prediction. We find that each XAI method evaluated has its own strengths and weaknesses, determined by the assumptions inherent in the explanation mechanism. Thus, though such methods are model-agnostic, we find significant differences in explanation quality across different technical setups. Given the numerous factors that determine the quality of explanations, including the specific explanation-generation procedures implemented by XAI methods, we suggest that model-agnostic XAI methods may still require expert guidance for implementation.

**Keywords** Predictive models · Explainable AI (XAI) · Performance evaluation of algorithms and systems · Explanation fidelity

## 1 Introduction

Modern machine learning and deep learning techniques have allowed the analysis and modelling of complex data to enable decision-making. However, these advanced machine learning techniques are often not human interpretable, and therefore, lack transparency and hamper responsible and accountable human decision-making [23]. Explainable AI (XAI) methods are used to improve the interpretability of these complex "black box" models, thereby increasing transparency and enabling informed decision-making [18]. A key category of XAI techniques is *post hoc explainable methods*; that is, external mechanisms that provide explanations for a given predictive model, rather than an interpretation provided directly by the model.

Despite the increasing use of post hoc XAI methods in the literature, evaluations and comparisons of post hoc explainable methods are so far under-explored. In particular, past evaluations of *explanation fidelity*, a measure to assess the correctness and completeness of an explanation with respect to the underlying model [59], are specific to particular classes of datasets or specific explainable methods. A number of easily accessible, open source XAI techniques, such as LIME [41] and SHAP [27] are commonly used, as they are model-agnostic and can be easily applied to pre-existing predictive models. However, the ability of such techniques to mimic the

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workings of predictive models of different classes remains unknown.

Evaluations of fidelity in the literature can generally be classified as one of the following: *external fidelity*, which assesses how well the prediction of the underlying model and the prediction implied by the explanation and/or explanation generation mechanism agree; and *internal fidelity*, which assesses how well the explanation matches the decision-making processes of the underlying model [32]. While methods to evaluate external fidelity are relatively common in the literature [19, 25, 33, 46], internal fidelity evaluations are generally limited to text and image data [13, 16, 26, 38, 44], rather than tabular data. Thus, the capability of an explanation to mimic the decision-making processes of the underlying model, particularly models built on tabular data, is unknown.

In this paper, we attempt to understand the strengths and limitations of explanations provided by four XAI methods. We generate explanations for fully transparent "glass box" predictive models and compare explanations directly against the underlying model. This facilitates the evaluation of an explanation's ability to reflect the behaviour of the model. In particular, we focus on local explanations, though the techniques used to generate these explanations often vary between, and even within, explainable method types. The goal of our evaluation is not to identify "the best" explainable method, but to understand the strengths and weaknesses of different explanation-generating mechanisms and the contexts that each explainable method can support.

The main contributions of this work are:

- the development of a method to compare local explanations against the decision-making of three classes of fully transparent predictive models;
- evaluation of four XAI methods in various experimental setups; and
- insights to support the choice of XAI methods to be used within a particular context.

This paper is structured as follows. We first present a number of related works, along with the motivations of this work, in Sect. 2. This is followed by a proposal of our evaluation method (Sect. 3) and design of experiments (Sect. 4). We present our results in Sect. 5 and analyse the results and draw insights in Sect. 6. Finally, we conclude our paper and discuss future work (Sect. 7).

## 2 Background and related works

### 2.1 Explainable AI

The "black box problem" of AI arises from the inherent complexity and sophisticated internal data representations of many modern machine learning algorithms [26]. Although more complex predictive models may produce more accurate results, this accuracy comes at the cost of human interpretability of algorithmic decisions [24]. XAI research attempts to provide human-understandable explanations for predictive models. This is necessary to ensure system quality and to facilitate informed human decision-making [8].

In this work, we consider "interpretability" to be the ability to provide meaning in terms understandable to a human and "explanations" to be the interface between the human and the predictive model [18]. Interpretability in machine learning is generally broken down into two categories: inherently interpretable predictive models and post hoc interpretation. Interpretable predictive models are those that are immediately interpretable by a human [18], though this often means that the models are simpler, and so may have reduced predictive power. This category includes simple decision trees and linear and logistic regression models, which are easily interpretable, and thus *fully transparent*.

Post hoc methods provide interpretations that are not inherent to the predictive model and are derived after its creation through an external mechanism [18]. Post hoc methods are usually applied to *opaque* models, which have complex internal mechanisms, such as a neural network, or are otherwise "black box". A sufficiently complex tree-ensemble model or a model protected by IP agreements may also fall into this category [43]. Thus, post hoc interpretability allows for the use of more complex and accurate predictive models without compromising interpretability. However, since post hoc interpretation is external to the predictive model, there is no guarantee that the explanations provided are fully and correctly representative of the underlying model [43].

### 2.2 Post hoc explainable methods

The purpose and scope of post hoc explanations often vary [29]. In this work, we assess four local (prediction–level) explanations, rather than global (whole–model) explanations. That is, when given an individual input $x \in X$ of length $n$ and a prediction function $f$, a local XAI method would return an explanation $\phi(x)$, rather than $\phi(f)$. In this work, we use four methods that generate local, post hoc explanations.

Moreover, we examine XAI methods that are model-agnostic. These methods typically use some explanation-generation mechanism external to the underlying predictive model, and/or use multiple mechanisms suited for examining multiple types of predictive models. We describe these in more detail below.

#### 2.2.1 LIME

Local Interpretable Model-Agnostic Explanations (LIME) is a local feature attribution method that highlights how important each feature is to a prediction [41]. Given an instance for which one wants to generate explanations, $x$, LIME first applies a set of permutations $x' \in X'$ and then, generates predictions over those permutations, $f(X')$, to train a more interpretable surrogate model. The surrogate model $g \in G$ that mimics the behaviour of $f$ in the vicinity of $x$ is obtained by:

$$\underset{g \in G}{\operatorname{argmin}}\, \mathcal{L}(f, g, \pi_x) + \Omega(g) \tag{1}$$

where $\Omega(g)$ is defined as the complexity of $g$ (the number of nonzero coefficients in the linear surrogate model), $\pi_x$ is a distance function centred on $x$ and the loss function $\mathcal{L}$ is defined as:

$$\mathcal{L}(f, g, \pi_x) = \sum_{x' \in X'} \pi_x(x')[f(x') - g(x')]^2 \tag{2}$$

The coefficients of $g$ are returned as the explanation $\phi(x)$.

#### 2.2.2 SHAP

Much like LIME, Shapley Additive Explanations (SHAP) is a feature attribution method and produces SHAP values as a measure of a feature's contribution [27]. SHAP values for a single feature $i$ can be calculated as:

$$\phi_i(f, x) = \sum_{z' \subseteq z} \frac{|z'|!(M - |z'| - 1)!}{|M|!}[f_x(z') - f_x(z' \backslash i)] \tag{3}$$

where:

- $z$ is a simplified representation of $x$,
- $z' \subseteq z$ represents all $z'$ vectors where the nonzero entries are a subset of the nonzero entries in $z$,
- $|z'|$ is the number of nonzero entries in $z'$, and
- $M$ is the number of features in $z$

Rather than attempting to estimate the importance of features to the model, SHAP defines the contribution of a feature $\phi_i$ such that:

$$f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i z_i \tag{4}$$

where $\phi_0$ is generally defined as $f(\varnothing)$ and calculated as the expected value of the prediction function $f$. Both model-agnostic [27] and model-specific [27, 28] approximations of SHAP values exist and are used in this work (see Sect. 4).

#### 2.2.3 LINDA-BN

Unlike LIME and SHAP, Local Interpretation-Driven Abstract Bayesian Network (LINDA-BN) provides a profile-like explanation instead of a feature attribution explanation [34]. Similarly to LIME, LINDA-BN creates a surrogate model in a neighbourhood of $x$ within a variance of $\epsilon$. However, instead of using Linear Regression to fit the permuted data points, LINDA-BN learns a Bayesian network using the Greedy Hill Climbing approach. This Bayesian network is returned as the explanation for a prediction. Therefore, this method expresses explanations as correlations between features, rather than assuming that each feature individually contributes to the prediction. LINDA-BN is considered a profile-like explanation, as the extracted Bayesian network enables the identification of four rules to inform a decision-maker about the potential correctness of a single prediction (Fig. 1) [34].

Since LINDA-BN's explanations are grounded in probabilistic graphical models, they fulfil the axioms of probability theory and the rules that derive from them. Thus, the structure of the explanation, i.e. the Bayesian network, can be used to determine the model's confidence in and the likely correctness of a prediction [34].

Although LINDA-BN returns only the surrogate model, a quantified impact of each feature can be calculated by querying the returned Bayesian network (see Sect. 4).

#### 2.2.4 ACV

Active Coalition of Variables (ACV) is a probabilistic method of explanations that aims to find some set of minimal suffi-

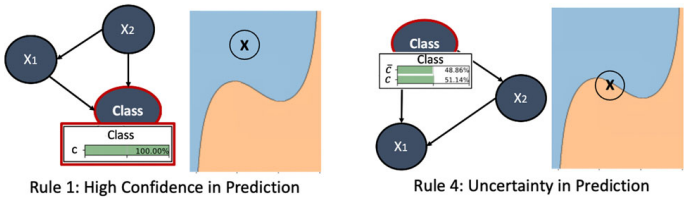



**Fig. 1** Outputs from LINDA-BN [34]: when the data instance to be explained falls in a well-defined region of the decision space, then there is high confidence in the generated predictions (Rule 1). When the data instance falls near the decision boundary, the returned networks will show uncertainty towards the prediction (Rule 4)

cient features (minimal sufficient explanations, M-SE) such that retaining only the M-SE in $x$ returns the same or similar $f(x)$ given a high probability $\pi$ [5]. That is, given an instance $x$, prediction $f(x)$ and threshold $t$, $x_s \subseteq x$ is a Sufficient Explanation for probability $\pi$ if the same decision probability $\text{SDP}_s(x, f(x), t) \geq \pi$ and no $x_z \subseteq x_s$ satisfies $\text{SDP}_z(x, f(x), t) \geq \pi$ where:

$$\text{SDP}_s(x, f(x), t) = \Pr((f(X) - f(x))^2 \leq t \mid X_s = x_s) \tag{5}$$

for regression, and for classification:

$$\text{SDP}_s(x, f(x)) = \Pr(f(X) = f(x) \mid X_s = x_s) \tag{6}$$

A M-SE is a Sufficient Explanation with minimal size, and the set of all Sufficient Explanations is denoted as A-SE.

ACV uses a Random Forest model as a global surrogate, i.e. a surrogate model that attempts to capture the entirety of model behaviour, instead of some subset of behaviour. The idea of Projected Forest, which is used to estimate $E[f(X) \mid X_s = x_s]$, is combined with a Quantile Regression Forest, which estimates $Pr(f(X) \leq f(x) \mid X = x)$ to compute the SDP [5]. As the number of possible subsets is exponential, to reduce the complexity of the computation, the search for Sufficient Explanations is contained to a maximum of ten features that are used most frequently in the surrogate model.

ACV is also distinct from the other XAI techniques used in this work, as it can also function as a self-explaining predictive model. When explaining a separate predictive model $f$, rather than taking the model as input, as LIME, SHAP and LINDA-BN do, ACV's surrogate model takes as input the training set $X$ and the original model's predictions $f(X)$. However, if given $X$ and the associated labels $Y$, ACV can also be trained as a predictive model, though we do not use it as such in this work.

### 2.3 Evaluating explanations

Evaluation methods for XAI can be categorised into three levels of evaluation [12]:

- *Application-grounded evaluation* wherein the evaluation is conducted with real end users in full context replicating a real-world application;
- *Human-grounded evaluation* wherein the evaluation is conducted with laymen in simpler simulated contexts or completing proxy tasks that reflect a target application context; and
- *Functionally-grounded evaluation* which requires no input from users and relies on evaluating the inherent abilities of the system via a formal definition of interpretability.

The former two categories are also often referred to as *cognitive metrics* of evaluation, and functionally-grounded evaluations fall into the complementary category of *computational metrics* [26].

Evaluations of this last category are often the first step in determining the quality of an XAI method [12]. This includes evaluation of *explanation fidelity* with respect to the original predictive model. The term "fidelity" is generally defined as a measure of how faithful and accurate an explanation is to the underlying black box model [56]. Two characteristics of explanations are relevant to explanation fidelity [30, 59]: the *completeness* of an explanation in capturing the dynamics of the underlying model; and the *correctness* and truthfulness of the explanation with respect to the underlying model.

Explanation fidelity can severely impact the safety and usefulness of a predictive system. The authors of [21] find that, when the user has deep knowledge of the context, local explanations enabled the identification of prediction errors and the predictive model's error boundaries. The study also finds that explanations affect the user's perception of model correctness. Similarly, the authors of [47] find that model transparency affects causability—the ability of the user to understand what features affect the prediction. In turn, causability is found to determine the user's trust in and perception of model performance. We must note that explanation fidelity does not play a direct role in causability or user perception; these are generally determined by explanation presentation and content [21, 58]. However, if a user's knowledge or perception is based on incorrect or incomplete explanations of the model, it may engender trust in a false prediction, or mistrust in a correct prediction, leading to unsafe or incorrect usage of a predictive system. Thus, explanation fidelity is necessary for predictive model safety and usefulness.

Although a number of methods have been proposed to assess fidelity, they are often highly specific to particular XAI methods or datasets. Fidelity evaluation approaches fall into two general measures: measurements of *external fidelity* and *internal fidelity* [32]. External fidelity approaches are those that compare how often the decision implied by the explanation and the decision made by the underlying predictive model agree. For example, when evaluating surrogate models that approximate the underlying predictive model, this method determines the accuracy of the surrogate model's predictions using the predictions of the underlying model as ground truth [46]. Explainable methods that produce decision rules are also often evaluated through this approach, and the predictions of the decision rule or decision rule sets are compared against the black box model's predictions [19, 25, 33].

Evaluating external fidelity provides an assessment of consistency between the explanation's predictions and the predictions of the underlying model, i.e. that the predictive model and the explanation mechanism can reach the same

decision. However, this does not guarantee that the explanations are faithful to the computations, or "decision-making", of the black box model, for example, weights given to each feature, or the decision path followed in a tree [32]. Such information may be necessary in order to change the predicted outcome to a desired outcome. For example, in the case of a risk prediction model, taking the most appropriate action to reduce the risk. Lack of full transparency can also hamper model inspection and diagnosis, which may be necessary to ensure that the model relies on appropriate features when computing a prediction [41]. Moreover, some explainable methods do not produce an output suitable for external fidelity evaluation approaches, including LINDA-BN and some optimisations of SHAP.

As such, we aim to develop a method to assess the internal fidelity of feature attribution explainable methods, where the fidelity of the explanation regarding the underlying model's decision-making process is evaluated [32]. As noted in [6], given that the internal workings of a black box model are opaque and there is no "ground truth" of the model, it is near impossible to assess explanation correctness. Significant effort is required to assess the internal fidelity of an explanation or explainable method. Potential internal fidelity evaluation approaches include:

1. Generating explanations for a fully transparent, inherently interpretable "glass box" model, and comparing the explanation to the glass box model's decision-making [4, 41];
2. Using a post hoc explanation approach to explain both a surrogate model and the underlying predictive model, and comparing how often the explanations concur [32]; and
3. "Removing" or permuting the features identified as relevant by the explanation and measuring the change in model output [4, 13, 16, 26, 37, 38, 44].

It is important to note that the method in [32] is an evaluation of a surrogate model using a post hoc technique, rather than an evaluation of the post hoc technique. Thus, it is not suitable for this work. As another note, while the method of permuting features is relatively common in the literature, it is mostly applied to text [13, 38] or image data [16, 26, 44], in which case, the relevant features are often removed or blurred from the original input prior to encoding. This method has been applied to tabular data [4, 31, 37], typically by permuting rather than removing features at the input level [4, 37] or removing the feature as a whole from the dataset [31]. Moreover, we note that these works typically used few, standard benchmark datasets and/or limited predictive models. In our previous work, we adopted a local, perturbation method to evaluate explanations for models built on time series tabular data, which, however, provided poor results [49]. We theorise that the results of our previous work were likely influenced by the difficulty in choosing and permuting an appropriate subset of features given the complexity of time series data, and lack of understanding of model and XAI technique behaviour with respect to tabular data.

Tabular datasets are highly heterogeneous with respect to the quantity and type of features and may present challenges such as lack of locality and sparsity in data, as well as the need to manage mixed feature types [48]. Furthermore, there may be fewer correlations between features in tabular data than in other types of data such as images, as well as lack of semantic meaning and connection between features such as may be present in natural language data [7]. Although it is noted in [17] that simpler, glass-box models can generally provide accurate predictions for tabular data, the authors of the study also note that in some cases, particularly when the dataset is "noisy", more complicated black box models may be necessary. These factors affect the predictive techniques that may be used [7], but, as we noted in [49], the characteristics of tabular data also does not allow for use of explanation evaluation methods developed for image and text data.

### 2.4 Motivation

In this work, we use fully transparent, glass-box models to better understand interactions between data, model and XAI technique. This provides a "ground truth" for model decision-making, with which to assess explanations. Furthermore, we extend the approach used in [41], which relied on measuring only the completeness of the explanation in capturing a hard-coded, arbitrary number of features. Moreover, to the best of our knowledge, no work compares the performance of XAI techniques in different technical contexts to understand the factors that affect explanation quality. In particular, we aim to examine whether easily accessible, model-agnostic XAI methods, such as those described, work reliably and consistently in all technical contexts. Thus, in this work, we attempt to take a more holistic approach to evaluation, as described in Sect. 3.2.

We investigate properties of post hoc XAI methods by addressing the following questions:

- Local surrogate models are commonly used to generate local, post hoc explanations. *To what degree do permutation and the choice of local surrogate model affect the quality of explanation generated by the surrogate model?*
- Model-agnostic XAI methods often use some theoretical construct to define and calculate explanations. For example, principles from game theory are used to derive SHAP explanations [27, 28] and statistical models are applied by LINDA-BN [34]. However, some assumptions and design decisions may be needed to make these theoretical foundations suitable for application. *How do the design*

*choices made in the implementation of the explanation generation mechanism affect explanation quality?*

- While many local explanation methods use local surrogate models (at the data point level), a few also use global surrogate models (i.e. at the dataset level). *How does the use of a global surrogate model affect explanation quality?*

# 3 Method

## 3.1 Evaluation method

In this work, we aim to investigate the internal fidelity of model-agnostic explanations, where the faithfulness of the explanation to the underlying model's computational process is evaluated [32]. Given that the internal workings of a black box model are opaque, significant effort is required to assess the internal fidelity of an explanation or explainable method. In our previous work [49], we attempted to assess explanation fidelity for black box models trained on tabular data, by replicating the methods used to test the internal fidelity of explanations for models trained on image and text data. Our results suggested that explanation fidelity is poor when explaining tabular data, though we found that this may be a consequence of the evaluation method and the complexity of the datasets used. This lack of evaluation method for tabular data is notable given the volume and diversity of tabular data used in modern data analytics.

Thus, in this work, we evaluate explanations using fully transparent glass-box models. We choose models from which the relevance of a feature to a prediction can be determined, to provide a reference for evaluating the internal fidelity of post hoc explanations. As the internal workings of a glass-box model are apparent, it can be compared against an explanation to determine the fitness of the explainable method. It must be noted that high explanation fidelity for a relatively simple glass box method does not necessarily imply high fidelity for a more complicated black box model. As such, our aim in using this method is not to determine some "best" XAI method. Rather, we aim to more precisely understand the specific workings of each XAI method, how different types of predictive models affect explanation quality, and any patterns of quality that could inform the selection of XAI methods for a given setup and context. Thus, we focus on behaviours and characteristics of models and XAI algorithms. Moreover, given the ease of availability of model-agnostic XAI techniques, we also wish to determine the level of expertise needed to select an XAI technique for use.

In this work, we choose to examine four local, post hoc XAI methods of differing characteristics in order to answer the questions posed:

- **LIME** uses permutation, then a local, linear surrogate model to derive an explanation. Previous works have suggested that LIME's permutation of input $x$ affects explanation stability, particularly as the length of the input increases [46, 50]. In addition to the effect of permutation, in this work we will examine the effects of the choice of local surrogate model.
- While **SHAP** is grounded in game theory, it uses both model-specific and model-agnostic computations to determine feature attribution [27, 28].
- **LINDA-BN** is grounded in statistical modelling and provides a queryable explanation.
- **ACV** uses a global, rather than local, surrogate model to generate explanations. Moreover, ACV provides explanations of multiple types, including feature subsets and feature attribution.

We have chosen datasets and predictive models in such a way as to facilitate answers to these questions. Our experimental setup and choice of algorithms are introduced in 4.

## 3.2 Evaluation metrics

In this work, the goal of our evaluation is to understand the strengths and limitations of several local, post hoc XAI mechanisms. In particular, we examine these mechanisms with respect to the characteristics of the underlying models and datasets. Thus, in this work, we use a range of datasets, XAI techniques and predictive models for a comprehensive evaluation. We evaluate four XAI techniques, all of which use different explanation-generation mechanisms, with three predictive models of different classes. We also selected datasets by considering a breadth of different characteristics, including the prediction problem, the volume of the data, types of variables present, and the number of features that can be encoded from the data. A detailed description of techniques and datasets used is presented in Sect. 4.

We evaluate three properties when measuring fidelity:

- The correctness of the explanation in identifying the most impactful features;
- The completeness of the explanation in identifying the most impactful features; and
- The correctness of the importance ranking of all features, which is particularly key for feature attribution methods.

The evaluation method for the first two properties use two subsets of features. The basic approach of this method is to compare the subset of features determined to be most relevant by the model ($x_t \subseteq x$) against the features determined to be the most relevant by the explanation ($x_e \subseteq x$). A similar approach was used in [41], where the recall metric was used

to determine fidelity in the form of *completeness* as follows:

$$\text{Recall}(x_t, x_e) = \frac{|x_t \cap x_e|}{|x_t|} \tag{7}$$

Additionally, we also use the precision metric to capture the *correctness* of the explanation as follows:

$$\text{Precision}(x_t, x_e) = \frac{|x_t \cap x_e|}{|x_e|} \tag{8}$$

To evaluate the third property, we extract the importance ranking of features by the model ($r_t$) and the ranking of features by the explanation ($r_e$), where $r$ is a reordering of $x$ indicating feature importance. We then use a correlation measure to determine the similarity between $r_t$ and $r_e$, where a higher similarity would indicate higher *correctness* of the explanation's ranking. We use Kendall's Tau-B to measure rank correlation between $r_t$ and $r_e$.

As XAI methods, particularly methods relying on feature permutations, often produce unstable explanations [46, 50, 51], we generate five $\phi(x)$ for each $x$, compute an adjusted or average explanation $\varphi(x)$, and determine $x_e$ and $r_e$ based on $\varphi(x)$.

These metrics are applied at the local level (i.e. when testing individual inputs), but can be averaged out at the dataset level, as in Sect. 5.

Note that, of the chosen predictive methods, only the decision tree model provides a natural subset to be used as $x_t$. Similarly, only the ACV provides a subset of features as $\phi(x)$. Thus, a heuristic is necessary to identify $x_t$ and $x_e$ for other methods. The procedure for determining this heuristic, as well as the results, are summarised in Sect. 4.4.

## 4 Design of experiments

### 4.1 Datasets

We focus our evaluations on tabular data, which has been relatively unexplored in the literature on explanation fidelity. In order to better understand the effect of dataset properties on the results, we chose datasets with specific characteristics, rather than datasets from any particular domain. The datasets all vary in terms of the variable types present, the number of variables present once the dataset is encoded as described below, the volume of the data, and, in the case of the regression datasets, the distribution of the target variable.

A total of 14 open source datasets are used, which include seven *classification* datasets:

1. Adult Income [22]
2. Breast cancer [55]
3. COMPAS [39]
4. Diabetes [2]
5. Iris [15]
6. Mushroom [45]
7. Nursery [40]

and seven *regression* datasets:

1. Bike Rentals [14]
2. Facebook [35, 36]
3. Housing [1]
4. Real Estate [57]
5. Solar Flare [3]
6. Student Scores [10]
7. Wine Quality [11]

A brief profile of each dataset is provided in Tables 1 and 2.

Some pre-processing was necessary for all datasets. All classification datasets were balanced through downsampling, to ensure parity between classes. Where a classification dataset had more than two prediction classes, the target variable was binarized before downsampling. All categorical variables in the datasets were one-hot encoded, and min-max scaling was applied to all numeric variables (excluding the target variable in regression datasets). A train-test split of 70–30 was used, and a sample of the testing set was used to evaluate the XAI methods.

### 4.2 Predictive models

In this work, we use predictive model types for which extracting $x_t$ is unambiguous. As such, we choose to exclude techniques for which $x_t$ cannot be easily derived, or are unclear. For example, attention mechanisms are an increasingly popular technique used to interpret models without post hoc methods. This is a form of input selection within some neural networks, which allows the model to focus on inputs most relevant to the output [23]. While attention mechanisms are sometimes used as explanation and have been shown to be successful in some domains [53], the validity of attention as explanation is still debated [20, 31, 54]. Thus, we choose not to use it in this work.

We use four different algorithms of three different classes to create predictive models. Firstly, we use a simple CART decision tree to create both classification and regression models, covering all datasets. For all classification datasets, we also create logistic regression and Naïve Bayes models, and for all regression datasets, we create linear regression models. In summary, we have conducted our evaluations using a total of 35 models: 21 classification models and 14 regression models. The accuracy of these models is shown in Tables 3 and 4.

**Table 1** A brief profile of the classification datasets used in this work

| Dataset | Variable types | Num variables | Training instances | Class balance (%) |
|---|---|---|---|---|
| Adult Income | Mixed | 104 | 10,977 | 50.47 |
| Breast Cancer | Continuous | 30 | 296 | 51.01 |
| COMPAS | Mixed | 20 | 2793 | 50.13 |
| Diabetes | Continuous | 8 | 375 | 50.4 |
| Iris | Continuous | 4 | 70 | 52.86 |
| Mushroom | Discrete | 117 | 5842 | 50.09 |
| Nursery | Discrete | 27 | 6048 | 50.39 |

**Table 2** A brief profile of the regression datasets used in this work

| Dataset | Variable types | Num variables | Training instances | Target distribution |
|---|---|---|---|---|
| Bike Rentals | Mixed | 62 | 12,165 | Exponential |
| Facebook | Discrete | 49 | 349 | Exponential |
| Housing | Mixed | 23 | 354 | Normal |
| Real Estate | Continuous | 6 | 289 | Normal |
| Solar Flare | Discrete | 32 | 972 | Exponential |
| Student Scores | Mixed | 58 | 454 | Normal |
| Wine Quality | Continuous | 11 | 3428 | Normal |

**Table 3** Model accuracy on all classification datasets

| Dataset | Decision tree | Logistic regression | Naïve Bayes |
|---|---|---|---|
| Adult Income | 0.82 | 0.82 | 0.81 |
| Breast Cancer | 0.88 | 0.98 | 0.95 |
| COMPAS | 0.71 | 0.73 | 0.72 |
| Diabetes | 0.69 | 0.71 | 0.68 |
| Iris | 1.00 | 1.00 | 1.00 |
| Mushroom | 1.00 | 1.00 | 1.00 |
| Nursery | 1.00 | 1.00 | 1.00 |

**Table 4** Model R-squared error on all regression datasets

| Dataset | Decision tree | Linear regression |
|---|---|---|
| Bike Rentals | 0.88 | 0.68 |
| Facebook | 0.42 | 0.26 |
| Housing | 0.63 | 0.71 |
| Real Estate | 0.55 | 0.45 |
| Solar Flare | 0.14 | 0.17 |
| Student Scores | 0.79 | 0.90 |
| Wine Quality | 0.28 | 0.29 |

Given their differences, the extraction of $x_t$ from each type of model varies. We provide a summary of the used methods in Table 5.

#### 4.2.1 Decision tree

Given that a decision tree model is extremely structured, identifying $x_t$ is relatively simple. When using a decision tree model, we traverse the tree to identify the decision path for input $x$ and take as $x_t$ the unique set of features that fall along the decision path (see Fig. 2). We then calculate the ranking of all features $r_t$ based on each feature's position and frequency on the decision path.

#### 4.2.2 Linear and logistic regression

Given a linear regression model:

$$f(x) = w_0 + w_1 x_1 + \cdots + w_n x_n \tag{9}$$

and a logistic regression model:

$$f(x) = \frac{1}{1 + \exp(-(w_0 + w_1 x_1 + \cdots + w_n x_n))} \tag{10}$$

where $w_0$ is the intercept and $w \ni w_1 \ldots w_n$ are the coefficients applied to each feature. We consider the weights in the top 5% of the range of $w' \ni |w_1| \ldots |w_n|$ to be the most relevant to the model and take $x_t \ni x_i$ for $i \in n$ if $|w_i| \geq p$ where:

$$p = \max w' - (\max w' - \min w') \times 0.05 \tag{11}$$

We order the features in $x$ by the absolute values of their coefficients $|w_1| \ldots |w_n|$ to determine $r_t$.

**Table 5** A brief summary of the predictive models used in this work

| Model | Applied To | Feature extraction | Ranking extraction |
|---|---|---|---|
| Decision tree | All Datasets | Features along the decision path were used as true features | Features were ranked in order and frequency of appearance on the decision path |
| Logistic regression | All Classification Datasets | Features with coefficients in the top 5% of the range of coefficients | Features were ranked in order of the absolute values of their coefficients |
| Linear regression | All Regression Datasets | Features with coefficients in the top 5% of the range of coefficients | Features were ranked in order of the absolute values of their coefficients |
| Naïve Bayes | All Classification Datasets | Features for which the difference in likelihoods given each class were in the top 5% of the range of likelihoods | Features were ranked in order of the absolute values of the difference in likelihoods given each class |

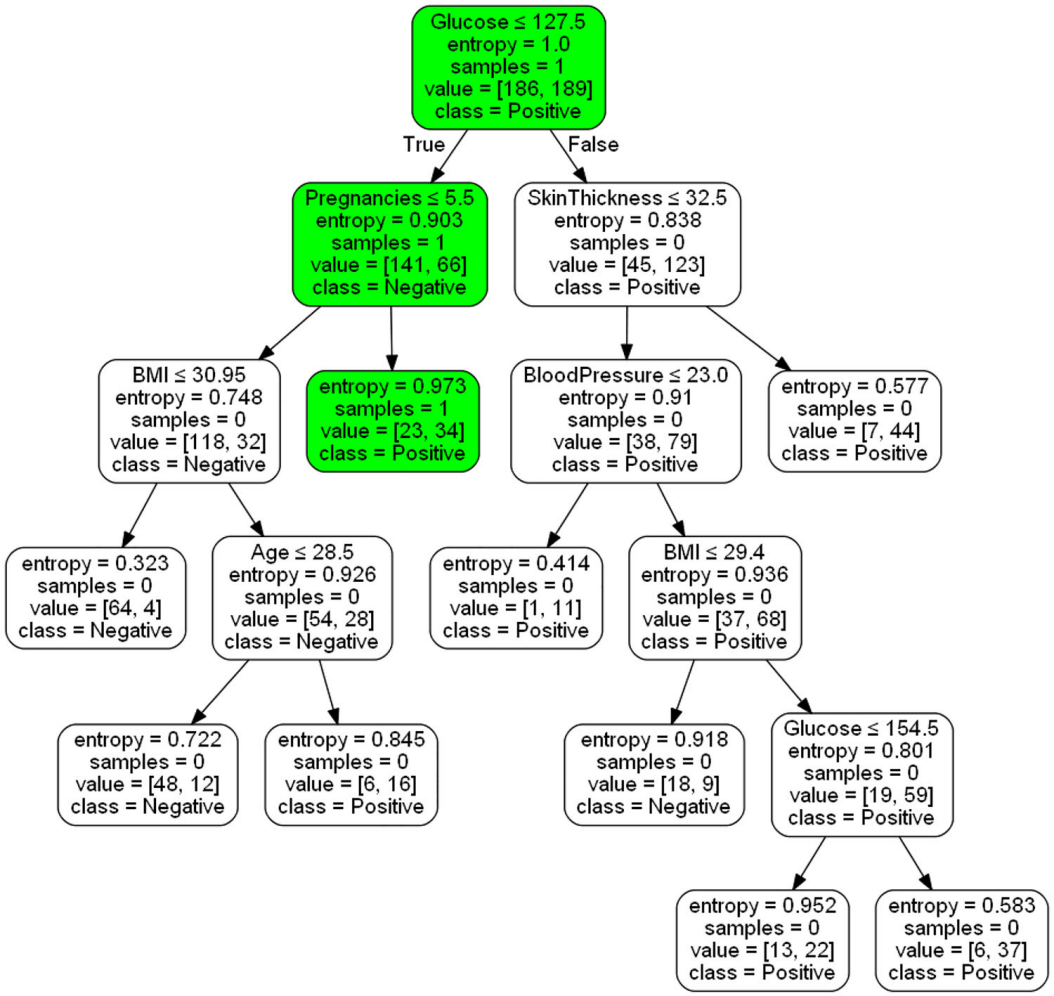



**Fig. 2** Prediction of Diabetes, using a decision tree. The highlighted decision path shows all the True Features $x_t$ that are impacting the prediction for a single instance

#### 4.2.3 Naïve Bayes

Since a Naïve Bayes algorithm applies Bayes' theorem with the assumption that all features are conditionally independent from one another, for all $c \in C$ classes in a dataset and given an instance $x$ of length $n$, $\Pr(c \mid x)$ is calculated as:

$$\Pr(c \mid x) = \Pr(c) \prod_{i=1}^{n} \Pr(x_i \mid c) \tag{12}$$

to determine the most probable $c$.

As such, we calculate the importance of each feature $i$ given a Naïve Bayes model $f$ and instance $x$ as:

$$w_i = \Pr(x_i \mid f(x)) - \Pr(x_i \mid 1 - f(x)) \quad (13)$$

We calculate $w_1 \ldots w_n \in w'$ and take $x_t \ni x_i$ for $i \in n$ if $|w_i| \geq p$ where $p$ is calculated using Eq. 11. Since we use a Gaussian Naïve Bayes model, the distribution of $Pr(x_i \mid f(x))$ is assumed to be Gaussian and easily calculated using the training set. We order the features in $x$ by the absolute values of $w$ (i.e. $|w_1| \ldots |w_n|$) to determine $r_t$.

## 4.3 XAI techniques

We use and evaluate four XAI methods in this work, each of which use different underlying mechanisms. The format of the provided explanations also varies across the explainable methods. The method by which we extract $x_e$ from each method is described in detail below. A summary of the XAI techniques used, the feature extraction method and the experiments that it is used for is provided in Table 6.

As specified, one of the goals of this work is to understand the effect of explanation mechanisms on explanation fidelity. Thus, we examine XAI techniques with different underlying theoretical principles and explanation-generation procedures. LIME uses linear, local surrogate models to determine feature importance [41], while SHAP is grounded in game theory and differs its approach to determining the contribution of a feature depending on the underlying model [27, 28]. LIME and SHAP are also among the most commonly used local, post hoc techniques for tabular data, and thus, important candidates for evaluation. Although LINDA-BN also uses local surrogate models to generate explanations, unlike LIME, it is grounded in probability theory and uses a Bayesian Network as surrogate model [34]. Moreover, unlike LIME and SHAP, LINDA-BN does not attempt to determine feature importance to the prediction, but examines the conditional dependence between all variables, including the features. Similarly, ACV is also not a feature attribution method, but attempts to identify which subsets of features are necessary for the prediction [5]. Thus, we choose LINDA-BN and ACV in contrast to the more "conventional" LIME and SHAP.

### 4.3.1 LIME

As a model-agnostic method, LIME was used across all experiments. For a user-defined number of features $s$, $x_s \subseteq x$, LIME returns weights for each feature in $x_s$, i.e. $w_1..w_s \in \phi(x)$. For each $x$, we set $s = n$, generate five explanations, and calculate $|\overline{w_1}| \ldots |\overline{w_s}| \in \varphi(x)$.

For $i \in n$, we take $x_e \ni x_i$ if $\overline{w_i} \geq p$ where:

$$p = \max \varphi(x) - (\max \varphi(x) - \min \varphi(x)) \times 0.05 \quad (14)$$

We order the features in $x$ by the weights in $\varphi(x)$ to extract $r_e$.

### 4.3.2 SHAP

Although SHAP is model-agnostic, specific optimisations have also been developed for tree-based and linear and logistic regression models. Thus, we use TreeSHAP when explaining the decision tree models and LinearSHAP when explaining the logistic regression and linear regression models. Model-agnostic computations of SHAP are dependent on the number of features present in the dataset; thus, we use two different SHAP optimisations for the Naïve Bayes models. We use the ExactExplainer to explain the Naïve Bayes classifiers for the Iris and Diabetes datasets, and the PermutationExplainer to explain all other Naïve Bayes models.

SHAP provides its explanations as SHAP values, where $\phi(x)$ is single vector of length $n$ for regression models. When explaining classification models, $\phi(x)$ is a vector of SHAP values for each possible class $c \in C$, where each $\phi_c(x)$ is of length $n$. Though the underlying mechanisms of LIME and SHAP are different, SHAP's explanation is similar in structure to that of LIME, such that $\phi(x) \ni v_1 \ldots v_n$. Therefore, we repeat our feature extraction method for LIME, calculating $|\overline{v_1}| \ldots |\overline{v_n}| \in \varphi(x)$ from five explanations. For $i \in n$, we take $x_e \ni x_i$ if $\overline{v_i} \geq p$ where $p$ is determined by Eq. 14. As with LIME, we order the features based on $\varphi(x)$ to determine $r_e$.

### 4.3.3 LINDA-BN

Although LINDA-BN is also a model-agnostic method, as the explanation produced is a probabilistic model, it can currently only be used to explain classifiers. Thus, it was evaluated using only the classification models. The explanation $\phi(x)$ returned by LINDA-BN is a probabilistic graphical model that captures conditional dependencies between variables (including the prediction) and provides a posterior probability for any given prediction class $c$. That is, $Pr(c)$, given no knowledge about the feature values in $x$. Given the instance $x$ and $f(x)$ being explained, we can query $\phi(x)$ to determine the impact ($\pi_i \in \pi$) of each feature $x_i \in x$ to $f(x)$ as follows:

$$\pi_i(\phi(x)) = |\Pr(f(x)) - \Pr(f(x) \mid x_i)| \quad (15)$$

In this way, we can create something approximating feature attribution explanations using LINDA-BN. As with

**Table 6** A summary of the XAI methods used in this work

| XAI method | Explanation-generation mechanism | Used for | Feature extraction | Ranking extraction |
|---|---|---|---|---|
| LIME | Local, linear surrogate model to determine feature weights | All models | Features with weights in the top 5% of the range of weights | Features ranked in order of absolute value of weights |
| SHAP | TreeSHAP: Tree traversal to determine contribution of each feature | All decision tree models | Features with contributions in the top 5% of the range of contributions | Features ranked in order of absolute value of contributions |
| | LinearSHAP: Examining feature coefficients and means to determine contribution | All linear and logistic regression models | | |
| | ExactExplainer and PermutationExplainer: Use of gray codes to determine feature contribution | All Naïve Bayes models | | |
| LINDA-BN | Local Bayesian Network surrogate model to determine conditional dependence between all variables, including the target variable | All classification models | Features with the greatest impact on the target variable in the surrogate model | Features ranked in order of impact on the target variable in the surrogate model |
| ACV | Global, random forest surrogate model to determine sufficient features for prediction | All models | Smallest set of sufficient features returned as explanation | Features ranked by percentage of occurrence across all sufficient feature sets |

LIME and SHAP, we repeat this procedure five times, generating a new Bayesian Network each time and calculating $\pi$ for each network. When using LINDA-BN, $\varphi(x) \ni \overline{\pi_1} \ldots \overline{\pi_n}$, and we take $x_e \ni x_i$ if $\overline{\pi_i} \geq p$ where $p$ is determined by Eq. 14. We then order the features in $x$ based on $\varphi(x)$ to determine $r_e$.

#### 4.3.4 ACV

ACV offers explanations in many formats, including the set of all possible sufficient explanations (A-SE), the smallest possible set of sufficient explanations (minimal sufficient explanations, M-SE) and a Local Explanatory Importance (LEI) score for each feature. The LEI is not directly determined by the prediction, but is a measure of how often any given feature is present across all sets in the A-SE. Thus, we take the A-SE as $\phi(x)$, and the smallest set in the A-SE (i.e. the MSE) as $\varphi(x)$. When evaluating ACV, $x_e = \varphi(x)$, and the LEI is used as $r_e$.

We performed hyperparameter optimisation of ACV's surrogate model to ensure external explanation fidelity.

### 4.4 Identifying feature subsets

To test the completeness and correctness of the most impactful features, both in the model and as determined by the explanation, subsets of features are required. However, only one predictive method and one XAI method used in this work rely on feature subsets. For all other methods, these subsets must be determined and extracted from $x$, based on the "weights" attached to each feature.

Thus, a certain percentile of the Top-K features must be chosen as subsets from methods that do not provide subsets. To determine the most appropriate percentile, we conducted tests to identify a percentile from 0.05 . . . 0.5 that would provide the most faithful explanations. For each combination of dataset, model and XAI technique, we extracted $x_t$ and $x_e$ using each of the ten percentiles and calculated the $F1$-score to determine the match between the subsets. As with the fidelity evaluation, this resulted in 126 experiments. In 38 experiments, the subsets produced for all percentiles were equally faithful, i.e. the top-K percentile had no impact on the fidelity of the subsets, and these results were set aside. Out of the remaining 88 experiments, 0.05 produced the most faithful subsets in the majority of experiments (see results in Fig. 3). Of these, in 49 experiments, 0.05 produced the most faithful subsets. In another 38, the subsets produced for all percentiles were equally faithful, i.e. the top-K percentile had no impact on the fidelity of the subsets. Thus, 0.05 was chosen to determine $x_t$ and $x_e$.

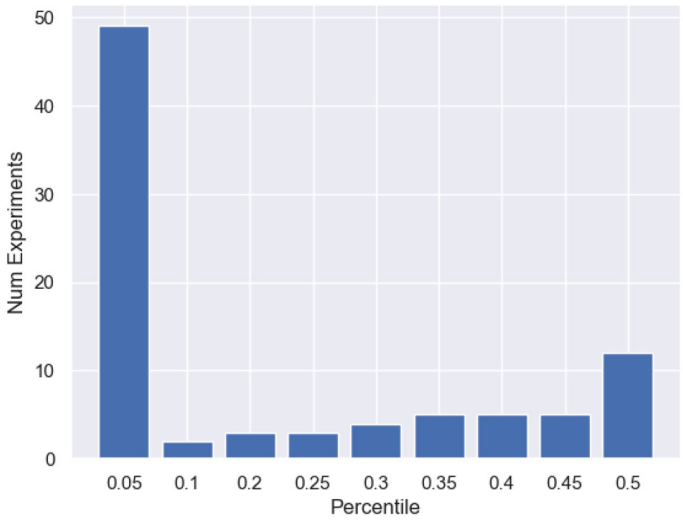



**Fig. 3** Distribution of the number of experiments that produces the most faithful explanations, over the percentile used to calculate the top-K features. Taking the top 5% of features as the true and explanation features generally produced the most faithful results in a significant proportion of the 126 experiments conducted

## 5 Results and analysis

The average results for each combination of model and dataset are presented in Tables 7 and 8.

### 5.1 Analysis of fidelity results

There are a number of notable observations to be made from these results.

LIME's results are highly inconsistent across the different experimental settings. LIME explanations generally have high precision coupled with low-to-moderate recall scores, indicating that a large number of relevant features are missing from the subset, but also that few extraneous features are included. Feature ranking correlation is also generally moderate. There are no significant differences in LIME performance between classification and regression model. However, model type appears to affect results, and LIME generally performs poorly when explaining Naïve Bayes models. Dataset characteristics also affect LIME explanation quality, and explanations are more faithful for datasets with a larger proportion of categorical features.

SHAP explanations also change significantly under different experimental conditions. Feature subsets derived based on SHAP contributions show high precision in the results, but poor recall and generally low-to-moderate feature rank correlation. SHAP shows no difference in performance between classification and regression datasets, but does show poor performance when explaining logistic and linear regression models, and exceptional precision when explaining decision tree models. Dataset characteristics do not appear to affect SHAP's explanation fidelity.

**Table 7** Average precision (Pr), recall (Re) and correlation (Tau) results for XAI methods when explaining the decision tree (DT), logistic regression (LR) and Naïve Bayes (NB) classification models

| Dataset | Model | LIME | | | SHAP | | | LINDA-BN | | | ACV | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Pr | Re | Tau | Pr | Re | Tau | Pr | Re | Tau | Pr | Re | Tau |
| Adult Income | DT | 0.95 | 0.15 | 0.24 | **1.00** | 0.19 | **0.59** | 0.47 | **0.42** | 0.07 | 0.62 | 0.15 | 0.43 |
| | LR | **1.00** | **1.00** | **0.75** | 0.32 | 0.34 | 0.11 | 0.00 | 0.46 | 0.01 | 0.09 | 0.09 | 0.14 |
| | NB | 0.15 | 0.15 | 0.48 | **0.68** | **0.68** | **0.67** | 0.03 | 0.91 | 0.04 | 0.22 | 0.24 | 0.20 |
| Breast Cancer | DT | **1.00** | **0.43** | 0.35 | **1.00** | **0.43** | **0.62** | 0.09 | **0.43** | 0.10 | 0.26 | 0.25 | 0.30 |
| | LR | **0.75** | 0.24 | 0.52 | 0.60 | 0.17 | **0.53** | 0.24 | **0.57** | 0.06 | 0.38 | 0.20 | 0.29 |
| | NB | **0.41** | 0.42 | 0.61 | 0.40 | 0.37 | **0.69** | 0.05 | **0.53** | −0.07 | 0.18 | 0.20 | 0.44 |
| COMPAS | DT | 0.96 | 0.26 | 0.45 | **1.00** | 0.26 | **0.59** | 0.33 | 0.11 | 0.13 | 0.70 | **0.35** | 0.34 |
| | LR | **0.56** | **0.57** | **0.69** | 0.50 | 0.55 | 0.39 | 0.01 | 0.24 | −0.04 | 0.32 | 0.55 | 0.18 |
| | NB | 0.06 | 0.05 | 0.35 | **0.82** | **0.82** | **0.52** | 0.11 | 0.55 | 0.01 | 0.25 | 0.38 | 0.21 |
| Diabetes | DT | 0.89 | 0.30 | 0.49 | **1.00** | 0.35 | **0.61** | 0.49 | **0.42** | 0.05 | 0.52 | 0.35 | 0.28 |
| | LR | **0.53** | 0.54 | 0.53 | 0.50 | 0.51 | **0.55** | 0.25 | **0.66** | 0.02 | 0.36 | 0.48 | 0.23 |
| | NB | 0.64 | 0.64 | 0.63 | **0.82** | **0.83** | **0.80** | 0.24 | 0.60 | −0.07 | 0.27 | 0.42 | 0.20 |
| Iris | DT | **1.00** | **1.00** | 0.71 | **1.00** | **1.00** | **1.00** | 0.23 | 0.88 | 0.03 | 0.75 | **1.00** | 0.24 |
| | LR | **0.85** | 0.88 | **0.83** | 0.71 | 0.81 | 0.72 | 0.24 | **0.96** | −0.01 | 0.25 | 0.50 | 0.74 |
| | NB | 0.81 | 0.75 | 0.83 | **0.94** | 0.94 | **0.88** | 0.27 | **0.96** | −0.01 | 0.44 | 0.63 | 0.58 |
| Mushroom | DT | 0.47 | 0.06 | 0.20 | **0.94** | 0.34 | **0.54** | 0.05 | **1.00** | 0.00 | 0.24 | 0.06 | 0.15 |
| | LR | **1.00** | **1.00** | **0.86** | 0.48 | 0.58 | 0.33 | 0.01 | **1.00** | 0.00 | 0.29 | 0.58 | 0.11 |
| | NB | 0.06 | 0.20 | **0.44** | **0.86** | 0.60 | 0.34 | 0.02 | **0.97** | 0.00 | 0.14 | 0.11 | 0.24 |
| Nursery | DT | **1.00** | **1.00** | 0.27 | **1.00** | **1.00** | **1.00** | 0.04 | **1.00** | 0.00 | 0.13 | 0.26 | 0.18 |
| | LR | **1.00** | **1.00** | 0.30 | **1.00** | **1.00** | **0.81** | 0.04 | **1.00** | 0.00 | 0.14 | 0.27 | 0.10 |
| | NB | **1.00** | **1.00** | 0.78 | **1.00** | **1.00** | **0.88** | 0.04 | **1.00** | 0.00 | 0.14 | 0.29 | 0.11 |

Bold values indicate the best performance for each metric in each row

**Table 8** Average precision (Pr), recall (Re) and correlation (Tau) results for XAI methods when explaining the decision tree (DT) and linear regression (LR) regression models

| Dataset | Model | LIME | | | SHAP | | | ACV | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | Pr | Re | Tau | Pr | Re | Tau | Pr | Re | Tau |
| Bike Rentals | DT | 0.74 | **0.07** | 0.38 | **0.99** | 0.06 | **0.51** | 0.47 | 0.06 | 0.22 |
| | LR | **1.00** | **0.88** | **0.78** | **1.00** | 0.14 | 0.63 | 0.06 | 0.01 | −0.16 |
| Facebook | DT | **1.00** | **0.18** | 0.50 | 0.92 | 0.17 | **0.59** | 0.34 | 0.03 | 0.37 |
| | LR | **1.00** | **1.00** | **0.74** | **1.00** | 0.33 | 0.59 | 0.00 | 0.00 | 0.17 |
| Housing | DT | 0.81 | 0.11 | 0.34 | **1.00** | **0.14** | **0.55** | 0.78 | 0.12 | 0.43 |
| | LR | **0.53** | **0.33** | 0.44 | 0.44 | 0.23 | **0.45** | 0.18 | 0.12 | 0.41 |
| Real Estate | DT | **1.00** | **0.33** | 0.48 | **1.00** | 0.31 | **0.56** | 0.84 | 0.27 | 0.12 |
| | LR | **0.40** | **0.48** | 0.44 | 0.30 | 0.30 | **0.45** | **0.40** | 0.43 | 0.10 |
| Solar Flare | DT | 0.89 | 0.16 | 0.51 | **0.96** | **0.22** | **0.84** | 0.21 | 0.03 | 0.33 |
| | LR | **1.00** | **1.00** | **0.92** | 0.16 | 0.16 | 0.51 | 0.00 | 0.00 | −0.02 |
| Student Results | DT | 0.96 | 0.56 | 0.19 | **1.00** | **0.57** | **0.48** | 0.77 | 0.43 | 0.43 |
| | LR | **1.00** | **1.00** | **0.91** | 0.08 | 0.08 | 0.61 | 0.00 | 0.00 | −0.30 |
| Wine Quality | DT | 0.99 | **0.33** | 0.59 | **1.00** | 0.31 | **0.70** | 0.45 | 0.18 | 0.10 |
| | LR | **0.43** | **0.52** | 0.65 | 0.34 | 0.39 | **0.68** | 0.20 | 0.22 | −0.08 |

Bold values indicate the best performance for each metric in each row

**Table 9** The Wilcoxon signed-rank test was applied to the results of the work on the classification datasets

| XAI techniques | Precision | | | Recall | | | $F1$-score | | | Rank correlation | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $T$ | $z$ | $p$ | $T$ | $z$ | $p$ | $T$ | $z$ | $p$ | $T$ | $z$ | $p$ |
| LIME & SHAP | 64,472 | −6.6 | 0.00 | 62,995 | −4.9 | 0.00 | 81,245 | −5.4 | 0.00 | 542,663 | −12.9 | 0.00 |
| LIME & LINDA-BN | 101,122 | −28.8 | 0.00 | 111,778 | −11.5 | 0.00 | 124,550 | −27.6 | 0.00 | 9730 | −37.1 | 0.00 |
| LIME & ACV | 73,746 | −25.5 | 0.00 | 97,228 | −15.4 | 0.00 | 155,344 | −20.9 | 0.00 | 198,713 | −29 | 0.00 |
| SHAP & LINDA-BN | 47,991 | −31.6 | 0.00 | 108,833 | −8.1 | 0.00 | 62,908 | −30.8 | 0.00 | 3148 | −37.4 | 0.00 |
| SHAP & ACV | 43,307 | −29.1 | 0.00 | 74,658 | −19.6 | 0.00 | 109,395 | −25.4 | 0.00 | 46,797 | −35.5 | 0.00 |
| LINDA-BN & ACV | 385,472 | −11.1 | 0.00 | 82,189 | −23.1 | 0.00 | 392,239 | −11.6 | 0.00 | 83,384 | −29.7 | 0.00 |

We find that there is a statistically significant difference in explanation fidelity across all techniques, and when using all metrics

**Table 10** The Wilcoxon signed-rank test was applied to the results of the work on the regression datasets

| XAI techniques | Precision | | | Recall | | | $F1$-score | | | Rank corrrelation | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $T$ | $z$ | $p$ | $T$ | $z$ | $p$ | $T$ | $z$ | $p$ | $T$ | $z$ | $p$ |
| LIME & SHAP | 19,874 | −9.8 | 0.00 | 11,408 | −13.5 | 0.00 | 16,542 | −13.1 | 0.00 | 190,176 | −17.9 | 0.00 |
| LIME & ACV | 31,400 | −22 | 0.00 | 34,864 | −19.6 | 0.00 | 39,536 | −20.2 | 0.00 | 118,162 | −24.6 | 0.00 |
| SHAP & ACV | 13,599 | −18.5 | 0.00 | 28,498 | −13.3 | 0.00 | 29,000 | −14.4 | 0.00 | 36,792 | −30 | 0.00 |

We find that there is a statistically significant difference in explanation fidelity across all techniques, and when using all metrics

The faithfulness of LINDA-BN's explanations appears to be related to model confidence in predictions. LINDA-BN's explanations generally had correctness, with respect to both correctness of the subset and the correctness of the feature ranking. However, recall scores for LINDA-BN's explanations were generally high. Closer inspection shows that the set $x_e$ returned by LINDA-BN contains numerous features. Thus, simply by volume, this set includes many features that are also present in $x_t$. This does not appear to be related to model type, or dataset characteristics, but the prediction probability associated with a prediction. This phenomenon was most common when explaining models trained on the Iris, Mushroom and Nursery datasets, all of which produced predictions with high prediction probabilities.

ACV shows little to no consistency in performance, especially when explaining the classification models. ACV explanations generally have low-to-moderate scores for all measures. In particular, ACV generally performs poorly when explaining linear regression models, but otherwise shows little consistency in performance.

We examine these phenomena further in Sect. 6.

## 5.2 Comparison of XAI techniques

There were notable differences in XAI technique performance for each of the three fidelity criteria examined. We applied the Wilcoxon Signed Rank test with a significance level of $p < 0.05$, to determine whether this difference in performance is statistically significant. Our results Tables 9 and 10) show that there is a significant difference in the performance of the four techniques. Additionally, we also compute the $F1$-score from recall and precision. We find that the differences between XAI techniques on precision and recall do not appear to be a trade-off, as the $F1$-scores are also distinct across the XAI techniques.

Various factors appear to underlie these differences. LIME's and SHAP's performance differs significantly given the underlying model, particularly when the precision metric is used (i.e. in *correctness* of the subset of most important features). While SHAP performs best for the decision tree model, it must be noted that LIME's performance is comparable. In reverse, LIME's explanations for the logistic and linear regression models are more faithful than that of SHAP. Similarly, SHAP clearly outperforms LIME when explaining Naïve Bayes models. This is noteworthy, as it suggests that the type of predictive model used should be taken into consideration when choosing a post hoc XAI technique. Though LIME and SHAP are model-agnostic methods, the characteristics of the underlying model affect explanation fidelity when using these techniques. LINDA-BN and ACV generally produce less precise explanations than LIME and SHAP.

LINDA-BN performed best with respect to the completeness of the subsets. The low precision scores that generally pair with high recall scores, and close examination of LINDA-BN's explanations suggest that this is likely because LINDA-BN's feature "weights" are often similar. This is especially true when the model is confident regarding a prediction. The other XAI methods generally produce less complete explanations than LINDA-BN. However, the consistently low correlation between $r_t$ and $r_e$ suggests that

LINDA-BN simply does not perform adequately as a feature attribution technique. Thus, while LINDA-BN generally has higher recall than the other methods, this cannot be said to be an advantage of LINDA-BN.

Feature ranking correlation across all XAI methods was poor. These results suggest that none of the XAI methods could fully capture the true importance of all features to the model. This is particularly noteworthy when considering LIME and SHAP as they were intended to generate feature attribution explanations, which is closely tied to feature ranking. ACV's low feature ranking correlation suggests that the LEI scores, which were used to determine $r_e$, are not fully aligned with the importance of the model. It must be noted that the LEI is suggested by the creators of ACV to show the necessity of each feature to the prediction [5], and thus, it may not fully align with its importance to the model.

Overall, we note significant differences in the performance of the four XAI methods. ACV was the most consistent method, though it generally performed poorly across all measures, indicating that it could not fully capture the features most important to the model. LINDA-BN's explanation fidelity is more consistent across model types than that of LIME or SHAP, but generally does not provide precise explanations. While LIME and SHAP generally provide precise explanations, the explanations they offer may not be complete. Moreover, the characteristics of the underlying model and data appear to affect the faithfulness of LIME and SHAP.

## 6 Discussion

There are two key insights to be derived from these results. Firstly, the explanation mechanism itself affects the explanation quality. This is particularly noticeable when using the different optimisations of SHAP. Though they all share a similar theoretical grounding, the implementation of the explanation mechanism greatly affects explanation quality. Thus, it is key to note that choice of XAI method should consider the potential implications of the explanation mechanism, not simply its grounding. Moreover, as noted in section 5.2, it is apparent that there are significant differences between the fidelity of different post hoc techniques. This suggests that, though a number of model-agnostic techniques are open source and easily accessible, *machine learning expertise may be needed to choose the most appropriate XAI method for the task*. It may also be more appropriate to use model-specific XAI techniques, rather than the model-agnostic techniques applied in this work.

Secondly, *explanation quality is heavily impacted by all aspects of the technical setting, including data and the chosen predictive model*. The strengths and weaknesses of the different XAI mechanisms, as well as their causes, are explored in Sects. 6.2, 6.3, 6.4 and 6.5. While these methods are all model-agnostic, their performance across different predictive model types is highly inconsistent.

These findings indicate clear implications for the use of XAI in real-life settings. In this work, the class of the model used is known and, the models themselves are fully transparent. In cases where the predictive model is considered to be intellectual property, the category of the predictive model may be unknown to end users [43]. Given the clear impacts of model type on explanation quality, it will likely become difficult to choose the most appropriate XAI method for the task without knowledge of model type. This then becomes a consideration not only for the choice of an XAI method but also for the choice of the predictive model.

### 6.1 Correctness versus completeness

There is a clear distinction between the precision and recall scores for almost all XAI techniques evaluated. In particular, precision results for LIME and SHAP are generally higher than recall. That is, while the features most highly weighted by LIME and SHAP do not necessarily include all relevant features (i.e. the explanation is not complete), in general, all features weighted highly by the explanation are relevant to the model (i.e. explanation correctness is high). Thus, a user applying LIME and SHAP explanations can be confident that highly weighted features are truly relevant to the model, but must also be aware that other factors could have impacted on the model prediction.

LINDA-BN generally has low precision, often paired with high recall. This is a characteristic of the permutation method of LINDA-BN. Because LINDA-BN was intended to determine the conditional dependence between variables within a small neighbourhood, the variance $\epsilon$ of permutations is generally low. Thus, in a neighbourhood where small changes in feature values do not necessarily affect the model's prediction, the Bayesian Network returned by LINDA-BN reflects this. When $x_e$ is extracted from the network using our method, $\Pr(f(x) \mid x_i)$ is the same or similar for a large number of features, and a large set of features is returned as $x_e$. Such a result indicates that a user can have strong confidence in the original model's prediction [34]. However, a faithful feature attribution explanation cannot be extracted from LINDA-BN for such a prediction.

All XAI techniques generally show a low-to-moderate correlation between $r_t$ and $r_e$. This is particularly noteworthy when using LIME and SHAP, which explicitly attempt to explain the importance of each feature to the prediction. Combined with the findings regarding precision and recall, we can come to the conclusion that while a user of LIME and SHAP can be confident that highly ranked features are relevant, they must be aware that the ranking of features as a whole may not be correct. Moreover, neither can they assume that they have identified all relevant features.

It must also be noted that LINDA-BN and ACV were not intended to provide feature attribution explanations. Their internal explanation-generation mechanisms reflect this, and thus, it is not surprising that both show low correlation scores. We explore this further in Sects. 6.4 and 6.5.

## 6.2 LIME

### 6.2.1 Impact of surrogate model

Notably, LIME performed poorly when explaining Naïve Bayes models. It is important to note that Gaussian Naïve Bayes was used in this work, which assumes that the relationship between features and the target variable has a Gaussian, i.e. nonlinear, distribution. We demonstrate this using ICE plots in Fig. 4.

This issue stems from the choice of surrogate model for LIME. LIME uses a Ridge Regression model as a surrogate to mimic the computations of the original model, returning the coefficients of the surrogate model as $\phi(x)$. When fitting nonlinear relationships between variables, Linear Regression and associated methods increase the coefficients of less important features in order to capture complex interaction and cancellation effects [28]. Thus, the coefficients of the model become more nuanced and can no longer be interpreted purely as feature importance. Therefore, when using such models as a surrogate for a predictive model, if the original model assumes a nonlinear relationship between variables, the coefficients of the surrogate model can no longer be interpreted simply as importance. One exception to this rule is in cases where the dataset contains all or mostly categorical features. When explaining Naïve Bayes, LIME's precision was generally poorest for the Breast Cancer and Diabetes datasets, which have only continuous variables. LIME permutes categorical features and continuous features differently, resulting in smaller variation in feature values, and thus in the prediction, when permuting categorical variables. And so, LIME is able to compute their importance more accurately.

### 6.2.2 Impact of permutation

As noted, LIME's permutation procedure differs for categorical and continuous features. Therefore, LIME explanations are more sound for datasets with a greater proportion of categorical features. This is especially true when explaining Linear Regression models, for which LIME performs best given that both the predictive model and the surrogate model are of the same class. When explaining Linear Regression models, LIME's precision falls below 0.5 for only two datasets, both of which have only continuous features (Real Estate and Wine Quality). The models trained on the Facebook and Solar Flare datasets, which have all categorical features, and the Student Score dataset, which has a high proportion of categorical features, have both perfect precision and perfect recall.

We also note that the length of the input does not affect the result. LIME's explanation stability is known to be related to input length, due to the nature of LIME's permutation method [46, 50]. It is reasonable to expect that a decrease in explanation consistency would affect the correctness of the explanation. However, when considering LIME's fidelity results, particularly explanation correctness, such a relationship is absent. This could suggest that our strategy of taking an average explanation from multiple explanations reduced the impacts of instability. The datasets with the longest inputs in this study are also composed primarily of categorical variables, due to the encoding method. It is also possible that the impact of the permutation method was reduced because of the large proportion of categorical variables, for which there is smaller variability in permutations.

Overall, our results suggest that the fidelity of LIME explanations are determined by the types of variables used, as well as the linearity of the relationship between variables and model prediction.

## 6.3 SHAP

It is important to note that SHAP values are not defined as the importance of a feature to the model. Rather, SHAP values indicate the contribution of that feature in moving the actual prediction away from the expected value of the prediction, i.e. the average value of the target variable in the training set. In our evaluations, we compared SHAP values against the importance of the feature to the model. In the case of decision trees, we found that SHAP values aligned closely with the model regarding feature importance, likely because position in the tree matters both to the prediction and TreeSHAP's calculation of SHAP values [28]. On the other hand, LinearSHAP assumes that the model treats features in the dataset as conditionally independent, and calculates SHAP values as for a given feature $x_i \in x$ as [27]:

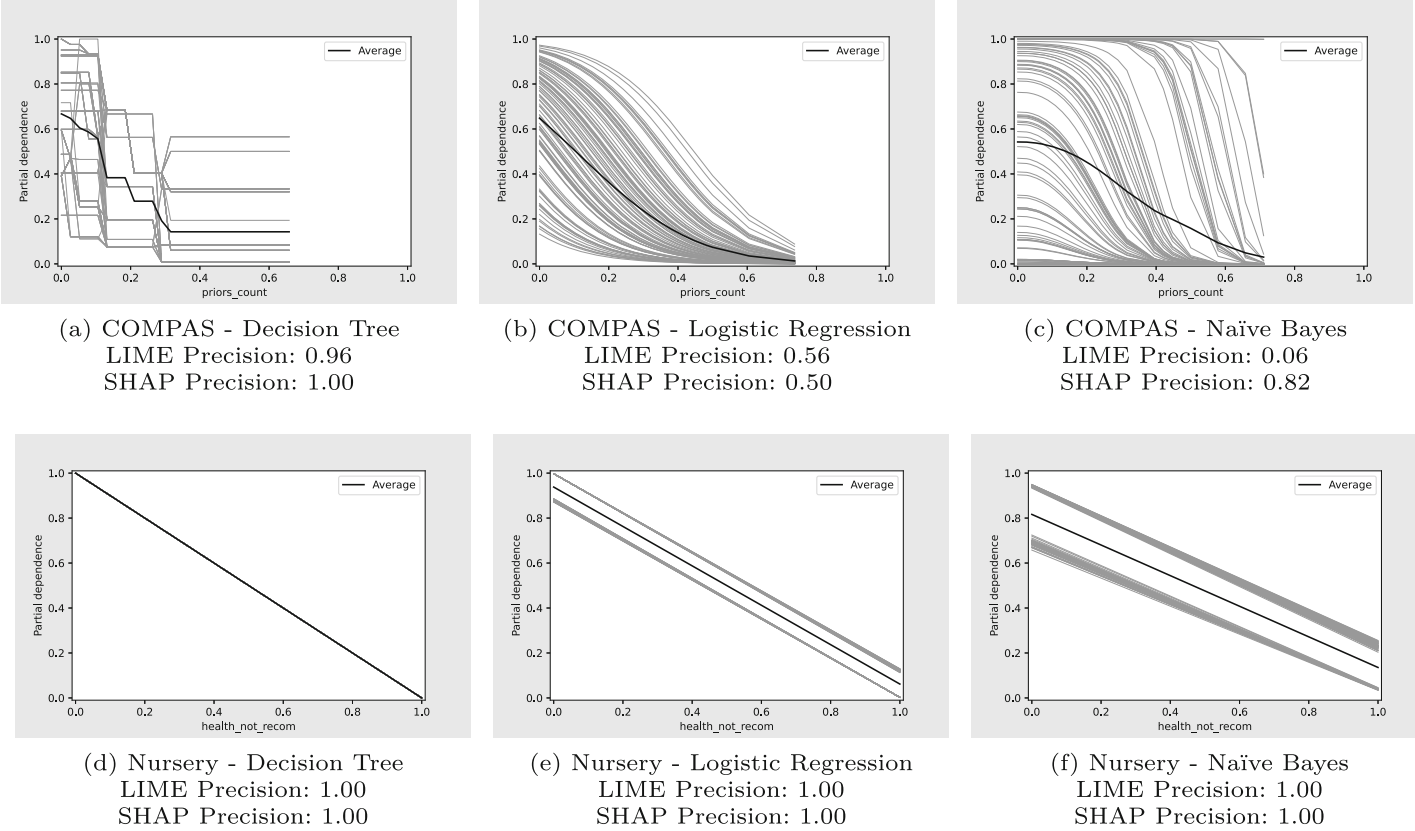
$$\phi_i(f, x) = w_i(x_i - E[x_i]) \tag{16}$$

Therefore, SHAP values, when produced by LinearSHAP, do not fully align with model importance.

Given this, we suggest that SHAP may be more useful for supporting end user decision-making, rather than data engineers or data scientists investigating the quality of a model.

## 6.4 LINDA-BN

LINDA-BN was designed to aid in understanding the dependencies between variables and determining the truthfulness of a prediction. Thus, rather than returning a static explanation, it returns a Bayesian Network describing the relation-




(a) COMPAS - Decision Tree
LIME Precision: 0.96
SHAP Precision: 1.00

(b) COMPAS - Logistic Regression
LIME Precision: 0.56
SHAP Precision: 0.50

(c) COMPAS - Naïve Bayes
LIME Precision: 0.06
SHAP Precision: 0.82

(d) Nursery - Decision Tree
LIME Precision: 1.00
SHAP Precision: 1.00

(e) Nursery - Logistic Regression
LIME Precision: 1.00
SHAP Precision: 1.00

(f) Nursery - Naïve Bayes
LIME Precision: 1.00
SHAP Precision: 1.00

**Fig. 4** ICE plots showing the relationship between the model prediction and a single feature for the COMPAS (**a**–**c**) and the Nursery (**d**–**f**) datasets. As the nonlinearity of the relationship between variables increases, the fidelity of the explanation decreases

ships between features within a small neighbourhood. This network can be further queried and explored to generate more insights into a prediction [34]. Generalising this explanation to a simple feature attribution explanation over-simplifies the information that this Bayesian Network provides, and so should not be the sole purpose of LINDA-BN.

However, in some cases, approximating the impact of each feature on the target variable does provide useful insights, even when the results suggest that the explanation is not sound. For example, as previously noted, the feature attribution scores extracted from the network returned by LINDA-BN could be used to determine the certainty of an explanation. Therefore, we suggest that the greatest strength of LINDA-BN is not in attempting to extract a single type of explanation from the Bayesian Network, but in querying and exploring the relationships between the variables using various methods, including the one presented in this work.

It must be noted, however, that understanding LINDA-BN's explanations requires statistical knowledge and an understanding of Bayesian Networks. While a technical expert may have this knowledge, a domain expert in another field may not. Thus, LINDA-BN's use for an end-user must also be evaluated prior to implementation.

## 6.5 ACV

There are several notable insights that can be derived from the described results. Firstly, extensive hyperparameter optimisation was conducted when training the surrogate model for ACV, resulting in relatively high surrogate model accuracy with respect to the original model's predictions. However, our results suggest that this high external fidelity does not translate into high internal fidelity, particularly at the local level.

Secondly, in cases where no feature subset meets $\pi$, no A-SE explanations are returned by ACV. As we use the A-SE to derive both the M-SE (which we take as $\phi(x)$) and the LEI (which is used to determine $r_e$), in cases where no explanation meets the $\pi$-level, we get poor scores for all three measures.

Finally, it is important to note that, although we used ACV as a post hoc explanation method in this work, ACV can function as predictive model in its own right. If used in this way, ACV would no longer have to mimic a distinct and separate model and can provide faithful explanations for its own behaviour. So, while ACV may underperform as a local, post hoc method in comparison with LIME and SHAP, its application as a self-explaining predictive model can eliminate the need for post hoc explanations altogether, which is generally more desirable [43].

**Table 11** A brief summary of the strengths and weaknesses of the four XAI methods tested

| | Strengths | Weaknesses | Should be used for |
|---|---|---|---|
| LIME | Generally precise in identifying features | Generally, does not identify all important features | Model types with relatively distinct relationships between features and prediction |
| | Performs well for decision tree and linear and logistic regression models | Does not correctly rank features based on importance | Datasets with mostly categorical variables |
| | More accurate for datasets with higher proportion of categorical variables | Cannot accurately work with Naïve Bayes model - relationship between prediction and features are too nonlinear | Contexts in which not all relevant variables may be important (for example, not in medical decision-making) |
| | | More continuous variables result in poorer explanation correctness | |
| | | Explanation quality is subject to dataset characteristics, and, in some cases, model characteristics | |
| SHAP | Generally precise in identifying features | Generally, does not identify all important features | Not for investigations of model quality or model fairness - performance may be inconsistent across model types |
| | Performs well for decision tree and Naïve Bayes models | Does not correctly rank features based on importance | End user decision-making |
| | | Cannot accurately work with linear and logistic regression models | Contexts in which not all relevant variables may be important (for example, not in medical decision-making) |
| | | Explanation quality is subject to model type and SHAP implementation for model type | |
| LINDA-BN | Fidelity is relatively consistent across model types | Cannot always distinguish between most important and least important features | Not feature attribution |
| | | Explanations cannot be taken as accurate feature attribution/feature ranking | Model debugging and determining model confidence with no ground truth (i.e. in context) |
| ACV | Can show feature necessity to prediction (i.e. features without which predictions cannot be accurately made) | Cannot always distinguish between most important and least important features | A self-explaining model - does not function well post hoc |
| | | Explanations cannot be taken as accurate feature attribution/feature ranking | |

## 6.6 Summary of insights

Based on the insights identified in this section, we consolidate the strengths and weaknesses of the XAI techniques in Table 11.

These strengths and weaknesses can assist a data scientist or a practitioner in choosing an XAI technique to use. While these consolidated findings are not recommendations or guidelines for use, to the best of our knowledge, this is the first known attempt at moving towards such guidelines. Future work performing benchmarks of XAI performance can be useful in producing more concrete recommendations for XAI use in practice and in data science.

### 6.7 Limitations and future work

There are a number of limitations associated with our work. Firstly, we have evaluated XAI methods using only tabular data. While evaluating explanation fidelity for this form of data is relatively under-explored in the literature, it is unclear how our findings in this work may be generalised for other dataset types. In particular, Naïve Bayes models are commonly used in text classification. It is currently unclear how well our findings regarding the fidelity of linear explanation-generation mechanisms would apply for Naïve Bayes models trained on text data, rather than tabular data.

Secondly, in this work, we evaluate explanations by using fully transparent predictive models. Thus, we focus on the behaviours of the models and XAI techniques applied, rather than the classes of the models. It is key to note that high explanation fidelity for the relatively simple models we have used in this work does not necessarily translate into high explanation fidelity for other, more complicated predictive models. However, the findings of this work can provide guidance in selecting an appropriate XAI method, if considering the model or dataset behaviours, rather than model class. For example, if a model is known to assume a nonlinear relationship between features and target variable, such as with Naïve Bayes models, linear surrogate models may not be able to accurately capture those relationships. It is noted in [6] that an adversarial party could manipulate choice of explainable technique or explanation generation parameters in order to produce an explanation of benefit to them—a method which may be difficult for an external auditor or examiner to detect. However, with a clearer understanding of how explanation generation mechanisms interact with predictive techniques, such manipulations can be detected.

However, this work and the fidelity evaluation method used within could serve as the first step in developing a fidelity evaluation method for other contexts. As noted earlier, current fidelity evaluation methods at the local level performed poorly when applied to more complex tabular data than those used in this work. Future work could use the findings of this work in order to develop a new fidelity evaluation method compatible with the more complex time series tabular data, and which can examine explanations of black box predictive models. Our findings in this work regarding the behaviour of XAI techniques under certain situations can be used to validate such a method. We also suggest that glass box methods can be used as a proxy for black box models in developing a new evaluation method.

Similarly, we must also note that the evaluation methods outlined in this work necessitates that explanations be provided or can be reduced to some form of feature attribution or ranking. Though we found that LINDA-BN and ACV generally showed poor fidelity, this is because we simplified or focused on one aspect of the explanation, rather than assessing the explanation as a whole. Therefore, we suggest that future work should also explore methods to evaluate the fidelity of other forms of explanations. This includes not only non-static, queryable explanations, such as LINDA-BN's explanations, but also counterfactuals, among others.

Furthermore, we suggest that the findings of this work could be used to motivate and inform future XAI development. In this work, we have outlined the strengths and weaknesses of each technique and identified potential causes for these. Future XAI development could use the insights presented in this work to develop XAI techniques that address these weaknesses.

Finally, it is important to note that the findings in this work are only the first step in determining the suitability of an XAI technique for a specific situation. When evaluating the XAI techniques used, we consider only the technical context of the explanation, such as dataset characteristics and the type of model used. This fails to consider how an explanation might be understood or perceived by a user. For example, past user studies have demonstrated that feature attribution explanations are often not helpful for users performing specific tasks [42, 52, 58], instead suggesting the rule-based or counterfactual-based explanations are most useful [9, 42, 52]. Moreover, some explainable methods, such as LINDA-BN, produce an explanation that a user must have a certain level of statistical knowledge to accurately understand, and so may not be useful for an end user. Thus, the use of cognitive metrics, used to gauge explanation quality with respect to a human user [26], also becomes necessary given a context.

## 7 Conclusion

Given the increasing use of explainable AI methods to understand the decision-making of opaque predictive models, it becomes necessary to understand how *well* an explainable method can interpret any given model. However, evaluation methods to achieve this still remain an open question in the field of XAI, particularly for tabular data. In this work, we examine and evaluate the explanations of four model-agnostic, open-source XAI methods, using fully transparent "glass box" models trained on fourteen open-source datasets. We examined the various dataset characteristics and model behaviours that affected the faithfulness of an explanation to the underlying model.

Overall, our results suggest that model-agnostic XAI methods show significant differences in explanation quality under different technical contexts. This is likely to necessitate technical expert involvement and close examination of techniques when deploying XAI techniques. Our key findings are as follows:

- the explanation mechanism, i.e. the engineering of the XAI method, has a strong effect on explanation quality,

which may once again require a technical expert to examine and assess whether a given mechanism is suitable for the technical context; and
- there is no one "best", most faithful XAI method, even for a single dataset or model type—all of the methods show significant differences in performance across datasets and models.

In addition to these insights, we also specifically highlight some characteristics of all four methods that affect explanation quality. The insights we present in this paper are simply the first step in understanding the strengths and weaknesses of XAI methods in certain contexts. More extensive experiments can help in understanding the suitability of XAI techniques when used with different datasets and model types, and, importantly, for different user groups.

**Acknowledgements** Computational resources and services used in this work were provided by HPC and Research Support Group, Queensland University of Technology, Brisbane, Australia. We thank the Australian Research Training Program for the first author's scholarship.

**Funding** Open Access funding enabled and organized by CAUL and its Member Institutions

## Declarations

**Conflict of interest** The authors have no competing interests to declare that are relevant to the content of this article.

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## References

1. (1978) Housing. UCI Machine Learning Repository, https://archive.ics.uci.edu/ml/machine-learning-databases/housing/
2. (1988) Pima indians diabetes. UCI Machine Learning Repository, https://archive.ics.uci.edu/ml/datasets/pima+indians+diabetes
3. (1989) Solar Flare. UCI Machine Learning Repository, https://archive.ics.uci.edu/ml/datasets/solar+flare
4. Agarwal, C., Krishna, S., Saxena, E., et al.: OpenXAI: Towards a Transparent Evaluation of Model Explanations, (2022) arXiv:2206.11104
5. Amoukou, S.I., Brunel, N.J.B.: Consistent Sufficient Explanations and Minimal Local Rules for explaining regression and classification models, (2021) https://arxiv.org/abs/2111.04658
6. Bordt, S., Finck, M., Raidl, E., et al.: Post-hoc explanations fail to achieve their purpose in adversarial contexts. In: 2022 ACM Conference on Fairness, Accountability, and Transparency, (2022) https://doi.org/10.1145/3531146.3533153
7. Borisov, V., Leemann, T., Seßler, K., et al.: Deep Neural Networks and Tabular Data: A Survey, (2021) https://arxiv.org/abs/2110.01889
8. Carvalho, D.V., Pereira, E.M., Cardoso, J.S.: Machine learning interpretability: a survey on methods and metrics. Electronics (2019). https://doi.org/10.3390/electronics8080832
9. Chou, Y.L., Moreira, C., Bruza, P., et al.: Counterfactuals and causability in explainable artificial intelligence: theory, algorithms, and applications. Inf. Fus. (2022). https://doi.org/10.1016/j.inffus.2021.11.003
10. Cortez, P.: Student Performance. UCI Machine Learning Repository, (2014) https://archive.ics.uci.edu/ml/datasets/Student+Performance
11. Cortez, P., Cerdeira, A., Almeida, F., et al.: Wine Quality. UCI Machine Learning Repository, (2009) https://archive-beta.ics.uci.edu/ml/datasets/wine+quality
12. Doshi-Velez, F., Kim, B.: Towards A Rigorous Science of Interpretable Machine Learning, (2017) https://arxiv.org/abs/1702.08608
13. Du, M., Liu, N., Yang, F., et al.: On attribution of recurrent neural network predictions via additive decomposition. In: The World Wide Web Conference - WWW'19, (2019) https://doi.org/10.1145/3308558.3313545
14. Fanaee-T, H.: Bike Sharing Dataset. UCI Machine Learning Repository, (2013) https://archive-beta.ics.uci.edu/ml/datasets/bike+sharing+dataset
15. Fisher, R.: Iris. UCI Machine Learning Repository, (1988) https://archive.ics.uci.edu/ml/datasets/iris
16. Fong, RC., Vedaldi, A.: Interpretable explanations of black boxes by meaningful perturbation. In: 2017 IEEE International Conference on Computer Vision (ICCV), (2017) https://doi.org/10.1109/iccv.2017.371
17. Goethals, S., Martens, D., Evgeniou, T.: The non-linear nature of the cost of comprehensibility. J. Big Data (2022). https://doi.org/10.1186/s40537-022-00579-2
18. Guidotti, R., Monreale, A., Ruggieri, S., et al.: A survey of methods for explaining black box models. ACM Comput. Surv. (2018). https://doi.org/10.1145/3236009
19. Guidotti, R., Monreale, A., Giannotti, F., et al.: Factual and counterfactual explanations for black box decision making. IEEE Intell. Syst. **34**(6), 14–23 (2019). https://doi.org/10.1109/mis.2019.2957223
20. Jain, S., Wallace, B.C.: Attention is not explanation. In: Proceedings of the 2019 Conference of the North, (2019) https://doi.org/10.18653/v1/D19-1002
21. Kenny, E.M., Ford, C., Quinn, M., et al.: Explaining black-box classifiers using post-hoc explanations-by-example: the effect of explanations and error-rates in XAI user studies. Artif. Intell. **294**(103), 459 (2021). https://doi.org/10.1016/j.artint.2021.103459
22. Kohavi, R., Becker, B.: Adult. UCI Machine Learning Repository, (1996) https://archive.ics.uci.edu/ml/datasets/Adult
23. Konstantinov, A.V., Utkin, L.V.: Attention-like feature explanation for tabular data. Int. J. Data Sci. Anal. (2022). https://doi.org/10.1007/s41060-022-00351-y
24. Koska, C., Filipović, A.: Blackbox AI: State regulation or corporate responsibility? Digitale Welt (2019). https://doi.org/10.1007/s42354-019-0208-5
25. Lakkaraju, H., Bach, S.H., Leskovec, J.: Interpretable decision sets. In: Proceedings of the 22nd ACM SIGKDD international conference on knowledge discovery and data mining, (2016) https://doi.org/10.1145/2939672.2939874

# A Synthetic Ground-Truth Framework for the Evaluation of Explainable AI Methods*

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ARTICLE INFO

*Keywords*:
Explainable Artificial Intelligence
Synthetic Ground Truth
Explainability Metrics
Model Interpretability
Intervention-Based Methods
XAI Benchmarking
Local Explanation Methods

ABSTRACT

Evaluating explainable Artificial Intelligence (XAI) methods is a challenging task due to the lack of reliable evaluation procedures and, in particular, the absence of ground truth explanations. In the literature, existing evaluation approaches typically assess explanations by measuring their fidelity with respect to the predictions of a black-box model. However, such evaluation strategies only quantify the degree to which an explanation reproduces the model's output, without ensuring that the explanation correctly reflects the underlying decision process. As a consequence, different explanations may achieve similar fidelity scores while providing inconsistent or misleading interpretations of the model behavior. In this paper, we propose a framework for the evaluation of XAI methods based on synthetic ground truth. The proposed approach relies on controlled interventions to generate synthetic datasets in which the importance of input components can be determined by design. This enables the construction of ground truth explanations that are directly aligned with the behavior of the model under analysis. The framework is instantiated across three data domains, namely binary images, tabular data, and time series, allowing a comprehensive assessment of explanation methods in heterogeneous settings. Experimental results obtained by evaluating nine widely used XAI methods show significant limitations in current techniques and highlight the importance of synthetic, intervention-based benchmarks for a reliable assessment of explanation quality.

## 1. Introduction

EXplainable Artificial Intelligence (XAI) has emerged as a broad and rapidly evolving field, with applications spanning numerous domains [30, 38], including healthcare, finance, transportation, and public policy, where understanding the reasoning behind automated decisions is crucial for ensuring transparency, trust, and accountability [5]. Despite the remarkable success of modern machine learning models, particularly deep neural networks, these systems are often regarded as *black boxes*, as their internal decision-making processes remain largely opaque to human interpretation [16]. In response, a wide range of XAI methods has been proposed to provide insights into model behavior. However, a fundamental open challenge lies in the lack of agreement among different explanation techniques. This phenomenon, commonly referred to as the *disagreement problem* [25], raises critical concerns about the reliability, consistency, and ultimately the trustworthiness of explanations produced by XAI methods.

Despite this growing body of work, there is still a lack of standardized evaluation measures that enable a systematic and reliable assessment of explanation quality. A common strategy to address this limitation is the systematic evaluation of XAI methods, with the goal of identifying which explanations more faithfully reflect the behavior of the underlying model. Several authors [5, 11, 39, 57] have proposed taxonomies to categorize XAI evaluation techniques. Broadly, these approaches can be divided into two main classes: *human-grounded* and *functionally-grounded* evaluations. Human-grounded approaches rely on user studies to assess explanations, focusing on how humans interpret, understand,

*This work has been partially supported by the Italian Project Fondo Italiano per la Scienza FIS00001966 "MIMOSA", by the European Community Horizon 2020 programme under the funding schemes G.A. 101120763 "TANGO", by the European Innovation Council project "EMERGE" (Grant No. 101070918) and project PID2023-149079OB-I00 funded by MICIU/AEI/10.13039/501100011033 and by ERDF/EU.

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or interact with them. In contrast, functionally-grounded methods evaluate explanation quality through quantitative metrics without human involvement, and are therefore often referred to as machine-centered evaluations.

In this work, we focus on functionally-grounded evaluation techniques. Recent surveys [8, 39] highlight both the rapid growth and the increasing fragmentation of this research area. In particular, Nauta *et al.* [39] identify twelve explanation qualities, six of which can be assessed using functionally-grounded approaches. Building on this perspective, Dembinsky *et al.* [8] conduct a large-scale systematic review, identifying more than 400 evaluation metrics, which are organized into 41 metric families and five major categories. A central conclusion of their analysis is the key role of *fidelity* (also referred to as faithfulness), described as the "foundation of the entire evaluation process". At the same time, the large number and diversity of metrics reflect a lack of consensus on how explanation quality should be assessed. Given this wide variety of evaluation approaches, several studies [20, 35, 36, 55] have undertaken *meta-evaluations* of existing metrics, investigating whether they effectively measure the properties they are intended to capture. Although these works consider different subsets of metrics, they reach a consistent conclusion: many widely used evaluation measures exhibit limited reliability and unstable behavior across experimental settings. This limitation largely stems from the absence of an *external* reference, such as a ground truth, forcing evaluation to rely on internal criteria, similarly to what occurs in the unsupervised evaluation of clustering algorithms [53]. As a result, these approaches attempt to assess explanation quality based solely on intrinsic properties, without a direct notion of correctness. Consequently, metrics based exclusively on *internal* properties of explanations often provide only partial and potentially misleading insights into their validity.

An alternative to meta-evaluation approaches is the use of *a priori* constraints that define a known ground truth for explanations. According to the taxonomy proposed by Dembinsky *et al.* [8], these approaches can be broadly divided into two categories: human-annotated and synthetic datasets. In the latter case, controlled experimental settings are constructed so that the importance of each input feature is known by design. This enables the generation of ground truth (GT) explanations that can be directly compared with the outputs of XAI methods. This class of approaches has recently been referred to as *Synthetic Artificial Intelligence Ground Truth* (SAIG) [37]. Within the SAIG paradigm, datasets are artificially generated to allow systematic manipulation of the contribution of specific input components. As a result, these methods provide reliable ground truth explanations without requiring human annotations, overcoming a key limitation of existing evaluation strategies. Among these approaches, the framework proposed by Hesse *et al.* [21] has attracted attention due to its *intervention-based design*, which enables the construction of ground truth explanations for the quantitative evaluation of XAI methods. However, the original formulation is restricted to color image data.

Building on these considerations, in this work we start from such intervention-based SAIG framework, and we extend it beyond the image domain. In particular, we investigate how the principles of intervention-based ground truth generation can be generalized to other data modalities commonly used in machine learning. To this end, we propose a unified methodology for constructing SAIG frameworks across binary images, tabular data, and time series, enabling the systematic and controlled evaluation of XAI methods in heterogeneous settings. The main contributions of this work are threefold. First, we define the intervention-based SAIG framework for multiple data modalities, including binary images, tabular data, and time series. Second, we introduce, to the best of our knowledge, the first faithful *local* ground truth construction for the evaluation of XAI methods in tabular data, addressing the limitations of existing approaches based on global feature importance. Third, we show how the SAIG framework can be tailored to time series data, enabling the evaluation of XAI methods in a domain where reliable ground truth has so far been largely unavailable.

The remainder of this paper is organized as follows. Section 2 reviews the relevant literature on XAI evaluation. Section 3 introduces the key concepts underlying the proposed framework. Section 4 presents the methodology for constructing synthetic ground truth across binary images, tabular data, and time series. Section 5 describes the experimental setup adopted to evaluate nine XAI methods across these data modalities. Section 6 reports and discusses the experimental results. Finally, Section 7 concludes the paper and outlines directions for future research.

## 2. Related Work

We review here existing approaches for the evaluation of XAI methods based on synthetic ground truth. These approaches construct controlled settings in which ground truth explanations are available by design, enabling a direct assessment of explanation quality. We focus on methods developed for tabular and image data, as, to the best of our knowledge, no SAIG frameworks for time series data have been proposed in the literature.

Several approaches have been proposed to construct synthetic ground truth for *tabular data*. In [7] is introduced one of the first SAIG methodologies by generating simple synthetic datasets with a limited number of input features. It

relies on predefined functions $a : \mathbb{R}^M \to \mathbb{R}$ that assign weights to input variables, which are interpreted as feature importance. However, this formulation provides a *global* notion of importance rather than instance-specific local explanations. Similar limitations are shared by subsequent methods that define ground truth through synthetic data generation processes [1, 4, 29], where feature relevance is determined globally by construction. Other approaches aim to evaluate specific explanation techniques. For instance, in [2] proposed a SAIG framework tailored to LIME [43]. While this method enables targeted evaluation, it lacks generality, as it is tightly coupled to a single explanation technique and relies on the assumption that LIME produces correct explanations, a claim that has been widely debated in the literature [12, 28, 49]. A different perspective is in [14], who defines synthetic classifiers using rule-based systems instead of trained machine learning models. This setting allows full control over the decision process and provides a known ground truth for feature importance. However, such approaches are inherently limited to model-agnostic explanation methods that infer importance solely from input-output relationships, without access to the internal behavior of learned models. Overall, existing methods for tabular data exhibit two main limitations: either *(i)* they define ground truth independently of the model, potentially reducing adherence to its actual behavior [1, 4, 7, 29], or *(ii)* they are restricted to specific explanation techniques or settings, limiting their general applicability [2, 14].

Recent work has explored synthetic ground truth generation for *image data*. In [37] is provided a comprehensive overview of this area, identifying several representative approaches [3, 21, 34, 58]. In [58] is constructed composite images by combining foreground objects from MSCOCO [27] with backgrounds from MiniPlaces [59], associating each image with dual labels corresponding to object and scene classification. In this setting, faithful explanations are expected to highlight different regions depending on the prediction task. The authors of [3] propose a mosaic-based approach, where each image is formed by combining patches from multiple classes. They introduce the "Focus!" score to quantify whether explanations concentrate on regions relevant to the predicted class. The paper [34] extends earlier tabular-based approaches to images by defining synthetic functions over image-derived features, such as region area or pattern frequency. However, these methods present several limitations, including a reliance on global rather than local importance [34], an emphasis on spatial localization rather than attribution magnitude [3], and the use of comparative evaluations across models instead of intrinsic performance measures [58]. A notable advancement is the intervention-based framework introduced in [21], which is evaluated on the *FunnyBirds* dataset. This dataset consists of synthetic bird images composed of semantic parts, such as beaks and wings, and the class labels are determined by specific components. By systematically intervening on these parts and observing the resulting changes in model predictions, the framework derives ground truth importance directly from model behavior. This intervention-based approach ensures a high degree of fidelity between the ground truth and the model's decision process.

## 3. Preliminaries

In this section we introduce the notation and fundamental concepts used throughout this work. In particular, we formalize the machine learning model under analysis, the notion of local feature attribution explanations, and the intervention-based definition of feature importance used to construct ground truth explanations. We begin by defining our input data. In particular, we consider three different data modalities, binary images, time series, and tabular data. For ease of notation each instance, independent of its data type, is represented as a vector $\mathbf{x} = [x_1, \dots, x_D]$, where $D$ denotes the total number of input features and $x_j$ represents the $j$-th scalar feature.

**Definition 1** (Dataset)**.** *A dataset,* $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$*, is a collection of N input samples, where each instance* $\mathbf{x} \in \mathcal{X}$ *belongs to an input space* $\mathcal{X}$*.*

More specifically, for black and white images, each sample, $\mathbf{x}$, is represented as a $H \times W$ matrix, where $H$ and $W$ denote the image height and width, and $D = H \cdot W$ is the number of pixels. For time series, each instance is represented as a sequence $\mathbf{x} \in \mathbb{R}^T$, where $D = T$ denotes the number of timestamps. For tabular data, each sample corresponds to a feature vector $\mathbf{x} \in \mathbb{R}^M$, where $D = M$ denotes the number of tabular features. Based on the dataset definition above, we formalize the predictive model considered in this work, i.e., classification models.

**Definition 2** (Classification Model)**.** *Let* $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$ *be a dataset with instances* $\mathbf{x} \in \mathcal{X}$ *and labels* $y \in \mathcal{Y}$*. A classification model is a function* $f : \mathcal{X} \to \mathcal{Y}$ *that maps an input instance* $\mathbf{x}$ *to a predicted output* $\hat{y} = f(\mathbf{x})$*.*

In practice, $f(\mathbf{x})$ typically represents a vector of class scores or probabilities over the label space $\mathcal{Y}$, and the predicted class corresponds to the label with the highest score. Independently of the data domain, state-of-the-art approaches for classification tend to be black-box models, not interpretable from a human standpoint [5, 33, 50, 54]. For this reason,

XAI approaches are increasingly used to shed light on the black-box decision-making. Given a model $f$ and an input instance $\mathbf{x}$, the most common XAI methods explain individual predictions by assigning an importance score to each input feature via a so-called *local feature attribution explanations*, as they describe the contribution of each feature to a single prediction [5].

**Definition 3** (Local Feature Attribution Explanation)**.** *A local feature attribution explanation for the prediction $f(\mathbf{x})$ is a vector $\mathbf{e} = [e_1, \dots, e_D] \in \mathbb{R}^D$ where each component $e_j$ represents the estimated importance of feature $x_j$ for the prediction associated with $\mathbf{x}$.*

The explanation vector $\mathbf{e}$ is produced by an attribution method $E$ applied to the model $f$ and the instance $\mathbf{x}$. In this work, we focus on *post-hoc*, *local*, *model-agnostic* attribution methods, i.e., explanation techniques that operate externally to the black-box model and can therefore be applied to any predictive model regardless of its internal structure [5]. A common strategy for estimating feature importance is based on interventions on the input [31, 43]. The basic idea is to measure how the model prediction changes when a particular input component is modified or removed. Importantly, these components do not necessarily correspond to individual scalar features, but may represent groups of features such as superpixels in images, temporal windows in time series, or subsets of variables in tabular data. Let $\mathcal{S} = \{S_1, \dots, S_K\}$ denote a partition (or collection) of input components, where each $S_j \subseteq \{1, \dots, D\}$ is a subset of feature indices. Let $\mathbf{x} \setminus S_j$ denote an intervened version of the input where all features in $S_j$ have been altered according to a predefined intervention rule. The importance of $S_j$ can then be estimated as the change in the model output

$$e_j = f(\mathbf{x}) - f(\mathbf{x} \setminus S_j).$$

This formulation estimates the marginal contribution of the component $S_j$ to the prediction and is used in a wide range of *post-hoc*, model-agnostic attribution methods. For instance, in image data, methods such as LIME [43] and KernelSHAP [31] define components $S_j$ as superpixels obtained through image segmentation. Perturbation-based methods such as RISE [42] implicitly define $S_j$ through random masks applied over the input. In time series, components $S_j$ typically correspond to contiguous temporal windows, as in methods such as T-SHAP [40], or masked temporal segments as used in perturbation-based approaches. For tabular data, components $S_j$ are usually defined at the level of individual features or subsets of variables, as in LIME [43] and KernelSHAP [31].

When evaluating explanation methods, it is important to evaluate how good are such local explanation, comparing them to a so-called ground truth. For better clarity, it is important to distinguish between two notions of ground truth. The first notion relates to the data-generating process, where certain features are assumed to be intrinsically responsible for the target label. The second notion, which is the focus of most XAI evaluation frameworks, concerns the behavior of the model itself. In this case, a ground truth explanation corresponds to the features that actually influence the model's prediction, regardless of whether the model has learned the correct underlying relationships. In this work we focus on this second notion, under which the objective of an explanation method is to faithfully describe how the model produces its prediction, even if the model relies on spurious or undesirable correlations. We denote the corresponding ground truth attribution as $\mathbf{g} = (g_1, \dots, g_D)$, where each component $g_j$ represents the true contribution of feature $x_j$ to the prediction of the model.

Obtaining such ground truth explanations is challenging for real-world datasets. Performing interventions on natural data is often non-trivial, as removing a feature typically requires replacing it with an estimated value or perturbation, which may introduce artifacts or out-of-distribution samples. In this work, we exploit controlled synthetic settings in which such interventions can be performed without introducing artifacts, enabling the construction of reliable ground truth explanations aligned with the behavior of the model.

## 4. Methodology

In this section, we introduce the intervention-based approach for generating Synthetic Artificial Intelligence Ground Truth (SAIG) datasets across multiple data modalities, including binary images, tabular data, and time series. Figure 1 illustrates the intervention procedure for each data typology: the first row shows the same original data with color, binary images and time series; the second row depicts the *wings removal intervention* on all three typologies.

First, we recall the intervention-based framework proposed for color images in [21]. Let represent an input image $\mathbf{x}$ as a collection of semantically meaningful components $\mathcal{S} = \{S_1, S_2, \dots, S_K\}$, where each component $S_j$ corresponds to a subset of pixels forming a distinct object part. These components are assumed to be disjoint, i.e., $S_i \cap S_j = \emptyset$ for

$i \neq j$. The importance of each component $S_j$ is then computed through an intervention-based formulation:

$$g_j = f(\mathbf{x}) - f(\mathbf{x} \setminus S_j), \tag{1}$$

where $g_j$ denotes the importance of component $S_j$, $f$ is the predictive model, and $\mathbf{x} \setminus S_j$ represents the intervened input obtained by removing the component $S_j$. In this setting, the intervention consists of replacing the pixels belonging to $S_j$ with the original background that the component occludes. Consequently, all pixels within a given component are assigned the same importance value.

In the specific case of the FunnyBirds dataset [21], each input $\mathbf{x}$ is composed of *six* predefined semantic parts, namely *beak*, *eye*, *feet*, *tail*, *wings*, and *body*, each corresponding to a component $S_j \in \mathcal{S}$. The class label $y$ is determined by a specific combination of these parts. M ultiple variants are defined for each component, e.g., different shapes and colors. Figure 2 illustrates three real examples on how these variants combine, generating a large combinatorial space, from which a subset of 50 classes was selected. The predictive model $f$ is trained to map each input $\mathbf{x}$ to its corresponding class label $y$. Specifically, the model is trained to predict classes directly from raw pixel data. Since no explicit semantic part information is provided, the model must autonomously learn to detect individual components and infer their relation to the predicted class. A key aspect of this framework is that the training distribution is explicitly augmented with samples in which one or more components are removed. This design ensures that intervened inputs, such as $\mathbf{x} \setminus S_j$, remain within the training distribution, thereby mitigating potential out-of-distribution effects during evaluation.

In the following, we define the intervention-based SAIG framework to different binary images, tabular data, and time series. For each case, we define how to construct semantically meaningful components and how to perform interventions to preserve consistency with the underlying data generation process. This enables the computation of ground truth importance scores that remain faithful to the model behavior while ensuring that intervened samples lie within the training distribution. The proposed formulations maintain the core principles of the original framework while adapting them to the structural characteristics of each data modality.

## 4.1. Binary Images

We define here the intervention-based SAIG framework for *binary images*, which serves as a simplified domain for validating our methodology before extending it to more complex data modalities. In this setting, each input is represented as a binary mask $\mathbf{x} \in \{0,1\}^{H \times W}$, where foreground pixels (value 1) indicate the presence of object components, and background pixels (value 0) denote their absence. These binary representations can be interpreted as silhouette-like abstractions that preserve the geometric structure of the object while removing color information.

The construction of the binary dataset involves two main steps: binary mask generation and class redundancy pruning. First, we obtain binary images by aggregating the ground truth segmentation masks provided by the FunnyBirds framework, effectively mapping the original multi-channel representation into a single foreground-background mask. Formally, let $\mathbf{x}^{\text{rgb}}$ denote the original image and $\mathcal{S} = \{S_j\}_{j=1}^{K}$ the set of semantic components; the binary representation $\mathbf{x}$ is defined such that $x_{u,v} = 1$ if $(u,v) \in \bigcup_{j=1}^{K} S_j$, and $x_{u,v} = 0$ otherwise. This transformation preserves the spatial arrangement of components while eliminating color as a discriminative factor.

Second, we address the ambiguity introduced by the loss of color information. In the original dataset, several classes differ only in terms of color attributes and become indistinguishable after binarization. Figure 3 illustrates representative examples of such cases. To resolve this issue, we remove redundant classes while retaining a single representative for each group of shape-equivalent instances. Rather than merging classes, we adopt a pruning strategy to preserve a balanced class distribution. As a result, the total number of classes is reduced from 50 to 47.

The importance of each component is then computed using the same intervention-based formulation described previously. Given a component $S_j \in \mathcal{S}$, we construct the intervened input $\mathbf{x} \setminus S_j$ by replacing the corresponding pixels with the background value, i.e., setting them to zero. The contribution of $S_j$ is defined as the difference between the model outputs for the original and intervened inputs, consistently with the formulation in Eq. (1).

## 4.2. Time Series

We extend the intervention-based SAIG framework to time series data by introducing a transformation that maps structured visual inputs into a one-dimensional sequential representation, while preserving the semantics of intervention-based importance. In [21], explanations are defined over spatial components, i.e., object parts, whose importance is quantified through controlled interventions. Our objective is to translate this paradigm into the temporal domain without losing the correspondence between components and their contributions.

Figure 4 depicted the proposed two-step procedure to construct a time series from the original image. First, we convert the original color images into binary representations, as described in the previous subsection, isolating the geometric structure of the object. Second, we transform the contour of the binary shape into a time series representation. Specifically, we extract the outer contour $C = \{(u_t, v_t)\}_{t=1}^{T}$ of the object, where $(u_t, v_t)$ denotes the coordinates of the $t$-th contour point, and encode it as a univariate sequence by computing the Euclidean distance between each contour point and a reference centroid $\mathbf{c}$. The resulting time series $\mathbf{x} \in \mathbb{R}^T$ is defined as:

$$x_t = \|(u_t, v_t) - \mathbf{c}\|_2, \quad t = 1, \dots, T, \tag{2}$$

where $T$ denotes the length of the contour sequence. This representation is inspired by classical shape-to-sequence transformations in time series analysis [23]. In our setting, the maximum sequence length $T$ is 1251 timestamps. This transformation enables the reinterpretation of spatial components as contiguous temporal segments, allowing the application of the same intervention-based framework. Importantly, interventions are performed in the original input space by removing components from the image, and are subsequently propagated to the time series representation. Both the original input $\mathbf{x}$ and the intervened version $\mathbf{x} \setminus S_j$ are transformed using the same procedure, ensuring consistency between the resulting sequences. To guarantee alignment across samples, we fix the centroid $\mathbf{c}$, computed as the mean of all object contour coordinates, from the original shape as the reference point for all corresponding interventions. This ensures that each temporal index $t$ corresponds to the same geometric location across interventions, preserving the coherence of the resulting importance estimates. As a result, the proposed construction yields a time series SAIG in which the contribution of each temporal segment can be directly associated with the removal of a corresponding structural component in the original input.

### 4.3. Tabular Data

The evaluation of XAI methods for tabular data has been extensively studied in the literature, as discussed in Section 2. However, existing approaches exhibit two main limitations: they either provide a global notion of ground truth explanation, which is not suitable for assessing local methods [1, 4, 7, 29], or they are not generalizable across different explanation techniques [2, 14]. To address this gap, we propose an intervention-based SAIG framework for tabular data, combining it with the functional benchmark introduced in [7].

Cortez and Embrechts [7] proposed a synthetic benchmark based on attribution functions, i.e., deterministic mappings $a : \mathbb{R}^D \to \mathbb{R}$ from which feature importance can be derived *a priori*. A canonical example is a weighted linear function:

$$a(\mathbf{x}) = \sum_{j=1}^{D} w_j \, x_j, \tag{3}$$

where $\mathbf{x} = [x_1, \dots, x_D]$ and $w_j$ denotes the weight associated with feature $x_j$. The output $a(\mathbf{x})$ is used as the target label for training a predictive model. In this formulation, feature importance is implicitly defined by the functional form of $a$. However, this assumption does not generally hold, particularly in non-linear settings where feature contributions may vary across instances. To overcome this limitation, we introduce an intervention-based formulation. In order to perform controlled interventions while avoiding out-of-distribution effects, we define a discretized input space:

$$\mathcal{X} = \left\{ \frac{k}{K} \;\middle|\; k \in \{0, 1, \dots, K\} \right\}^{D}, \tag{4}$$

where $K$ denotes the number of discrete levels per feature and $D$ is the dimensionality of the input space.

Given an input $\mathbf{x} \in \mathcal{X}$, we define an intervention on feature $x_j$ as a decrease of one discrete level, yielding an intervened instance $\mathbf{x}\backslash_j$. The importance of feature $x_j$ is then computed as:

$$g_j = f(\mathbf{x}) - f(\mathbf{x}\backslash_j). \tag{5}$$

This formulation estimates the marginal contribution of feature $x_j$ to the model output.

The proposed intervention strategy provides several advantages. First, it avoids the need to specify an external baseline value. Second, due to the discretization of $\mathcal{X}$, the perturbed instance $\mathbf{x}\backslash_j$ is guaranteed to remain within the valid input domain, preventing out-of-distribution artifacts. Third, the importance values are directly derived from the model behavior, ensuring consistency between the explanation and the underlying predictive mechanism. Fourth, this framework can be applied to all attribution functions introduced in [7], as well as to more complex functional benchmarks.

# 5. Experimental Setup

In the previous section, we introduced a set of intervention-based SAIG frameworks to construct ground truth explanations for binary images, time series, and tabular data. Building on these formulations, this section describes the experimental protocol adopted to evaluate the fidelity of different XAI methods with respect to the proposed ground truth. We detail the evaluation metrics, the predictive models used in each domain, and the set of explanation methods considered in our study.

## 5.1. Evaluation Metrics

The proposed SAIG framework provides ground truth explanations that enable a direct comparison with the outputs of XAI methods. To quantify this comparison, it is necessary to adopt evaluation metrics that are consistent with the nature of feature attribution explanations. While this work focuses on feature importance, the framework can be extended to other explanation modalities by selecting appropriate evaluation measures.

A wide range of metrics has been proposed in the literature to compare feature importance maps with a reference ground truth, ranging from simple overlap-based measures such as Intersection over Union [32] to task-specific metrics [3, 21, 58]. However, the problem of comparing attribution maps has been extensively studied in the context of visual saliency and human gaze prediction. In [45] is provided a comprehensive analysis of existing metrics and recommend a set of complementary measures based on their empirical properties.

Following these recommendations, we adopt three evaluation metrics. For each evaluation metric $e_j$ and $g_j$, as defined in Section 3, denotes the estimated importance of the component $j$ its actual importance.

*Similarity (*SIM*).* To quantify the agreement between ground truth and predicted importance distributions while preserving their spatial structure, we adopt the similarity metric (SIM) [22]. It is defined as:

$$\mathrm{SIM}(e, g) = \sum_{j=1}^{D} \min(e_j, g_j). \tag{6}$$

*Kullback–Leibler Divergence (*$\mathrm{KL}_{\mathrm{div}}$*).* To measure the discrepancy between ground truth and predicted importance distributions in terms of information loss, we adopt the Kullback–Leibler divergence ($\mathrm{KL}_{\mathrm{div}}$) [26]. Unlike SIM, this metric does not explicitly account for spatial structure. It is defined as:

$$\mathrm{KL}_{\mathrm{div}} = \sum_{j=1}^{D} g_j \cdot \log\left(\frac{g_j}{e_j + \epsilon} + \epsilon\right). \tag{7}$$

*Area Under the Curve (*$\mathrm{AUC}_{\mathrm{borji}}$*).* To evaluate the ability of attribution methods to discriminate relevant from non-relevant regions while mitigating center-bias effects, we adopt the $\mathrm{AUC}_{\mathrm{borji}}$ metric [6]. This measure is a variant of the traditional ROC-AUC [13] in which negative samples are drawn from random locations rather than uniformly across the domain. It is defined as:

$$\mathrm{AUC}_{\mathrm{borji}} = \int_0^1 \mathrm{TPR}(\tau) \cdot \mathrm{FPR}(\tau), \quad \mathrm{TPR}(\tau) = \frac{|\{j \in g \mid e_i \geq \tau\}|}{|g|}, \quad \mathrm{FPR}(\tau) = \frac{|\{j \in \mathbf{R} \mid e_j \geq \tau\}|}{|\mathbf{R}|}. \tag{8}$$

where $\mathbf{R}$ is a set of randomly sampled elements of the vectors. We employ SIM and $\mathrm{KL}_{\mathrm{div}}$ across all experimental domains. The $\mathrm{AUC}_{\mathrm{borji}}$ metric is used only for image and time series data, as tabular data lack an inherent spatial structure, making such metrics not applicable in this setting.

## 5.2. Predictive Models

To assess the effectiveness of the proposed SAIG framework, we train a predictive model for each data modality using the datasets generated as described in Section 4. These models serve as test beds for evaluating whether the ground truth explanations derived from our framework enable a reliable assessment of XAI methods. While the design of predictive models is not the primary focus of this work, their role is essential to ensure that the evaluation reflects the behavior of learned models across different data domains. We outline here the training procedures adopted in our study.

*Binary Images Model.* For the synthetically generated binary image datasets, we adopt a ResNet50 architecture [19] pre-trained on ImageNet [10]. To tailor the model to our task, we replace the original classification head with two fully connected layers and incorporate dropout regularization to mitigate overfitting. The first layer consists of 128 neurons, followed by a Rectified Linear Unit (ReLU) activation and a dropout layer with rate $p = 0.7$. The final output layer contains a single neuron with a Softmax activation function to produce class probabilities. During fine-tuning, the convolutional backbone is frozen, and only the newly introduced layers are updated.

The model is trained using the Adam optimizer with a learning rate of $10^{-5}$ and optimized with a cross-entropy loss function. Training is performed for 100 epochs with a batch size of 100. The training data consist of the binary image dataset generated according to the proposed methodology. In total, the dataset contains 47,470 images of size $256 \times 256$, of which 470 are reserved for validation (10 per class). The class distribution is consistent with that of the original dataset.

*Time Series Model.* For the synthetically generated time series datasets, we adopt a feature-based classification approach based on the MultiRocket transformation [52], a recent state-of-the-art method for efficient and accurate time series classification. MultiRocket extracts a large number of features by applying random convolutional kernels and pooling operations over temporal signals, resulting in a high-dimensional representation that captures diverse temporal patterns. As a preprocessing step, sequences of variable length are aligned through padding. Missing values at the end of each sequence are filled by propagating the last observed value forward, ensuring that all time series have a consistent length and can be processed uniformly. The MultiRocket transformation is then applied to the preprocessed data to generate a large set of features, which are subsequently normalized to ensure comparable contributions during classification.

For the classification stage, we employ a linear ridge classifier. Prior work on the ROCKET family of methods has shown that ridge-based linear models achieve strong performance when applied to such high-dimensional feature representations [9, 52]. Since ridge classifiers produce decision scores rather than probabilities, we apply a calibration step based on a sigmoid mapping, learned through cross-validation, to obtain calibrated probability estimates [17]. The regularization parameter is selected via cross-validation over a logarithmically spaced range between $10^{-3}$ and $10^{3}$. The resulting pipeline consists of four stages: sequence alignment via padding, feature extraction using MultiRocket, feature normalization, and classification with a regularized linear model followed by probability calibration. This configuration provides an efficient and competitive solution for large-scale time series classification while maintaining low computational complexity.

*Tabular Data Model.* For the tabular dataset, we adopt a Multilayer Perceptron (MLP) architecture. The model consists of three fully connected layers, each with 128 neurons, followed by Rectified Linear Unit (ReLU) activation functions to introduce non-linearity. Given the binary classification setting, the final output layer uses a sigmoid activation function to produce class probabilities. The network is trained using the Adam optimizer with an initial learning rate of $10^{-4}$. To improve convergence, we employ a step learning rate scheduler (StepLR), which reduces the learning rate by a factor of $\gamma = 0.85$ every 150 epochs. The model is optimized using a binary cross-entropy loss function over 5000 epochs with a full-batch training strategy.

The training dataset is generated according to the proposed SAIG methodology, using $K = 100$ different discrete levels, and consists of approximately 4 million samples with three input features. Of these, 10,000 samples are reserved for test, while the remainder are used for training. The target labels are derived from the *ssin-2c* attribution function introduced in [7]:

$$ssin\text{-}2c(x) = \left[ w_1 \cdot \sin\left(\frac{\pi}{2} x_1\right) + w_2 \cdot \sin\left(\frac{\pi}{2} x_2\right) + w_3 \cdot \sin\left(\frac{\pi}{2} x_3\right) \right] > 0.6, \tag{9}$$

where $w = [0.5, 0.25, 0.125]$. The output of $ssin\text{-}2c(x)$ is used as the training label.

*Predictive Performance.* The predictive performance obtained for each model is reported in Table 1. We observe near-perfect performance in the tabular setting, while lower accuracy is obtained for binary images and time series data. However, this does not limit the validity of our evaluation, as the intervention-based procedure provides ground truth importance independently of model accuracy. Therefore, even for imperfect models, the expected contribution of each input component remains well-defined[1].

[1]The implementation details, including code and trained model weights, are publicly available at `https://github.com/miquelmn/fb2ts`.

| Data Modality | Model | Precision | Recall | F1-Score |
|---|---|---|---|---|
| Binary images | ResNet50 | 0.91 | 0.88 | 0.89 |
| Time series | Multirocket w. Ridge | 0.64 | 0.63 | 0.63 |
| Tabular data | MLP | 0.98 | 0.98 | 0.98 |

**Table 1**
Performance measure results for each AI model used in each experiment.

| **Method** | **Hyperparameter** | **Value** |
|---|---|---|
| RISE | N Masks | 600 |
| | S Masks | 8 |
| | Probability remaining | 0.1 |
| LIME | Occlusion Value | 0 |
| | N Samples | 1500 |
| | Kernel Width | 0.25 |
| | Feature Selection | Highest Weights |
| | Kernel | Similarity kernel |
| | Distance Metric | Cosine |
| | Model Regressor | Ridge regression |
| Gradient | Absolute value | True |
| DeepLIFT | Baseline | 0 |
| | $\epsilon$ | $1 \cdot 10^{-10}$ |
| KernelSHAP | Occlusion Value | 0 |
| | N Samples | 25 |
| Integrated Gradients | N Steps | 50 |
| | Method | Gauss–Legendre quadrature |
| | Baseline | 0 |

**Table 2**
XAI method hyperparameter values for binary image experiments.

## 5.3. Explanation Methods

Following the definition of the evaluation methodology, we describe the XAI techniques considered in our comparative analysis. Our study focuses on local feature attribution methods, which explain individual predictions by assigning an importance score to each input feature, rather than characterizing the model at a global level. In total, we evaluate nine approaches: RISE [42], LIME [43], Gradient [48], DeepLIFT [47], KernelSHAP [31], Integrated Gradients (IG) [51], T-SHAP [40] (in both ROI and Window variants), and LORE [15]. Implementations for Gradient, DeepLIFT, KernelSHAP, and Integrated Gradients are based on the Captum library [24], while for RISE [41], LIME [44], LORE [46], and T-SHAP [40], we use the official implementations provided by the respective authors.

For the binary image experiments, we evaluate RISE, LIME, Gradient, DeepLIFT, KernelSHAP, and Integrated Gradients. Table 2 depicted each method hyperparameters used for binary images. RISE is configured with 600 random masks, a mask resolution of 8, and a retention probability of 0.1. For LIME and KernelSHAP, interpretable features are defined using the Quickshift algorithm [56] to generate superpixels. LIME is applied with 1500 perturbed samples, kernel width 0.25, cosine distance, ridge regression as surrogate model, and highest-weight feature selection. KernelSHAP uses an occlusion value of 0 and 25 samples. For gradient-based methods, Gradient is computed as the absolute value of the saliency map, DeepLIFT uses a zero baseline with $\epsilon = 10^{-10}$, and Integrated Gradients employs a zero baseline with 50 integration steps and Gauss–Legendre quadrature.

For the time series experiments, we evaluate RISE, LIME, KernelSHAP, and T-SHAP. Table 3 depicted each method hyperparameters used in this case. RISE is configured with 1000 random masks generated from a coarse binary grid of resolution 16, with each cell activated with probability 0.5, followed by linear upsampling and random shifting to match the original sequence length. This configuration produces smooth perturbations while preserving local temporal

| Method | Hyperparameter | Value |
|---|---|---|
| RISE | N Masks | 1000 |
| | Grid resolution | 16 |
| | Cell activation probability | 0.5 |
| | Mask generation | Linear upsampling + random shift |
| LIME | N Samples | 5000 |
| | Sampling | Gaussian |
| | Discretize continuous features | False |
| KernelSHAP | Background dataset | Balanced centroid |
| | Feature representation | Time steps as features |
| T-SHAP | Variants | ROI, Window |
| | Window length | 10% of time series length |
| | Stride | 5 |
| | Background dataset | Balanced centroid |

**Table 3**
XAI method hyperparameters values for time series experiments.

| Method | Hyperparameter | Value |
|---|---|---|
| LIME | Occlusion Value | 0 |
| | N Samples | 1500 |
| | Kernel Width | 0.25 |
| | Feature Selection | Highest Weights |
| | Kernel | Similarity kernel |
| | Distance Metric | Cosine |
| | Model Regressor | Ridge regression |
| Gradient | Absolute value | True |
| DeepLIFT | Baseline | 0 |
| | $\epsilon$ | $10^{-10}$ |
| KernelSHAP | Occlusion Value | 0 |
| | N Samples | 25 |
| Integrated Gradients | N Steps | 50 |
| | Method | Gauss–Legendre quadrature |
| | Baseline | 0 |
| LORE | Size | 1000 |
| | N Generation | 10 |
| | Other Class Ratio | 0.1 |

**Table 4**
XAI method hyperparameters values for tabular data experiments.

structure. LIME is applied using the default tabular configuration, generating 5000 perturbed samples via Gaussian sampling without discretization. KernelSHAP also follows the default tabular configuration, with a background dataset constructed using the *balanced centroid* strategy to ensure comparability with T-SHAP. T-SHAP is evaluated in both ROI and Window variants, with parameters set according to the original work [40]. In particular, the window length is set to 10% of the sequence length, the stride to 5, and the background dataset is constructed using balanced class centroids.

For the tabular experiments, we evaluate LORE, LIME, Gradient, DeepLIFT, KernelSHAP, and Integrated Gradients. LIME and KernelSHAP use the same configuration adopted for binary images, with an occlusion value of 0, 1500 perturbed samples for LIME, and 25 samples for KernelSHAP. Table 4 depicted each method hyperparameters used for this data typology. For gradient-based methods, Gradient is computed in absolute value, DeepLIFT uses a zero baseline

| | SIM ↑ | $AUC_{borji}$ ↑ | $KL_{div}$ ↓ |
|---|---|---|---|
| RISE [42] | $0.343 \pm 0.382$ | $0.573 \pm 0.273$ | $1.915 \pm 1.505$ |
| LIME [43] | $0.293 \pm 0.164$ | $\mathbf{0.932 \pm 0.083}$ | $\mathbf{1.217 \pm 0.664}$ |
| Gradient [48] | $0.378 \pm 0.427$ | $0.500 \pm 0.026$ | $1.903 \pm 1.517$ |
| DeepLIFT [47] | $\mathbf{0.380 \pm 0.429}$ | $0.555 \pm 0.123$ | $1.895 \pm 1.510$ |
| KernelSHAP [31] | $0.342 \pm 1.210$ | $0.848 \pm 0.165$ | $1.551 \pm 1.210$ |
| Integrated Gradients [51] | $\mathbf{0.380 \pm 0.430}$ | $0.510 \pm 0.098$ | $1.905 \pm 1.519$ |

**Table 5**
Metrics results to compare XAI methods output and proposed Ground Truth in the Binary images task. All reported value are the mean and standard deviation for the test set. Best values for each metric in bold.

with $\epsilon = 10^{-10}$, and Integrated Gradients employs a zero baseline with 50 integration steps and Gauss–Legendre quadrature. LORE [15] is adapted for feature attribution using an impurity-based approach following [35], with neighborhood size set to 1000, 10 generations, and other-class ratio equal to 0.1.

## 6. Results

We present here a comparative analysis of XAI methods using the SAIG frameworks introduced in Section 4, under the experimental setup described in the previous section. The goal is to assess the reliability of explanation methods across different data modalities. We begin with binary images, and subsequently extend the analysis to time series and tabular data, highlighting the strengths and limitations of each method in terms of explanation fidelity.

*Binary Images* Table 5 reports the comparison between the considered XAI methods and the corresponding ground truth explanations for binary images, using the evaluation metrics introduced earlier. The reported values correspond to the mean and standard deviation computed over the validation set for each method and metric. Figure 5 provides qualitative examples of the generated explanations. As previously discussed, the SIM metric primarily evaluates the spatial agreement between saliency maps. According to this measure, DeepLIFT [47], Integrated Gradients [51], and Gradient [48] emerge as the top-performing methods, yielding nearly identical results. This behavior is consistent with their shared underlying mechanism, as all three methods rely on backpropagating the model output to the input space. Nevertheless, their performance remains significantly below the theoretical optimum, i.e., SIM = 1.0, and shows only a marginal improvement over the lowest-performing method, LIME [43]. In contrast, the $AUC_{borji}$ metric reveals a different behavior. All methods, except LIME [43] and KernelSHAP [31], obtain values close to that of a random saliency map, i.e., $AUC_{borji} = 0.5$. This can be explained by the fact that $AUC_{borji}$ penalizes center-biased saliency maps [45], a common characteristic of gradient-based methods. LIME and KernelSHAP, on the other hand, rely on the Quickshift segmentation algorithm [56] to isolate object regions from the background. As a result, they assign little or no importance to background pixels, which leads to higher $AUC_{borji}$ scores. However, as indicated by the SIM metric, these methods still fail to correctly attribute importance to specific object components.

Overall, the evaluated methods struggle to consistently align with the ground truth, underscoring the limitations of current XAI techniques even in relatively structured visual settings.

*Time Series* Figure 6 shows a representative example of time series attributions. From top to bottom, the figure reports the ground truth (GT) and the explanations produced by LIME, RISE, T-SHAP (ROI and Window), and KernelSHAP. Higher color intensity indicates greater importance. For comparability, all attribution maps are transformed by taking the absolute value and applying min–max normalization.

Table 6 presents the comparison between XAI methods and ground truth explanations for the time series task, reporting mean and standard deviation values over the validation set for each metric. According to the SIM metric, RISE [42] achieves the best overall performance. This suggests that perturbation-based methods are particularly well suited for sequential data, as they directly estimate the effect of masking temporal segments on the model prediction. LIME [43] also achieves competitive performance, indicating that local surrogate models can partially capture relevant temporal patterns. KernelSHAP [31] performs substantially worse than all other methods. The extremely low SIM value indicates that the resulting importance distribution is almost uncorrelated with the ground truth, which is further confirmed by the high $KL_{div}$ divergence. A possible explanation is that KernelSHAP treats each time step

| | SIM ↑ | $AUC_{borji}$ ↑ | $KL_{div}$ ↓ |
|---|---|---|---|
| RISE | $\mathbf{0.522 \pm 0.136}$ | $\mathbf{0.709 \pm 0.161}$ | $\mathbf{0.790 \pm 0.405}$ |
| LIME | $0.442 \pm 0.199$ | $0.646 \pm 0.134$ | $0.907 \pm 0.510$ |
| Kernel SHAP | $0.007 \pm 0.003$ | $0.500 \pm 0.003$ | $5.315 \pm 0.509$ |
| T-SHAP ROI | $0.386 \pm 0.170$ | $0.490 \pm 0.089$ | $1.784 \pm 0.767$ |
| T-SHAP Window | $0.424 \pm 0.191$ | $0.490 \pm 0.132$ | $1.055 \pm 0.562$ |

**Table 6**
Metrics results to compare XAI methods output and proposed Ground Truth in the Time series task. All reported value are the mean and standard deviation for the validation set. Best values for each metric in bold.

| | SIM ↑ | $KL_{div}$ ↓ |
|---|---|---|
| LORE [15] | $0.784 \pm 0.177$ | $1.204 \pm 1.855$ |
| LIME [43] | $0.560 \pm 0.269$ | $3.806 \pm 3.535$ |
| Gradient [48] | $\mathbf{0.980 \pm 0.099}$ | $\mathbf{0.092 \pm 0.618}$ |
| DeepLIFT [47] | $0.636 \pm 0.221$ | $2.011 \pm 2.559$ |
| Kernel SHAP [31] | $0.601 \pm 0.229$ | $2.315 \pm 2.701$ |
| Integrated Gradients [51] | $0.642 \pm 0.221$ | $1.983 \pm 2.559$ |

**Table 7**
Metrics results to compare XAI methods output and proposed Ground Truth in the Tabular data task. All reported value are the mean and standard deviation for the validation set. Best values for each metric in bold.

as an independent feature, failing to capture the strong temporal dependencies inherent to time series data. Also, the performance may be influenced by the choice of background dataset. In order to maintain computational feasibility, we adopt the balanced centroid strategy instead of a larger background set, which may reduce the representativeness of the reference distribution and negatively affect the explanations. The two variants of T-SHAP exhibit intermediate performance. Both the ROI and Window strategies achieve moderate SIM values, indicating that they are able to partially identify relevant temporal regions. However, their $AUC_{borji}$ values remain close to 0.5, suggesting a limited ability to correctly rank important and non-important time steps. Among the two variants, the Window strategy shows slightly better alignment with the ground truth distribution, as reflected by the lower $KL_{div}$ divergence. The $AUC_{borji}$ metric further confirms that RISE and LIME outperform the remaining methods, indicating that perturbation-based and local surrogate approaches are more effective in isolating relevant temporal segments. Nevertheless, none of the evaluated methods closely matches the ground truth distribution, as evidenced by the relatively high $KL_{div}$ values across all methods. Since $KL_{div}$ evaluates the full probability distribution of the saliency map, it captures discrepancies in both central and peripheral regions, highlighting the difficulty of accurately reproducing the complete ground truth distribution.

*Tabular Data* Table 7 reports the evaluation results for the tabular experiments. As discussed in Section 5, the $AUC_{borji}$ metric is not considered in this setting due to the absence of spatial structure. Compared to the binary image and time series tasks, we observe a general improvement in the SIM metric. This is likely due to the lower complexity of the tabular setting. LIME [43] obtains the worst performance across both evaluation metrics. In contrast, Gradient [48] achieve near-perfect SIM scores. The second-best method, according to SIM measure, is LORE [15]. Integrated The $KL_{div}$ metric confirms this trend, with the same methods obtaining the best results. However, KernelSHAP performs slightly worse than the other two methods, which achieve values very close to the optimum.

## 7. Conclusion

In this work, we have introduced a unified intervention-based framework for generating Synthetic Artificial Intelligence Ground Truth (SAIG) across multiple data modalities, namely binary images, tabular data, and time series. In particular, we presented, to the best of our knowledge, the first local ground truth SAIG construction for time series data tabular data. By extending intervention-based principles beyond the image domain, the proposed framework enables the systematic and controlled evaluation of XAI methods under settings where the true contribution of each

input component is known by design. The experimental results highlight several limitations of current XAI techniques. Across all modalities, none of the evaluated methods consistently achieves high fidelity with respect to the ground truth explanations. While gradient-based methods perform relatively well in structured settings such as images and tabular data, their performance degrades when evaluated using stricter distribution-based metrics. Perturbation-based approaches, such as RISE, show stronger performance in time series data, suggesting that different methodological families are better suited to different data characteristics. These findings reinforce the idea that explanation quality is highly dependent on the underlying data modality and evaluation setting. In this context, our results provide empirical support for the *No Free Lunch Theorem for explanations* [18], indicating that no single XAI method can be expected to perform optimally across all tasks.

Despite these contributions, the proposed approach presents some limitations. As with existing SAIG methodologies, the evaluation is conducted in synthetic environments, where the data generation process is controlled. While this design enables the definition of ground truth explanations, it may not fully capture the complexity and variability of real-world scenarios. Hence, the insights obtained from such benchmarks should be interpreted as complementary to, rather than a replacement for, evaluations performed in real applications. Future research directions include extending the proposed framework to more complex and realistic data generation processes, as well as exploring hybrid evaluation strategies that combine synthetic ground truth with real-world validation. Furthermore, the development of standardized SAIG benchmarks could facilitate more consistent and reproducible comparisons across XAI methods, ultimately contributing to the design of more robust and reliable explanation techniques.

## CRediT authorship contribution statement

**Miquel Miró-Nicolau:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization.. **Francesco Spinnato:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization.. **Riccardo Guidotti:** Conceptualization, Supervision, Writing – Original Draft, Writing – Review & Editing..

## References


[1] Agarwal, C., Krishna, S., Saxena, E., Pawelczyk, M., Johnson, N., Puri, I., Zitnik, M., Lakkaraju, H., . OpenXAI: Towards a Transparent Evaluation of Model Explanations 35, 15784–15799.

[2] Amiri, S.S., Weber, R.O., Goel, P., Brooks, O., Gandley, A., Kitchell, B., Zehm, A., . Data Representing Ground-Truth Explanations to Evaluate XAI Methods. arXiv:2011.09892.

[3] Arias-Duart, A., Pares, F., Garcia-Gasulla, D., Gimenez-Abalos, V., 2022. Focus! Rating XAI Methods and Finding Biases, in: IEEE International Conference on Fuzzy Systems. arXiv:2109.15035.

[4] Barr, B., Xu, K., Silva, C., Bertini, E., Reilly, R., Bruss, C.B., Wittenbach, J.D., . Towards Ground Truth Explainability on Tabular Data. arXiv:2007.10532.

[5] Bodria, F., Giannotti, F., Guidotti, R., Naretto, F., Pedreschi, D., Rinzivillo, S., 2023. Benchmarking and survey of explanation methods for black box models. Data Min. Knowl. Discov. 37, 1719–1778. URL: https://doi.org/10.1007/s10618-023-00933-9, doi:10.1007/S10618-023-00933-9.

[6] Borji, A., Sihite, D.N., Itti, L., 2013. Quantitative analysis of human-model agreement in visual saliency modeling: A comparative study. IEEE Trans. Image Process. 22, 55–69. URL: https://doi.org/10.1109/TIP.2012.2210727, doi:10.1109/TIP.2012.2210727.

[7] Cortez, P., Embrechts, M.J., 2013. Using sensitivity analysis and visualization techniques to open black box data mining models. Inf. Sci. 225, 1–17. URL: https://doi.org/10.1016/j.ins.2012.10.039, doi:10.1016/J.INS.2012.10.039.

[8] Dembinsky, D., Lucieri, A., Frolov, S., Najjar, H., Watanabe, K., Dengel, A., 2026. Unifying VXAI: A systematic review and framework for the evaluation of explainable AI. Trans. Mach. Learn. Res. 2026. URL: https://openreview.net/forum?id=wAvFLe7o0E.

[9] Dempster, A., Petitjean, F., Webb, G.I., 2020. ROCKET: exceptionally fast and accurate time series classification using random convolutional kernels. Data Min. Knowl. Discov. 34, 1454–1495. URL: https://doi.org/10.1007/s10618-020-00701-z, doi:10.1007/S10618-020-00701-Z.

[10] Deng, J., Dong, W., Socher, R., Li, L., Li, K., Fei-Fei, L., 2009. Imagenet: A large-scale hierarchical image database, in: 2009 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR 2009), 20-25 June 2009, Miami, Florida, USA, IEEE Computer Society. pp. 248–255. URL: https://doi.org/10.1109/CVPR.2009.5206848, doi:10.1109/CVPR.2009.5206848.

[11] Doshi-Velez, F., Kim, B., 2018. Considerations for Evaluation and Generalization in Interpretable Machine Learning, in: Explainable and Interpretable Models in Computer Vision and Machine Learning. Cham, pp. 3–17.

[12] Ghorbani, A., Abid, A., Zou, J.Y., 2019. Interpretation of neural networks is fragile, in: The Thirty-Third AAAI Conference on Artificial Intelligence, AAAI 2019, The Thirty-First Innovative Applications of Artificial Intelligence Conference, IAAI 2019, The Ninth AAAI Symposium on Educational Advances in Artificial Intelligence, EAAI 2019, Honolulu, Hawaii, USA, January 27 - February 1, 2019, AAAI Press. pp. 3681–3688. URL: https://doi.org/10.1609/aaai.v33i01.33013681, doi:10.1609/AAAI.V33I01.33013681.

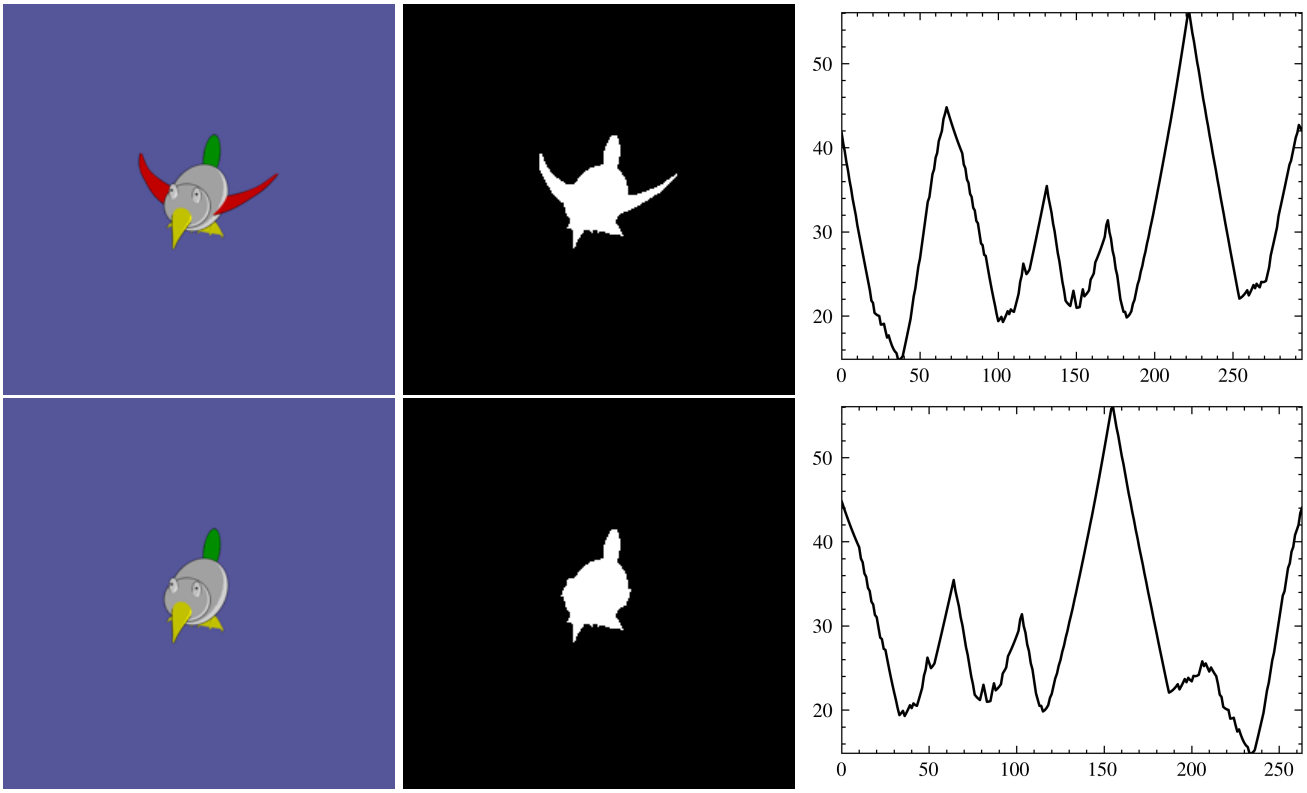



**Figure 1:** Illustration of the intervention procedure across data modalities: color images, binary images, and time series. The latter two are constructed from the original color image using the proposed framework.

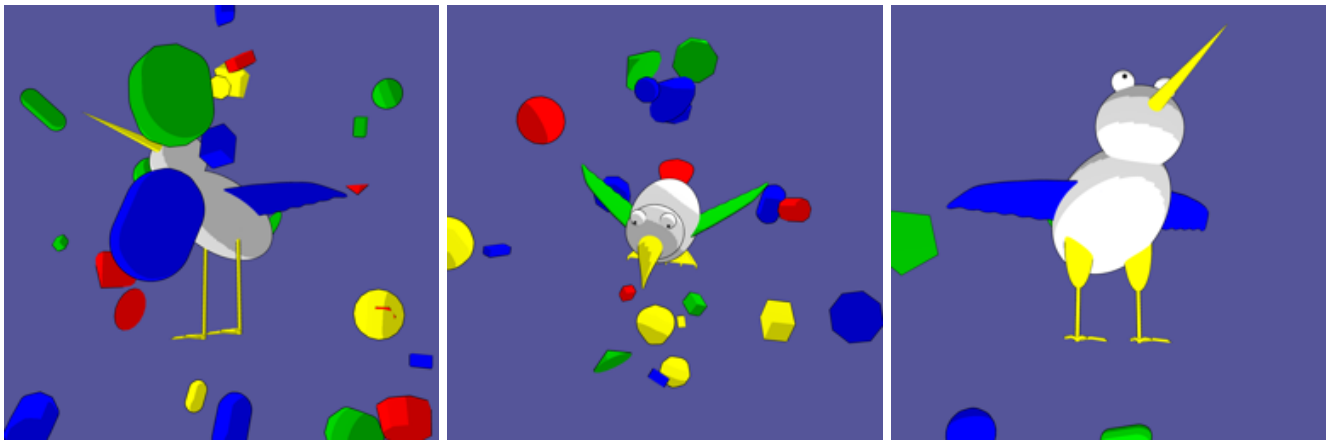


**Figure 2:** Representative samples from the FunnyBirds dataset across distinct classes. Each category is defined by a unique combination of geometric and color parts: wing color and foot shape (left image), wing and tail color (center image), and the foot shape (right image).

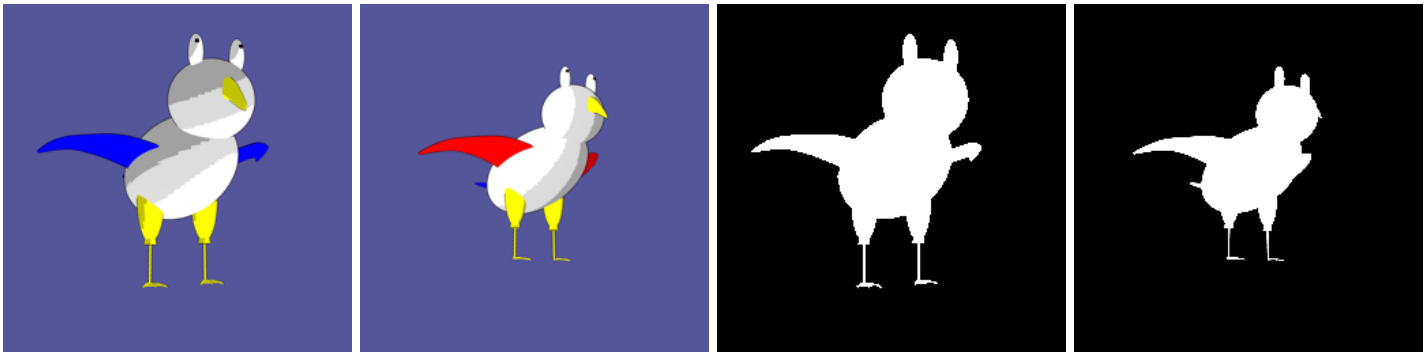


**Figure 3:** Illustration of class ambiguity induced by binarization. Two images belonging to different classes (first and second) become identical when converted to binary representations (third and fourth), as color information is removed.

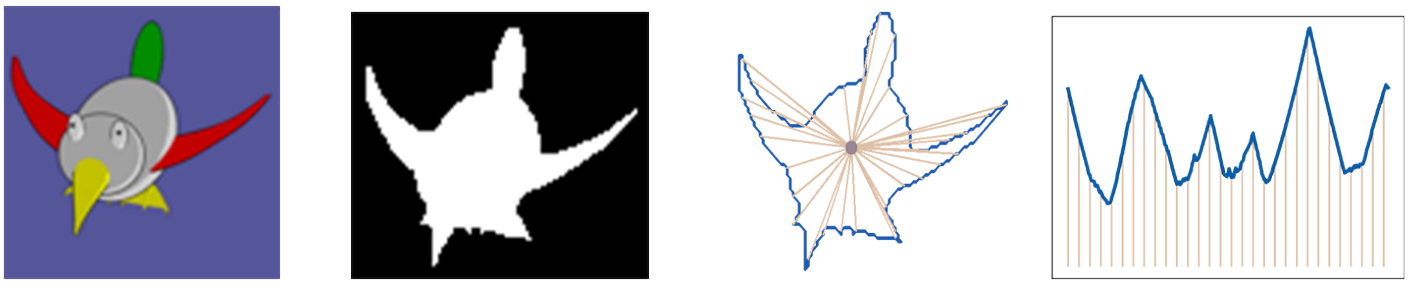
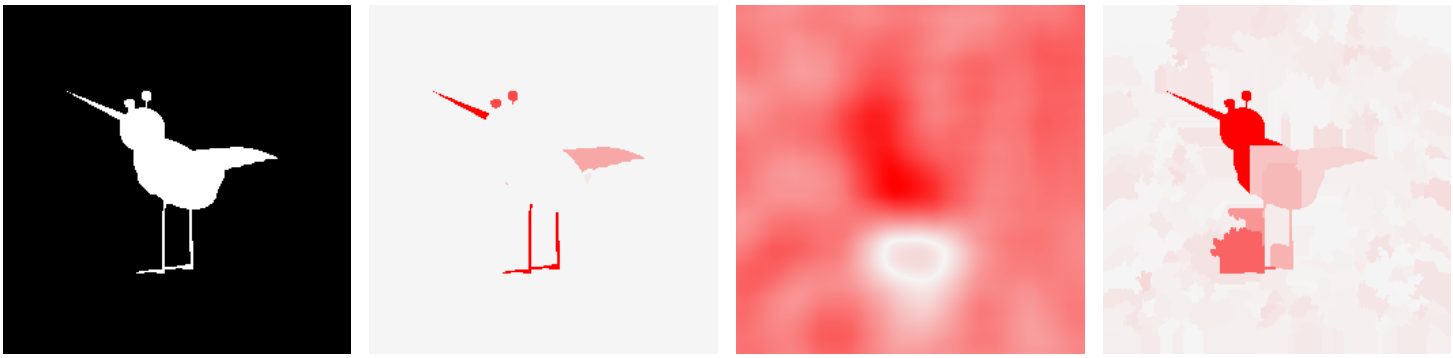
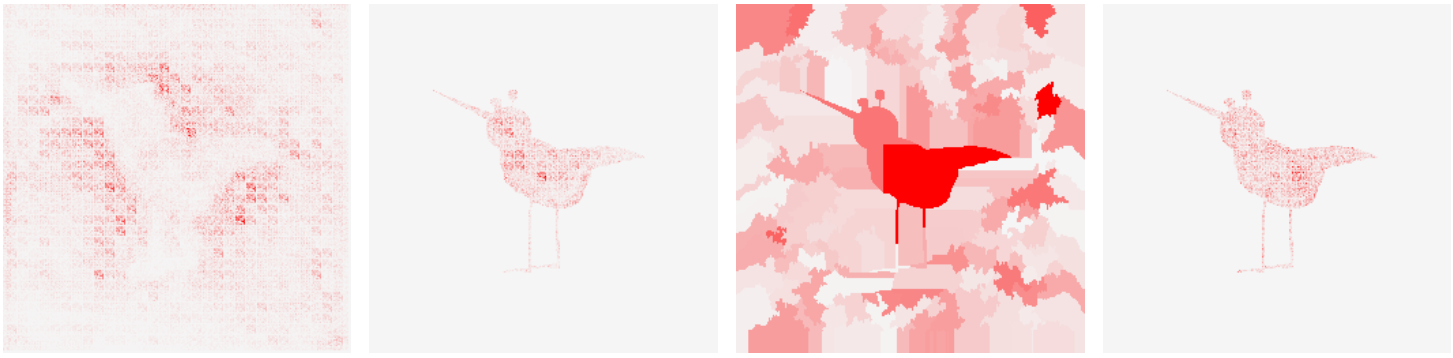


**Figure 4:** Step-by-step procedure for extracting a time series from a color image. From left to right: the original image, the binarized image, the measurement of radial distances from the object center to its contour, and the resulting time series plotted from these measurements



(a) Input Image (b) Pixel importance GT (c) RISE (d) LIME



(e) Grad-CAM (f) DeepLIFT (g) KernelSHAP (h) Int. Gradients

**Figure 5:** Comparison of different XAI saliency map techniques for the binary image dataset The more intense the color the more important the pixel.

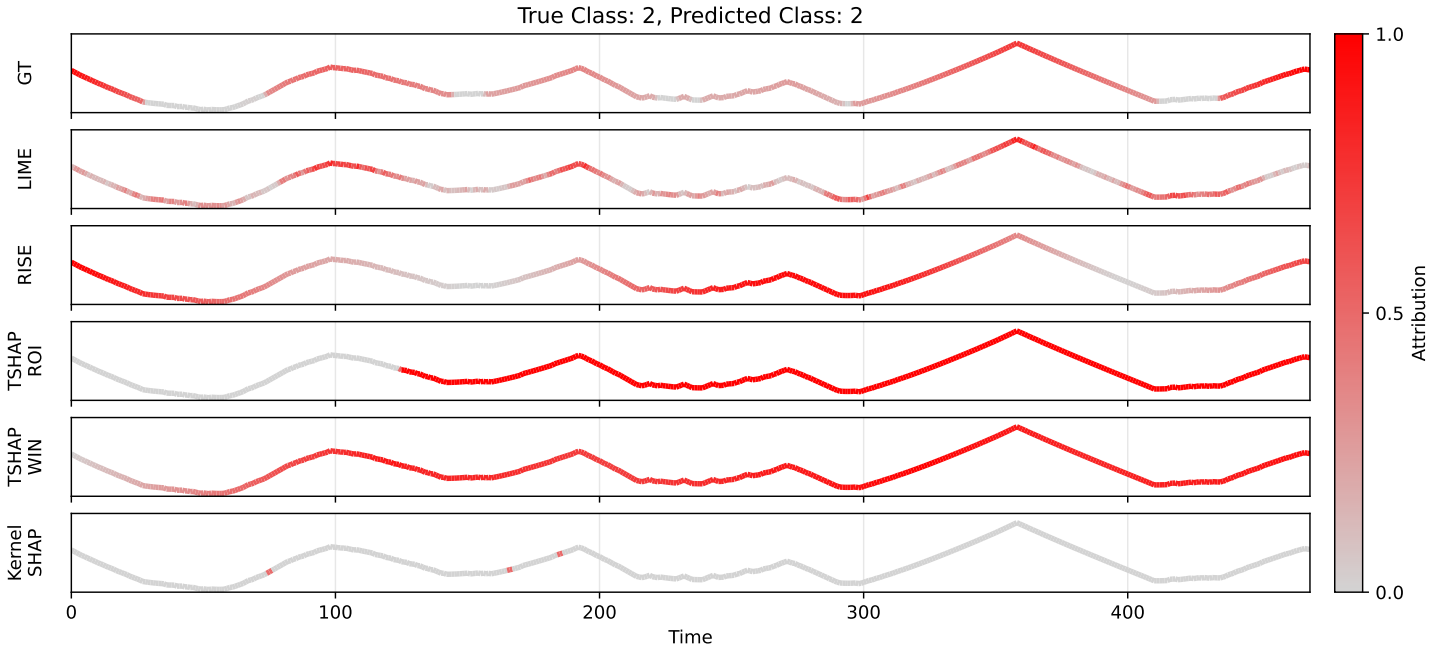



**Figure 6:** Time series attribution for an instance of class *2*. From top to bottom, the ground truth (GT), and the predicted attributions from LIME, RISE, TSHAP (ROI and WIN), and KernelSHAP. The more intense the color the more important the time series observation.